

The Heart

Richard N. Mitchell and Andrew J. Connolly

CHAPTER CONTENTS

Cardiac Structure and Specializations, 480

- Myocardium, 480
- Valves, 480
- Conduction System, 480
- Blood Supply, 480
- Effects of Aging on the Heart, 481

Overview of Cardiac Pathophysiology, 481

Heart Failure, 481

- Pathophysiology and Progression to Heart Failure, 482
- Left-Sided Heart Failure, 484*
- Right-Sided Heart Failure, 484*

Congenital Heart Disease, 485

- Incidence, 485
- Left-to-Right Shunts, 487
 - Atrial Septal Defect, 487*
 - Patent Foramen Ovale, 488*
 - Ventricular Septal Defect, 488*
 - Patent Ductus Arteriosus (PDA), 489*
- Right-to-Left Shunts, 489
 - Tetralogy of Fallot, 489*
 - Transposition of the Great Arteries, 490*
 - Tricuspid Atresia, 490*
- Obstructive Lesions, 491
 - Coarctation of the Aorta, 491*
 - Pulmonary Stenosis and Atresia, 491*
 - Aortic Stenosis and Atresia, 492*

Ischemic Heart Disease, 492

- Epidemiology, 492
- Pathogenesis, 493
 - Chronic Arterial Occlusion, 493*
 - Acute Plaque Change, 493*
- Angina Pectoris, 494
- Myocardial Infarction, 494
- Consequences and Complications of Myocardial Infarction, 501
- Chronic Ischemic Heart Disease, 502

Arrhythmias, 503

- Sudden Cardiac Death, 504

Hypertensive Heart Disease, 504

- Systemic (Left-Sided) Hypertensive Heart Disease, 504

- Pulmonary (Right-Sided) Hypertensive Heart Disease (Cor Pulmonale), 505

Valvular Heart Disease, 506

- Calcific Valvular Degeneration, 506
 - Calcific Aortic Stenosis, 506*
 - Calcific Stenosis of Congenitally Bicuspid Aortic Valve, 508*
 - Mitral Annular Calcification, 508*
- Mitral Valve Prolapse (Myxomatous Degeneration of the Mitral Valve), 508
- Rheumatic Fever and Rheumatic Heart Disease, 508
- Infective Endocarditis, 511
- Noninfected Vegetations, 513
 - Nonbacterial Thrombotic Endocarditis, 513*
 - Endocarditis of Systemic Lupus Erythematosus, 513*
- Carcinoid Heart Disease, 513
- Complications of Prosthetic Valves, 514

Cardiomyopathies, 515

- Dilated Cardiomyopathy, 516
- Arrhythmogenic Right Ventricular Cardiomyopathy, 519
- Hypertrophic Cardiomyopathy, 520
- Restrictive Cardiomyopathy, 521
 - Amyloidosis, 522*
- Myocarditis, 523
- Other Causes of Myocardial Disease, 525
 - Cardiotoxic Drugs, 525*
 - Radiation, 525*

Pericardial Disease, 525

- Pericardial Effusion and Hemopericardium, 525
- Pericarditis, 525
 - Acute Pericarditis, 525*
 - Chronic or Healed Pericarditis, 526*

Tumors of the Heart, 526

- Primary Cardiac Tumors, 526
- Metastatic Neoplasms, 527

Cardiac Transplantation, 528

Cardiac Devices, 529

- Ventricular Assist Devices, 529

The human heart is a remarkably efficient, durable, and reliable pump, propelling more than 7500 L of blood through the body each day, and beating more than 40 million times a year—the wellspring for tissue oxygenation, nutrition, and waste removal. In utero, the heart and vasculature are the first fully functional organ system (at roughly 8 weeks of gestation). Without a vascular supply and a beating heart,

further development cannot occur and fetal demise is inevitable. When the heart fails postnatally, the results are equally catastrophic. Cardiovascular disease (including coronary artery disease [CAD], stroke, and peripheral vascular disease) is the number-one cause of mortality worldwide. In the United States alone, cardiovascular disease accounts for roughly 1 in 5 deaths, totaling about 700,000 per

year—greater than the number of deaths caused by all forms of cancer combined.

This chapter begins with a brief review of the normal heart because most cardiac diseases manifest as structural and/or functional changes in one or more cardiac components. General principles underlying cardiac hypertrophy and failure—common end points of several forms of heart disease—are also discussed, before exploring the major categories of cardiac disease: congenital heart abnormalities, ischemic heart disease (IHD), hypertensive heart disease, diseases of cardiac valves, and primary myocardial disorders. The chapter concludes with a few comments about pericardial diseases and cardiac neoplasms, as well as cardiac transplantation and devices.

CARDIAC STRUCTURE AND SPECIALIZATIONS

Heart weight varies with body habitus, averaging approximately 0.4% to 0.5% of body weight (300 to 375 g in the average 75 kg adult US female and 360 to 450 g in the average 90 kg adult US male). Increased heart weight or ventricular thickness indicates hypertrophy, and an enlarged chamber size implies dilation; both reflect compensatory changes in response to volume and/or pressure overload (see later). Increased cardiac weight or size (or both)—resulting from hypertrophy and/or dilation—is called *cardiomegaly*.

Myocardium

The pumping function of the heart occurs through coordinated contraction (during systole) and relaxation (during diastole) of cardiac myocytes (the myocardium). Left ventricular myocytes are arranged in a spiral circumferential orientation that allows vigorous coordinated waves of contraction to spread from the cardiac apex to the base of the heart. In contrast, right ventricular myocytes are structured to provide contraction of longitudinal fibers to shorten the long axis, and inward movement of the free wall to cause a bellows-like effect; Although the heart is primarily a pump, it is worth remembering that it also has other functions. For example, atrial cardiomyocytes have cytoplasmic storage granules that contain atrial natriuretic peptide, and ventricular myocytes contain B-type natriuretic peptide. Both peptides are protein hormones that are released in response to increased stretch; they both promote arterial vasodilation and stimulate renal salt and water elimination (natriuresis and diuresis).

The myocardium of the heart is considered terminally differentiated and largely nonproliferative. While there are small numbers of myocardial stem cells in human hearts, these have very limited replicative potential after neonatal life and are not sufficient to repopulate even small areas of injury *in vivo*. *Ex vivo* expansion and subsequent administration of stem cell-derived myocytes into areas of myocardial injury has also been pursued, but integrating these cells into the preexisting conduction pathways of the host heart (lest there be autonomous arrhythmogenic foci) remains an unsolved problem.

Valves

The four cardiac valves—tricuspid, pulmonary, mitral, and aortic—maintain unidirectional blood flow. Valve function depends on the mobility, pliability, and structural integrity of the leaflets of the atrioventricular valves (tricuspid and mitral) or cusps of the semilunar valves (aortic and pulmonary).

The function of the semilunar valves depends on the integrity and coordinated movements of the cuspal attachments. Thus, dilation of the aortic root can result in valvular regurgitation. In contrast, the competence of the atrioventricular valves depends on the proper function not only of the leaflets but also the tendinous cords and the attached

papillary muscles of the ventricular wall, as well as the configuration of the ventricular chamber. Left ventricular dilation, a ruptured cord, or papillary muscle dysfunction can all interfere with mitral valve closure, causing valvular insufficiency.

Cardiac valves are lined by endothelium and are comprised of three layers, as follows:

- The *fibrosa* is a dense collagenous layer at the outflow surface, connected to the valvular supporting structures and providing mechanical integrity.
- The *spongiosa* is a central core of loose connective tissue.
- The *ventricularis* or *atrialis* (depending on which chamber it faces) is a layer rich in elastin on the inflow surface, providing leaflet recoil.

Because they are thin enough to be nourished by diffusion from the blood, normal leaflets and cusps have only scant blood vessels limited to the proximal portion of the valve. Valve endothelium also does *not express ABO or histocompatibility antigens*, so cryopreserved valvular tissues can be transplanted with relative impunity.

Conduction System

Coordinated contraction of the cardiac muscle depends on the initiation and rapid propagation of electrical impulses—accomplished through specialized myocytes in the conduction system. The frequency of electrical impulses in pacing elements is sensitive to neural inputs (e.g., vagal stimulation), extrinsic adrenergic agents (e.g., circulating adrenaline), hypoxia, and potassium concentration (i.e., hyperkalemia can block signal transmission altogether).

The components of the conduction system include:

- The *sinoatrial (SA) node* pacemaker, at the junction of the right atrial appendage and superior vena cava
- The *atrioventricular (AV) node*, located in the right atrium along the interatrial septum
- The *bundle of His (AV bundle)*, connecting the right atrium to the ventricular septum
- The *right and left bundle branch divisions* that stimulate their respective ventricles via further arborization into the *Purkinje network*

The cells of the cardiac conduction system can depolarize spontaneously, potentially enabling all of them to function as cardiac pacemakers. Because the normal rate of spontaneous depolarization in the SA node (60 to 100 beats/min) is faster than the other components, it normally sets the pace. However, if nodal tissues become dysfunctional, other cells in the conduction system can take over, generating, for example, a junctional escape rhythm (usually at a much slower rhythm). The AV node has a gatekeeper function; by delaying the transmission of signals from the atria to the ventricles, it ensures that atrial contraction precedes ventricular systole.

Blood Supply

Cardiac myocytes rely almost exclusively on oxidative phosphorylation for their energy needs. Besides a high density of mitochondria (20% to 30% of myocyte volume), myocardial energy generation also requires a constant supply of oxygenated blood, rendering myocardium extremely vulnerable to ischemia. Nutrients and oxygen are delivered via the coronary arteries, with ostia immediately distal to the aortic valve. These vessels initially course along the external surface of the heart (epicardial coronary arteries) and then penetrate the myocardium (intramural arteries). They subsequently branch into arterioles and a

dense network of capillaries, such that each myocyte contacts roughly three capillaries.

The right and left coronary arteries function as end arteries, although anatomically most hearts have numerous intercoronary anastomoses (connections called *collateral circulation*). Blood flow through the coronary arteries to the myocardium occurs during ventricular diastole, after closure of the aortic valve, and when the microcirculation is not compressed by cardiac contraction. At rest, diastole comprises approximately two-thirds of the cardiac cycle; with tachycardia (increased heart rate), the relative duration of diastole also shortens, thus potentially compromising cardiac perfusion.

Effects of Aging on the Heart

Most forms of heart disease become more prevalent with each advancing decade. Consequently, as the average populations in high-resource countries get progressively older, aging-associated changes in the cardiovascular system become ever more significant (Table 12.1).

The size of the left ventricular cavity, particularly in the base-to-apex dimension, is reduced in later life; this volume change is exacerbated by systemic hypertension as the basal ventricular septum protrudes into the left ventricular outflow tract (so-called *sigmoid septum*). Compared with younger myocardium, the “elderly” heart typically has fewer myocytes (due to degenerative attrition) and increased connective tissue; octogenarians (and older) also frequently have deposition of extracellular amyloid (most often transthyretin; Chapter 6) that stiffens the heart and reduces diastolic filling. Valvular aging changes are major contributors to significant valvular disease (see later), and progressive atherosclerosis, also strongly related to aging (Chapter 11), is the major cause of IHD.

TABLE 12.1 Changes in the Aging Heart

Chambers
Increased left atrial cavity size Decreased left ventricular cavity size Sigmoid-shaped ventricular septum
Valves
Aortic valve calcific deposits Mitral valve annular calcific deposits Fibrous thickening of leaflets Buckling of mitral leaflets toward the left atrium Lambl excrescences
Epicardial Coronary Arteries
Tortuosity Diminished compliance Calcific deposits Atherosclerotic plaque
Myocardium
Decreased mass Increased subepicardial fat Brown atrophy Lipofuscin deposition Basophilic degeneration Amyloid deposits
Aorta
Dilated ascending aorta with rightward shift Elongated (tortuous) thoracic aorta Sinotubular junction calcific deposits Elastic fragmentation and collagen accumulation Atherosclerotic plaque

OVERVIEW OF CARDIAC PATHOPHYSIOLOGY

Although a host of diseases can affect the cardiovascular system, the pathophysiologic pathways that result in a “broken” heart distill down to six principal mechanisms:

- **Pump failure.** In the most common type of failure, the cardiac muscle contracts weakly and the chambers cannot empty properly—so-called systolic dysfunction. In another common scenario, the myocardium cannot relax sufficiently to permit ventricular filling, resulting in diastolic dysfunction.
- **Obstruction to flow.** Lesions that prevent valve opening (e.g., calcific aortic valve stenosis) or cause increased ventricular chamber pressures (e.g., systemic hypertension or aortic coarctation) increase the work the myocardium must do to maintain cardiac output.
- **Regurgitant flow.** Valve pathology that allows backflow of blood results in increased volume workload and may overwhelm the pumping capacity of the affected chambers.
- **Shunted flow.** Defects (congenital or acquired) that divert blood inappropriately from one chamber to another, or from one vessel to another, lead to pressure and volume overloads.
- **Disorders of cardiac conduction.** Uncoordinated cardiac impulses or blocked conduction pathways can cause arrhythmias that slow contractions or prevent effective pumping altogether.
- **Rupture of the heart or major vessel.** Loss of circulatory continuity (e.g., a gunshot wound through the thoracic aorta or aortic dissection [Chapter 11]) may lead to massive blood loss, hypotensive shock, and death.

The pathologic basis of many heart disorders—from congenital heart disease through the various forms of injury including aging—involves an interplay between genetic and environmental factors. Certain polymorphisms are associated with risk of particular forms of heart disease, and, as discussed later, a number of adult-onset heart disorders have a genetic basis. Thus, molecular diagnostics is playing an increasingly important role in understanding the pathogenesis of cardiac disease in individual patients.

HEART FAILURE

Heart failure, often called congestive heart failure (CHF), is a common, usually progressive condition with a poor prognosis. In the United States, CHF affects more than 6.5 million individuals (approximately 2% of the population), requiring roughly 1 million hospitalizations annually, and contributes to the death of nearly 400,000 people each year. Roughly one-half of patients die within 5 years of receiving a diagnosis of CHF, and one in eight deaths in the United States includes heart failure as a contributory cause.

Heart failure is defined as the condition in which a heart cannot pump blood to adequately meet the metabolic demands of peripheral tissues or can do so only at elevated filling pressures. It is the common end stage of many forms of chronic heart disease, often emerging insidiously from the cumulative effects of chronic work overload (e.g., due to valve disease or hypertension) or IHD (e.g., after myocardial infarction [MI] with heart damage). However, acute hemodynamic stresses, such as fluid overload, abrupt valvular dysfunction, or MI, can all precipitate sudden CHF.

When cardiac workload increases or cardiac function is compromised, several physiologic mechanisms swing into action, and can at least initially maintain arterial pressure and organ perfusion:

- **Frank-Starling mechanism.** Increased filling volumes dilate the heart, stretching the myocyte to increase actin-myosin cross-bridge formation, thus causing a more forceful systolic contraction.

- **Activation of neurohumoral systems.** These augment heart function and/or regulate filling volumes and pressures (many of the therapies for CHF affect these systems when they become maladaptive).
 - **Release of norepinephrine** by adrenergic nerves of the autonomic nervous system, elevating heart rate, augmenting myocardial contractility, and increasing vascular resistance
 - **Activation of the renin-angiotensin-aldosterone system (RAAS)**, promoting water and salt retention (to augment circulatory volume) and increasing vascular tone
 - **Release of atrial natriuretic peptide**, counterbalancing the renin-angiotensin-aldosterone system through diuresis and vascular smooth muscle relaxation
- **Myocardial adaptations.** In many pathologic states, heart failure is preceded by cardiac hypertrophy, a compensatory response to increased mechanical work. Ventricular remodeling is the general term applied to the collective molecular, cellular, and structural changes that occur in response to injury or altered ventricular loading.

Although such adaptive mechanisms can potentially maintain adequate cardiac output in the face of acute perturbations, their capacity to do so may ultimately be overwhelmed. Heart failure can result from progressive deterioration of myocardial contractile function (*systolic dysfunction*)—reflected as a decrease in left ventricular ejection fraction (LVEF), the percentage of blood volume ejected from the ventricle during systole; normal is approximately 50% to 70% in a healthy heart). Reduction in EF can occur with ischemic injury, inadequate adaptation to pressure or volume overload due to hypertension or valvular disease, or ventricular dilation. Importantly, heart failure can also result from an inability of the heart chamber to expand and fill sufficiently during diastole (*diastolic dysfunction*), for example, due to left ventricular hypertrophy, myocardial fibrosis, constrictive pericarditis, or amyloid deposition. Clinically, heart failure patients are typically classified as having either *heart failure with reduced ejection fraction* (LVEF 40% or less; HFrEF), where systolic dysfunction dominates, or *heart failure with preserved ejection* (LVEF of 50% or more; HFpEF), where diastolic dysfunction is the dominant element. Patients with HFpEF tend to be older and female, with a higher prevalence of obesity and type 2 diabetes.

Whatever its basis, CHF is characterized by variable degrees of decreased cardiac output and tissue perfusion (forward failure), as

well as pooling of blood in the venous capacitance system (backward failure); the latter may cause pulmonary edema, peripheral edema, or both. As a result, many of the significant clinical features and morphologic changes noted in CHF are secondary to hypoxia and congestion in noncardiac peripheral tissues.

Pathophysiology and Progression to Heart Failure

A sustained increase in mechanical work of either ventricle due to pressure overload, volume overload, or trophic signals (e.g., those mediated through the activation of β -adrenergic receptors) causes myocytes to increase in size (cellular hypertrophy); cumulatively, this increases the size and weight of the heart (Fig. 12.1). Hypertrophy requires increased protein synthesis to form additional sarcomeres, as well as increased numbers of mitochondria. Hypertrophic myocytes typically exhibit enlarged or multiple nuclei, attributable to increased DNA ploidy resulting from DNA replication in the absence of cell division.

The pattern of hypertrophy reflects the nature of the stimulus.

- In **pressure-overload hypertrophy** (e.g., due to hypertension or aortic stenosis), new sarcomeres are predominantly assembled in parallel to the long axes of cells, expanding the cross-sectional area of myocytes in ventricles and causing a *concentric* increase in wall thickness.
- In contrast, **volume-overload hypertrophy** (e.g., due to valvular regurgitation) is characterized by new sarcomeres being assembled in series with existing sarcomeres, leading primarily to ventricular dilation. As a result, in dilation due to volume overload, or dilation that accompanies failure of a previously pressure-overloaded heart, the wall thickness may be increased, normal, or less than normal. Consequently, heart weight, rather than wall thickness, is the best measure of hypertrophy in dilated hearts.

Heart disease can lead to dramatic levels of cardiac hypertrophy. Patients with systemic hypertension, IHD, aortic stenosis, mitral regurgitation, or dilated cardiomyopathy frequently have heart weights double or triple the average; aortic regurgitation or hypertrophic cardiomyopathy can produce heart weights three- to four-fold greater than normal.

Important changes at the tissue and cell level occur with cardiac hypertrophy. Significantly, myocyte hypertrophy is not accompanied by a proportional increase in capillary density. As a result, the supply

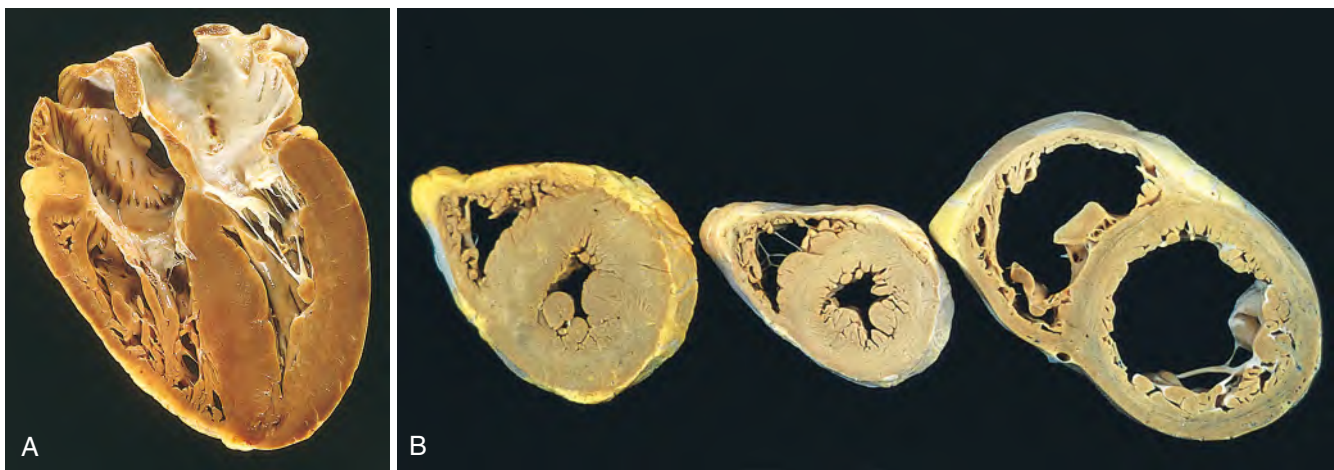


Fig. 12.1 Left ventricular hypertrophy. (A) Pressure hypertrophy due to left ventricular outflow obstruction. The left ventricle is on the lower right in this apical four-chamber view of the heart. (B) Left ventricular hypertrophy with and without dilation, viewed in transverse heart sections. Compared with a normal heart (center), the pressure-hypertrophied hearts (left and in A) have increased mass and a thick left ventricular wall, and the hypertrophied, dilated heart (right) has increased mass and an apparently normal wall thickness.

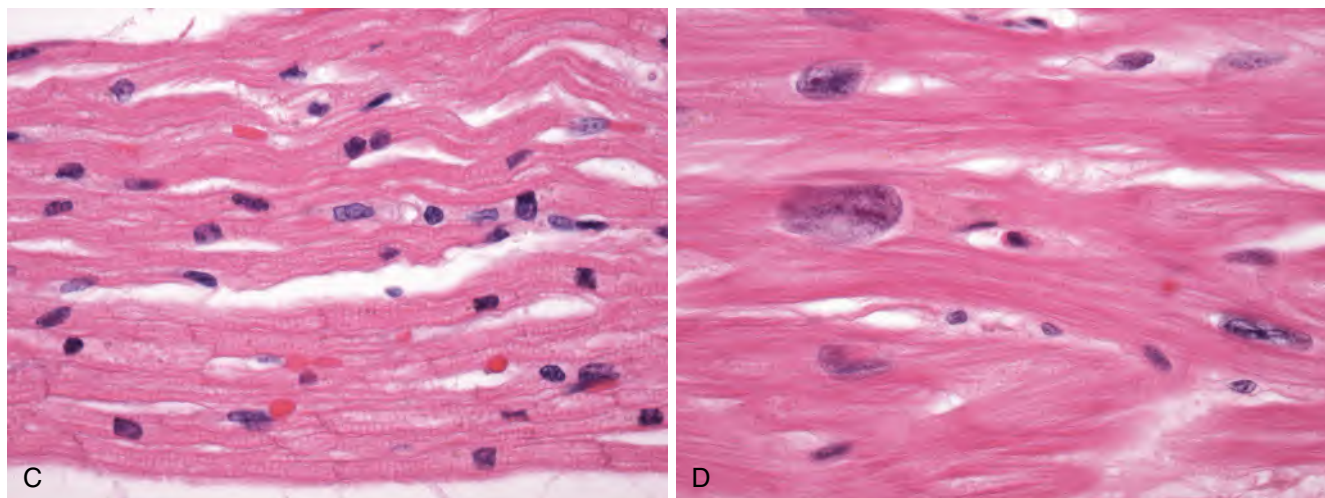


Fig. 12.1, cont'd (C) Normal myocardium. (D) Hypertrophied myocardium (C and D are photomicrographs at the same magnification). Note the increases in both cell size and nuclear size in the hypertrophied myocytes, and the interstitial cells remain small. (A and B, From Edwards WD. Cardiac anatomy and examination of cardiac specimens. In: Emmanouilides GC, ed. *Moss and Adams Heart Disease in Infants, Children, and Adolescents: Including the Fetus and Young Adults*. 5th ed. Williams & Wilkins; 1995:86.)

of oxygen and nutrients to the hypertrophied heart, particularly one undergoing pressure-overload hypertrophy, is more tenuous than in the normal heart. At the same time, cardiac hypertrophy is associated with heightened metabolic demands due to increases in mass, heart rate, and contractility (inotropic state, or force of contraction), all of which increase cardiac oxygen consumption. As a result of these changes, the hypertrophied heart is vulnerable to ischemia-related decompensation, which can evolve to cardiac failure. Adding insult to injury, the same mechanical forces that induce myocytes to hypertrophy also cause myocardial fibroblasts to increase extracellular matrix synthesis. Thus, hypertrophy is also typically accompanied by the deposition of fibrous tissue (interstitial fibrosis), causing increased resistance to diastolic filling.

Molecular changes in hypertrophied cardiomyocytes include the expression of immediate-early genes (e.g., *FOS*, *JUN*, *MYC*, and *EGR1*) (Chapter 2), which stimulate cellular growth and altered protein expression. With prolonged hemodynamic overload, myocytes even express genes that are normally only expressed during fetal cardiac development (including fetal forms of myosin, natriuretic peptides, and collagen). Recent single-cell analyses are providing new insight into gene expression in the numerous cell types and cell states in the myocardium, providing better understanding of the changes accompanying heart failure, including in stroma cells as well as myocardium.

The proposed sequence of initially adaptive—and later harmful—responses to increased cardiac work is summarized in Fig. 12.2. As illustrated, heart failure eventually supervenes. The degree of anatomic abnormality does not always reflect the severity of dysfunction; indeed, the gross appearance of the “failing heart” does not adequately convey the underlying structural, biochemical, and molecular basis for myocardial contractile failure. The hearts of patients with CHF are generally heavy and dilated, but can be relatively thin-walled, and histologically exhibit variable degrees of myocyte hypertrophy. Loss of myocardial mass due to myocardial infarction leads to work-related hypertrophy of the surrounding viable myocardium, whereas in valvular heart disease, the increased pressure or volume overloads the myocardium globally.

In contrast to the pathologic hypertrophy associated with persistent volume or pressure overload, regular exercise can promote potentially

beneficial physiologic hypertrophy. Aerobic exercise (e.g., long-distance running) tends to be associated with volume-load hypertrophy accompanied by increases in capillary density (unlike other forms of hypertrophy); regular aerobic activity also decreases the resting heart rate and blood pressure—effects that are beneficial. In comparison, purely static exercise (e.g., weightlifting) induces mild pressure hypertrophy (e.g., secondary to recurring Valsalva maneuvers) and less beneficial remodeling.

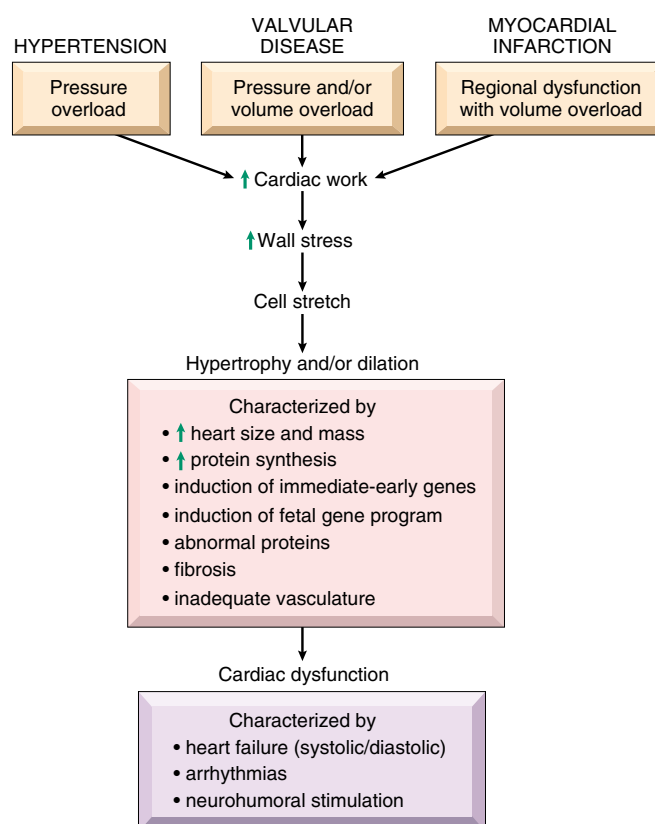


Fig. 12.2 Causes and consequences of cardiac hypertrophy.

The cardiovascular system is a closed circuit. Thus, although left-sided and right-sided failure can occur independently, failure of one side (particularly the left) often produces excessive strain on the other, terminating in global heart failure. Despite this interdependence, it is easiest to understand the pathology of heart failure by considering right- and left-sided heart failure separately.

Left-Sided Heart Failure

Left-sided heart failure is most often caused by one of the following:

- IHD
- Hypertension
- Aortic and mitral valvular disease
- Primary myocardial disease

The clinical and morphologic effects of left-sided CHF are a consequence of passive congestion (blood backing up in the pulmonary circulation), stasis of blood in the left-sided chambers, and inadequate perfusion of downstream tissues, leading to organ dysfunction.

MORPHOLOGY

Heart. The findings in the heart depend on the disease process, ranging from MIs, to stenotic or regurgitant valves, to intrinsic myocardial pathology. Except for failure caused by mitral valve stenosis or unusual restrictive cardiomyopathies (described later), the left ventricle is usually hypertrophied and often dilated, sometimes massively. Left ventricular diastolic dysfunction or ventricular dilation with mitral valve incompetence causes secondary dilation of the left atrium, increasing the risk of atrial fibrillation. This in turn results in stasis of blood, particularly in the atrial appendage, which is a common site of thrombus formation. The microscopic changes in the failing heart are nonspecific and consists of variable degrees of myocyte hypertrophy and interstitial fibrosis.

Lungs. Pulmonary congestion and **edema** produce heavy, wet lungs, as described elsewhere (Chapters 4 and 15). Pulmonary changes—from mildest to most severe—include (1) perivascular and interstitial edema, particularly in the interlobular septa; (2) progressive edematous widening of alveolar septa; and (3) accumulation of edema fluid in the alveolar spaces. Extravasated red cells and plasma proteins in the alveoli are phagocytosed and digested by macrophages; the accumulated iron is stored as hemosiderin. These hemosiderin-laden macrophages (also known as **heart failure cells**) are telltale signs of previous episodes of pulmonary edema. Pleural effusions (typically serous) arise from elevated pleural capillary and lymphatic pressure and the resultant transudation of fluid into the pleural cavities.

Early left-sided heart failure symptoms are related to pulmonary congestion and edema. Initially, cough and dyspnea (breathlessness) may occur only with exertion. As CHF progresses, worsening pulmonary edema may cause orthopnea (dyspnea when supine, relieved by sitting or standing) or paroxysmal nocturnal dyspnea (dyspnea usually occurring at night that is so severe that it induces a feeling of suffocation). Dyspnea at rest may follow. The respiratory symptoms are characteristically associated with fine rales at the lung bases, caused when edematous pulmonary alveoli snap open during inspiration. Other manifestations of left ventricular failure include an enlarged heart (cardiomegaly, apparent on imaging), tachycardia, a third heart sound due to volume overload (S3), or a fourth heart sound (S4) due to increased myocardial stiffness. If heart failure is associated with progressive ventricular dilation, the papillary muscles are displaced outward, causing mitral regurgitation. Subsequent chronic dilation of the left atrium can cause atrial fibrillation, and such uncoordinated, chaotic atrial contractions reduce the atrial contribution to ventricular filling, thus reducing the ventricular stroke volume.

In moderate CHF, a reduced ejection fraction leads to diminished renal perfusion, causing activation of the renin-angiotensin-aldosterone

system as a compensatory mechanism to correct the “perceived” hypotension. This leads to salt and water retention and expansion of the interstitial and intravascular fluid volumes (Chapters 4 and 11) exacerbating pulmonary edema. If the hypoperfusion of the kidney becomes sufficiently severe, impaired excretion of nitrogenous products may cause azotemia (called prerenal azotemia; Chapter 20). In far-advanced CHF, cerebral hypoperfusion can give rise to hypoxic encephalopathy (Chapter 28), marked by irritability, loss of attention span, and restlessness that can progress to stupor and coma with ischemic cerebral injury.

Left-sided heart failure can be divided into systolic and diastolic failure:

- **Systolic failure** is defined by reduced ejection fraction (pump failure; heart failure with reduced ejection fraction [HFrEF]) and can be caused by any of the many disorders that damage or derange the contractile function of the left ventricle.
- In **diastolic failure**, the left ventricle is abnormally stiff and cannot relax during diastole (heart failure with preserved ejection fraction [HFpEF]). Thus, although cardiac function is relatively preserved at rest, the heart is unable to increase its output in response to increases in the metabolic demands of peripheral tissues (e.g., during exercise). Moreover, because the left ventricle cannot expand normally, any increase in filling pressure is immediately transferred back into the pulmonary circulation, producing pulmonary edema. Hypertension is the most common underlying etiology; diabetes, obesity, and bilateral renal artery stenosis can also be causal. Reduced left ventricular relaxation may stem from myocardial fibrosis (e.g., in cardiomyopathies and IHD) or infiltrative disorders associated with restrictive cardiomyopathies (e.g., cardiac amyloidosis). Diastolic failure may appear in older patients without any known predisposing factors, possibly as an exaggeration of the normal fibrotic stiffening of the heart with age. Constrictive pericarditis (discussed later) can also limit myocardial relaxation and therefore mimics primary diastolic dysfunction.

Right-Sided Heart Failure

Right-sided heart failure is most commonly caused by left-sided heart failure, as any increase in pressure in the pulmonary circulation from left-sided failure inevitably burdens the right side of the heart. Consequently, the causes of right-sided heart failure include all the etiologies for left-sided heart failure. Isolated right-sided heart failure is infrequent and typically occurs in patients with one of a variety of disorders affecting the lungs; hence it is often referred to as *cor pulmonale*. Besides parenchymal lung diseases, *cor pulmonale* can also arise secondary to disorders that affect the pulmonary vasculature, for example, primary pulmonary hypertension (Chapter 15), recurrent pulmonary thromboembolism (Chapter 4), or conditions that cause pulmonary vasoconstriction (obstructive sleep apnea, altitude sickness). The common feature of these disorders is pulmonary hypertension (discussed later), which results in hypertrophy and dilation of the right side of the heart. In extreme cases, increased right ventricular pressures cause leftward bulging of the interventricular septum and left ventricular dysfunction. The major morphologic and clinical effects of primary right-sided heart failure differ from those of left-sided heart failure in that pulmonary congestion is minimal while engorgement of the systemic and portal venous systems is pronounced.

MORPHOLOGY

Heart. As in left heart failure, the cardiac morphology varies with cause. Rarely, structural defects such as tricuspid or pulmonary valvular abnormalities or endocardial fibrosis (as in carcinoid heart disease) may be present. However,

because isolated right heart failure is most often caused by lung disease, most cases exhibit only hypertrophy and dilation of the right atrium and ventricle.

Liver and Portal System. Congestion of the hepatic and portal vessels may produce pathologic changes in the liver, the spleen, and the gastrointestinal tract. The liver is usually increased in size and weight (**congestive hepatomegaly**) caused by **passive congestion**, greatest around the central veins (Chapter 4). Grossly, this is reflected as congested red-brown pericentral zones, with relatively normal-colored tan periportal regions, producing the characteristic “nutmeg liver” appearance (Chapter 4). In some instances, especially when left-sided heart failure with hypoperfusion is also present, severe centrilobular hypoxia produces **centrilobular necrosis**. With longstanding severe right-sided heart failure, the central areas can become fibrotic, culminating in **cardiac cirrhosis** (Chapter 18). Portal venous hypertension also causes enlargement of the spleen with platelet sequestration (**congestive splenomegaly**) and can also contribute to chronic congestion and edema of the bowel wall. The latter may be sufficiently severe as to interfere with nutrient (and/or drug) absorption.

Pleural, Pericardial, and Peritoneal Spaces. Systemic venous congestion can lead to fluid accumulation (**effusions**) in the pleural, pericardial, or peritoneal spaces (a peritoneal effusion is also called **ascites**). Large pleural effusions can impact lung inflation, causing atelectasis, and substantial ascites can also limit diaphragmatic excursion, causing dyspnea on a purely mechanical basis.

Subcutaneous Tissues. Edema of the peripheral and dependent portions of the body, especially foot/ankle (pedal) and pretibial edema, is a hallmark of right-sided heart failure. In chronically bedridden patients, sacral skin edema may predominate. Generalized massive edema (**anasarca**) can also occur.

The kidney and the brain are also prominently affected in right-sided heart failure. Renal congestion is more marked with right-sided than with left-sided heart failure, leading to greater fluid retention and peripheral edema, and more pronounced azotemia. Venous congestion and hypoxia of the central nervous system can also produce deficits of mental function akin to those seen in left-sided heart failure with poor systemic perfusion.

Although we have discussed right and left heart failure separately, it is again worth emphasizing that in many cases of chronic cardiac decompensation, patients develop biventricular CHF and symptoms reflecting both right-sided and left-sided heart failure. Besides a careful history and physical examination, serum levels of B-type (or brain) natriuretic factor (BNP) are a useful means to quantitatively assess the extent of CHF. Recall that BNP is released by ventricular cardiomyocytes during increased wall stress; thus, a low value has a high negative predictive value for CHF. Echocardiography is also an extremely valuable tool in following patients with CHF, providing a measure of ejection fraction, wall motion, valvular function, and possible mural thrombosis.

Treatment for CHF is initially focused on correcting any underlying cause, for example, a valvular defect or inadequate cardiac perfusion. Beyond that, the clinical approach includes salt restriction or pharmacologic agents that variously reduce volume overload (e.g., diuretics), increase myocardial contractility (so-called positive inotropes), or reduce afterload (via adrenergic blockade or inhibitors of angiotensin-converting enzymes [ACE]). Although many of these medications provide benefit through effects on neurohumoral pathways, ACE inhibitors also limit myocyte hypertrophy and cardiac remodeling. Although cardiac resynchronization therapy (exogenous pacing of both the right and left ventricles) and mechanical ventricular assist devices (VADs, discussed later) have also been added to the cardiologist's armamentarium, CHF remains a serious cause of morbidity and mortality.

KEY CONCEPTS

Heart Failure

- Heart failure occurs when the heart is unable to provide adequate perfusion to meet the metabolic requirements of peripheral tissues; inadequate cardiac output is usually accompanied by increased congestion of the venous circulation.
- Prolonged hemodynamic overload (pressure and volume), β -adrenergic signals, expression of immediate-early genes are likely to be associated with cardiac myocyte hypertrophy.
- Left-sided heart failure is most commonly due to IHD, systemic hypertension, mitral or aortic valve disease, and primary diseases of the myocardium; symptoms are mainly a consequence of pulmonary congestion and edema, although systemic hypoperfusion can cause secondary renal and cerebral dysfunction.
- Right-sided heart failure is most often due to left heart failure, and less commonly to primary pulmonary disorders; symptoms are chiefly related to peripheral edema and visceral congestion.

CONGENITAL HEART DISEASE

Congenital heart disease (CHD) refers to abnormalities of the heart or great vessels that are present at birth. Most CHD arises from faulty embryogenesis during gestational weeks 3 to 8, when major cardiovascular structures form and begin to function. The most severe anomalies preclude intrauterine survival, and significant heart malformations are common among stillborn infants. On the other hand, circumscribed defects affecting discrete regions of the heart or individual chambers can be compatible with live birth. In this category are:

- **Septal defects**, or “holes in the heart,” including atrial septal defects (ASDs) or ventricular septal defects (VSDs)
- **Stenotic lesions**, either at the level of valves, or the entire cardiac chamber as in hypoplastic left heart syndrome
- **Outflow tract anomalies** including inappropriate routing of the great vessels from the ventricles, or anomalous coronary arteries

Such “tolerated” forms of CHD usually produce clinically important manifestations only after birth—uncovered by the transition from fetal to perinatal circulation. Roughly one-half will be diagnosed in the first year of life, although some milder forms may not be discovered until adulthood (e.g., ASD).

Incidence

The incidence of CHD depends on what is counted as a defect. Thus, if echocardiography is performed routinely on neonates, small muscular VSDs or ASDs are detected in over 5% of live births. These typically close spontaneously in the first year of life, and they probably should not be added to the burden of CHD. Similarly, bicuspid aortic valve—with an incidence of 1% to 2%—persists beyond infancy, but often has modest manifestations and may not become evident until late adulthood. If the accounting is restricted to more serious defects, the worldwide incidence of congenital cardiovascular malformations is slightly less than 1%—still ranking CHD among the most prevalent birth defects. Twelve disorders account for about 85% of cases; their frequencies are listed in [Table 12.2](#).

The number of individuals who survive into adulthood with CHD is increasing at a rate of roughly 5% per year and is estimated at nearly 1.5 million people in the United States alone. Many have benefited from surgical advances that permit early postnatal repair of structural defects, and specialized management of adults with CHD is a burgeoning field.

Etiology and Pathogenesis

The diverse malformations seen in CHD are caused by errors during the complex migration of cells and reshaping of tissue that enables

TABLE 12.2 Frequencies of Congenital Cardiac Malformations^a

Malformation	Incidence (%) per Million Live Births
Ventricular septal defect	4482 (42)
Atrial septal defect	1043 (10)
Pulmonary stenosis	836 (8)
Patent ductus arteriosus	781 (7)
Tetralogy of Fallot	577 (5)
Coarctation of the aorta	492 (5)
Atrioventricular septal defect	396 (4)
Aortic stenosis	388 (4)
Transposition of the great arteries	388 (4)
Truncus arteriosus	136 (1)
Total anomalous pulmonary venous connection	120 (1)
Tricuspid atresia	118 (1)
Total	9757

^aPresented as upper quartile of 44 published studies. Percentages do not add up to 100% because of rounding. Does not include bicuspid aortic valves. Data from Hoffman JI, Kaplan S. The incidence of congenital heart disease. *J Am Coll Cardiol.* 2002;39(12):1890-1900.

cardiac morphogenesis. Derived from cells in the lateral mesoderm, the earliest cardiac precursors move to the midline in two migratory waves (called the first and second heart fields) within the first 15 days of fetal development. These multipotent progenitor cells, which can produce all the major cell types of the heart (endocardium, myocardium, and smooth muscle cells), differentiate to form two parts of the heart; cells in the first wave will form the developing left ventricle, whereas those in the second wave will form the outflow tract, right ventricle, and most of the atria. Thus, defects in this process can explain some of the CHDs involving discrete structures. By day 20 of development, the nascent heart has become a beating tube, which begins to form the basic heart chambers roughly 8 days later. At about the same time, neural crest-derived cells migrate into the outflow tract, where they participate in the septation of the aortic and pulmonic outflow tracts and the formation of the aortic arch; and interstitial connective tissue expands to become definitive endocardial cushions that will form the atrioventricular canals and outflow tracts. By day 50, further septation of the ventricles, atria, and atrioventricular valves produces a four-chambered heart.

Proper orchestration of these remarkable transformations depends on a network of transcription factors that are regulated by several signaling pathways, particularly the Wnt, Hedgehog, vascular endothelial growth factor (VEGF), bone morphogenetic protein, transforming growth factor- β (TGF- β), fibroblast growth factor, and Notch pathways (Chapter 1). Many of the inherited defects that affect heart development involve genes that encode components of these pathways or downstream transcription factors; these typically cause partial loss of function and are autosomal dominant. In addition, specific microRNAs play critical roles in cardiac development by coordinating patterns and levels of transcription factor expression.

Environmental exposures (e.g., congenital rubella infection, teratogens—including some therapeutic drugs, and gestational diabetes) and genetic factors are the best established causes of CHD but still account for a minority of cases. Nutritional factors can also influence risk; for example, folate supplementation during early pregnancy reduces CHD incidence.

Genetic factors include specific loci implicated in familial forms of CHD and certain chromosomal abnormalities (e.g., trisomies 13, 15, 18, and 21, and monosomy X/Turner syndrome). Indeed, the most common known genetic cause of CHD is trisomy 21 (Down syndrome); roughly 40% of patients with Down syndrome have one or more heart defects, most often affecting structures derived from the second migratory wave of cells (e.g., the atrioventricular septa). The role of chromosomal anomalies in CHD likely involves the dysregulated expression of multiple genes.

A notable example of a small chromosomal lesion causing CHD is deletion of chromosome 22q11.2, occurring in patients with DiGeorge syndrome. In this syndrome, the fourth branchial arch and the derivatives of the third and fourth pharyngeal pouches (which contribute to the formation of the thymus, parathyroid glands, and heart) develop abnormally. Of the 30 or so genes present on this chromosome segment, deletion of the *TBX1* transcription factor gene is probably the responsible culprit. *TBX1* regulates neural crest migration as well as the expansion of cardiac progenitors in the second migratory wave.

In the case of single-gene mutations, the affected genes encode proteins belonging to several different functional classes (Table 12.3); as mentioned earlier, of these involve signaling molecules or downstream transcription factors. Because affected patients are heterozy-

TABLE 12.3 Examples of Gene Defects Associated With Congenital Heart Disease^a

Disorder	Gene(s)	Gene Product Function
Nonsyndromic		
ASD or conduction defects	<i>NKX2.5</i>	Transcription factor
ASD or VSD	<i>GATA4</i>	Transcription factor
Tetralogy of Fallot	<i>ZFPM2</i> or <i>NKX2.5</i>	Transcription factors
Syndromic^b		
Alagille syndrome: pulmonary artery stenosis or tetralogy of Fallot	<i>JAG1</i> or <i>NOTCH2</i>	Notch signaling
Char syndrome: PDA	<i>TFAP2B</i>	Transcription factor
CHARGE syndrome: ASD, VSD, PDA, or hypoplastic right side of the heart	<i>CHD7</i>	Helicase-binding protein
DiGeorge syndrome: ASD, VSD, or outflow tract obstruction	<i>TBX1</i>	Transcription factor
Holt-Oram syndrome: ASD, VSD, or conduction defect	<i>TBX5</i>	Transcription factor
Noonan syndrome: pulmonary valve stenosis, VSD, or hypertrophic cardiomyopathy	<i>PTPN11</i> , <i>KRAS</i> , <i>SOS1</i>	Ras signaling

^aDifferent mutations can cause the same phenotype, and mutations in some genes can cause multiple phenotypes (e.g., *NKX2.5*). Many of these congenital lesions also can occur sporadically, without specific genetic mutation.

^bOnly the cardiac manifestations of the syndrome are listed; the other skeletal, facial, neurologic, and visceral changes are not.

ASD, Atrial septal defect; CHARGE, posterior coloboma, heart defect, choanal atresia, retardation, genital and ear anomalies; PDA, patent ductus arteriosus; VSD, ventricular septal defect.

gous for such mutations, it follows that a 50% reduction in the activity of these factors (or even less) may be sufficient to derange cardiac development. Even relatively minor decrements in the activity of particular genes can result in significant defects. Thus, transient environmental stresses during the first trimester of pregnancy that alter the synthesis or activity of these same genes can likely lead to acquired defects that mimic those produced by heritable mutations. In addition, many of the transcription factors act in concert, providing a rationale for why mutations in any one of several genes can produce similar defects. Thus, GATA4, TBX5, and NKX2.5, three transcription factors that are mutated in some patients with atrial and ventricular septal defects, all bind to one another and coregulate the expression of target genes required for proper cardiac development.

Other single-gene mutations can alter structural proteins or affect signaling pathway molecules. Thus, mutations in genes encoding various components of the Notch pathway (Chapter 1) are associated with a variety of congenital heart defects, including bicuspid aortic valve (*NOTCH1*, discussed later) and tetralogy of Fallot (TOF; *JAG1* and *NOTCH2*). As described in Chapter 5 and 11, fibrillin mutations underlie Marfan syndrome—associated with valvular defects and aortic aneurysms. Although fibrillin is an important structural protein in the extracellular matrix (ECM), it is also an important negative regulator of TGF- β signaling, and hyperactive TGF- β signaling contributes to the cardiovascular abnormalities in Marfan syndrome and the related Loeys-Dietz syndrome (Chapter 11).

Despite these advances, our understanding of the mechanisms underlying CHD remains rudimentary; the precise cause is unknown in almost 90% of cases. Most affected patients have no identifiable genetic defect, and even in those that do, the nature and severity of the defect are highly variable, suggesting interaction with environmental factors. Guidelines for genetic testing are evolving; patients with a broader noncardiac syndrome or a family history of CHD may warrant evaluation.

Clinical Features

Most of the various structural anomalies in CHD can be categorized as shunts and/or obstructions. A shunt is an abnormal communication between chambers or blood vessels. Abnormal channels permit blood flow down pressure gradients from the left (systemic) side to the right (pulmonary) side of the circulation or vice versa.

Right-to-left shunts occur when blood from the right side of the circulation flows directly into the left side; hypoxemia and *cyanosis* (a dusky blueness of the skin and mucous membranes) result because the pulmonary circulation is bypassed and poorly oxygenated venous blood flows directly into the systemic arterial supply. In addition, right-to-left shunts can allow emboli from the peripheral veins to bypass the lungs and enter the systemic circulation (paradoxical embolism). Severe, longstanding hypoxia/cyanosis also leads to increased numbers of circulating red cells (polycythemia), as well as a peculiar distal blunting and enlargement (“clubbing”) of the tips of the fingers and toes that can include bony changes (called *hypertrophic osteoarthropathy*).

In contrast, *left-to-right shunts* increase pulmonary blood flow but are not initially associated with cyanosis. However, left-to-right shunts chronically elevate both volume and pressure in the normally low-pressure, low-resistance pulmonary circulation. To maintain relatively normal distal pulmonary capillary and venous pressures, the muscular pulmonary arteries (<1 mm in diameter) initially respond by undergoing medial hypertrophy and vasoconstriction. However, prolonged pulmonary arterial vasoconstriction stimulates the development of irreversible obstructive intimal lesions analogous to the arteriolar

changes seen in systemic hypertension. The right ventricle also responds to the pulmonary vascular changes by undergoing progressive hypertrophy. Eventually, pulmonary vascular resistance approaches systemic levels, and the original left-to-right shunt becomes a right-to-left shunt that introduces poorly oxygenated blood into the systemic circulation (*Eisenmenger syndrome*).

Once irreversible pulmonary hypertension develops, the structural defects of CHD are considered irreparable; subsequent right heart failure leads to death unless combined heart-lung transplantation is performed. This provides the rationale for early intervention to close significant left-to-right shunts.

Obstructive CHD occurs when there is abnormal narrowing of chambers, valves, or blood vessels; these include coarctation of the aorta, aortic valvular stenosis, and pulmonary valvular stenosis. A complete obstruction may occur from failure of development (*atresia*). In some disorders (e.g., TOF), an obstruction (pulmonary stenosis) and a shunt (right-to-left through a VSD) are both present.

The altered hemodynamics of CHD usually cause cardiac dilation or hypertrophy (or both). However, some defects induce a decrease in the volume and muscle mass of a cardiac chamber; this is called *hypoplasia* if it occurs before birth and *atrophy* if it develops postnatally.

Left-to-Right Shunts

Left-to-right shunts are the most common CHD; these include ASD, VSD, and patent ductus arteriosus (PDA) (Fig. 12.3). ASDs typically increase only right ventricular and pulmonary outflow volumes, and VSDs and PDA increase both pulmonary blood flow and pressure. Depending on their size and location, manifestations of these shunts range from no symptoms at all to fulminant heart failure.

Atrial Septal Defect

ASDs are abnormal, fixed openings in the atrial septum that allows communication of blood between the left and right atria. ASDs are usually asymptomatic until adulthood (see Table 12.2 and Fig. 12.3A) and should not be confused with patent foramen ovale (PFO; see later), which represents the postnatal failure to close a foramen (channel) that is part of normal development. To understand how ASDs form, a brief review of the developmental stages of the atrial septum is warranted.

- The *septum primum* is a crescent-shaped membranous ingrowth that sits posteriorly between the right and left atria and partially separates them; the remaining transient anterior opening, called the *ostium primum*, allows movement of blood from the right to left atrium during early fetal development.
- Before the growing septum primum completely obliterates the ostium primum, it develops a second posterior opening called the *ostium secundum*.
- The *septum secundum* is a subsequent membranous ingrowth located to the right and anterior of the septum primum.
- The septum secundum grows to cover the ostium secundum, leaving only a small channel called the *foramen ovale* that is continuous with the ostium secundum—the foramen ovale/ostium secundum permits continued right-to-left shunting of blood during intrauterine development.

This flap of septum secundum tissue opens and closes in response to pressure gradients between the left and right atria; the valve opens only when the pressure is greater in the right atrium. In fetal life, the lungs are nonfunctional, and the pressure in the pulmonary circulation is greater than that of the systemic circulation; thus, the right

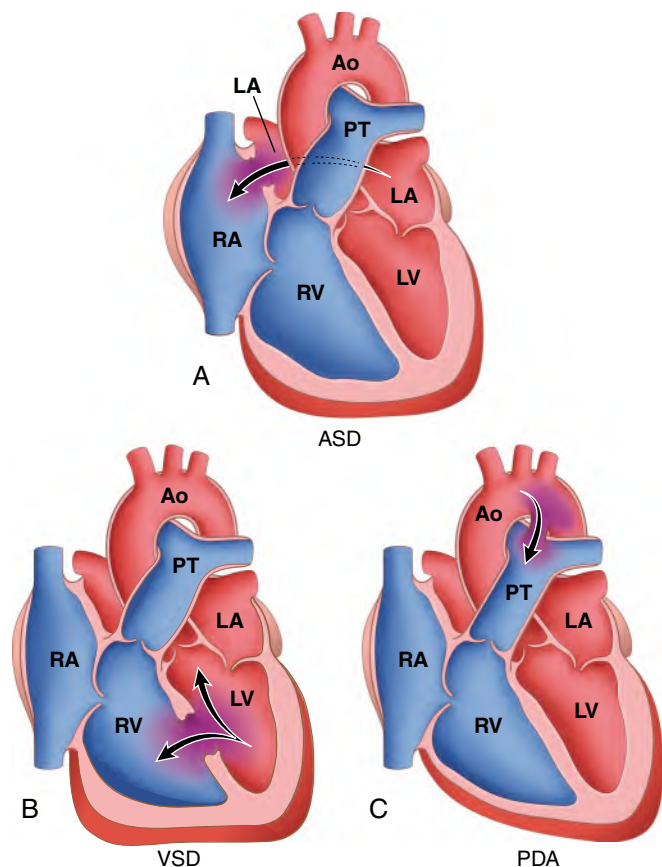


Fig. 12.3 Common congenital causes of left-to-right shunts (arrows indicate the direction of blood flow). (A) Atrial septal defect (ASD). (B) Ventricular septal defect (VSD). With VSD, the shunt is left-to-right, and the pressures are the same in both ventricles. Pressure hypertrophy of the right ventricle and volume hypertrophy of the left ventricle are generally present. (C) Patent ductus arteriosus (PDA). Ao, Aorta; LA, left atrium; LV, left ventricle; PT, pulmonary trunk; RA, right atrium; RV, right ventricle.

atrium is under higher pressures than the left atrium, and the valve of the foramen ovale is normally open. At birth, with lung expansion, the pulmonary vascular pressures drop, and the right atrial pressures fall below those in the left atrium. As a result, the valve of the foramen ovale closes—and is usually permanently sealed before adulthood (see later).

MORPHOLOGY

ASDs are classified according to their location. **Secundum ASDs** (90% of all ASDs) result from deficient septum secundum formation near the center of the atrial septum. These are usually not associated with other anomalies, can be of any size, and can be multiple or fenestrated. **Primum anomalies** (5% of ASD) occur adjacent to the AV valves and are often associated with AV valve abnormalities and/or a VSD. **Sinus venosus defects** (5%) are located near the entrance of the superior vena cava and can be associated with anomalous pulmonary venous return to the right atrium.

Clinical Features

ASDs are usually asymptomatic until adulthood. Although VSDs are more common, most close spontaneously. Consequently, ASDs—which

are less likely to close spontaneously—are the most common defects to be diagnosed in adults. ASDs result in a left-to-right shunt, largely because pulmonary vascular resistance is considerably less than systemic vascular resistance and because the compliance (distensibility) of the right ventricle is much greater than that of the left. The resulting pulmonary flow volumes may be two to eight times normal. A murmur is often present because of excessive flow through the pulmonary valve and/or through the ASD. Despite the right-sided volume overload, ASDs are generally well tolerated and usually do not become symptomatic before 30 years of age; irreversible pulmonary hypertension is unusual.

Surgical or intravascular ASD closure is performed to preempt the development of heart failure, paradoxical embolization, and irreversible pulmonary vascular disease. Mortality is low, and postoperative survival is comparable to that for an unaffected population.

Patent Foramen Ovale

The foramen ovale closes permanently in approximately 80% of people by 2 years of age. However, in the remaining 20%, the unsealed flap can open if right-sided pressures become elevated. Thus, sustained pulmonary hypertension or even transient increases in right-sided pressures (e.g., during a bowel movement, coughing, or sneezing) can produce brief periods of right-to-left shunting, with the possibility of paradoxical embolism.

Ventricular Septal Defect

VSDs are incomplete closures of the ventricular septum, allowing free communication of blood between the left-to-right ventricles; they are the most common form of CHD (see Table 12.2 and Fig. 12.3B).

MORPHOLOGY

VSDs are classified according to their location and magnitude. About 90% occur in the region of the membranous interventricular septum (**membranous VSD**; Fig. 12.4), and the majority are 2 to 3 cm in diameter. The remaining 10% occur below the pulmonary valve (**infundibular VSD**) or within the muscular septum. Although most VSDs are single, those in the muscular septum may be multiple.

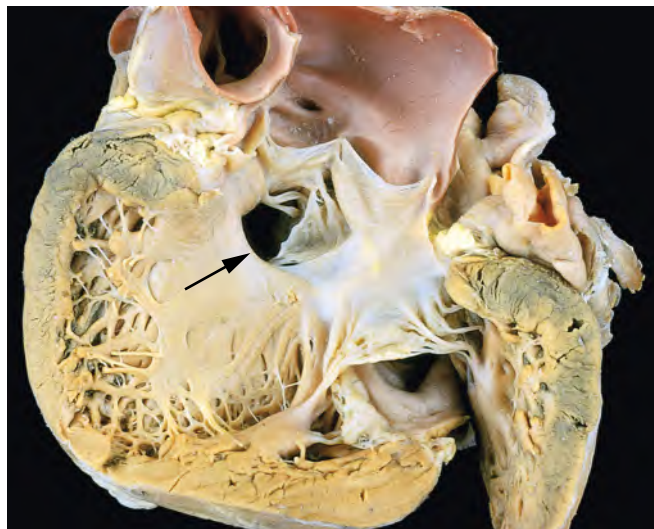


Fig. 12.4 Membranous type ventricular septal defect (arrow) just proximal to the aortic valve. (Courtesy William D. Edwards, MD, Mayo Clinic, Rochester, MN.)

Clinical Features

Most VSDs that manifest in the pediatric age group are associated with other congenital cardiac anomalies such as TOF; only 20% to 30% are isolated. Conversely, if a VSD is first detected only in an adult, it is usually an isolated defect. The functional consequences of a VSD depend on the size of the defect and whether there are associated right-sided malformations. Thus, large VSDs cause difficulties virtually from birth; smaller lesions are generally well tolerated for years and may not be recognized until much later in life. Moreover, approximately 50% of small muscular VSDs close spontaneously. Large defects generally cause significant left-to-right shunting, leading to early right ventricular hypertrophy and pulmonary hypertension. Over time, large unclosed VSDs almost universally lead to irreversible pulmonary vascular disease, ultimately resulting in shunt reversal and cyanosis. Closure of asymptomatic VSD is typically delayed beyond infancy, in hope of spontaneous closure. Early correction, however, must be performed for large defects to prevent the development of irreversible obstructive pulmonary vascular disease.

Patent Ductus Arteriosus (PDA)

The *ductus arteriosus* arises from the pulmonary artery and joins the aorta just distal to the origin of the left subclavian artery. During intra-uterine life, it permits blood flow from the pulmonary artery to the aorta, thereby bypassing the unoxygenated lungs. In healthy term infants, the ductus constricts and is functionally closed within 1 to 2 days of birth; this occurs in response to increased arterial oxygenation, decreased pulmonary vascular resistance, and declining local levels of prostaglandin E₂. Complete structural obliteration occurs within the first few months of extrauterine life, leaving behind the ligamentum arteriosum. Ductal closure is often delayed (or even absent) in infants with hypoxia (due to respiratory distress or heart disease) or when other congenital defects are present, particularly VSDs that increase pulmonary vascular pressures. PDAs account for about 7% of cases of CHD (see Table 12.2 and Fig. 12.3C), and 90% of these are isolated defects.

PDA produces a characteristic, continuous, harsh machinery-like murmur. The clinical impact of a PDA depends on its diameter and the cardiovascular status of the individual. PDA is usually asymptomatic at birth, and a narrow PDA may have no effect on the child's growth and development. Because the shunt is initially left-to-right, there is no cyanosis. However, with large shunts, the additional volume and pressure overloads eventually produce obstructive changes in small pulmonary arteries, leading to reversal of flow and its associated consequences. In general, isolated PDA should be closed as early in life as is feasible; therapy includes prostaglandin synthesis inhibitors and, if necessary, percutaneous or surgical interventions. Preservation of ductal patency (by administering prostaglandin E₁) may be life-saving for infants with various congenital malformations that obstruct the pulmonary or systemic outflow tracts. In CHD with aortic valve or pulmonary valve atresia, for example, a PDA may provide the entire systemic blood flow or pulmonary blood flow, respectively.

Right-to-Left Shunts

The diseases in this group cause cyanosis early in postnatal life (cyanotic CHD). TOF, the most common in this group, and transposition of the great arteries (TGA) are illustrated schematically in Fig. 12.5. The others include persistent truncus arteriosus, tricuspid atresia, and total anomalous pulmonary venous connection.

Tetralogy of Fallot

The four cardinal features of tetralogy of Fallot (TOF) are (1) VSD, (2) obstruction of the right ventricular outflow tract (subpulmonic stenosis),

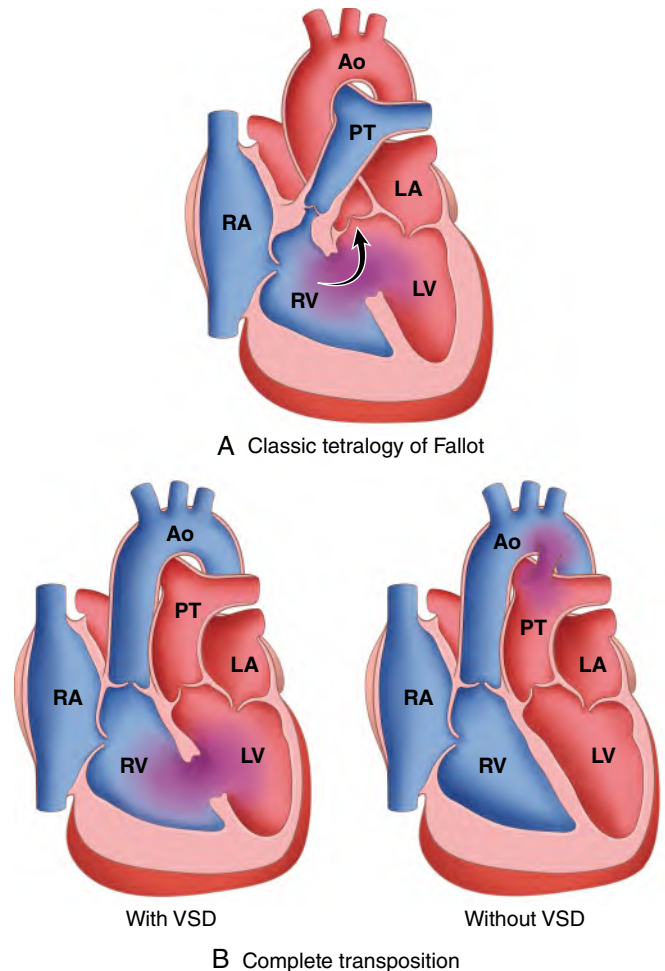


Fig. 12.5 Common congenital right-to-left shunts (cyanotic congenital heart disease). (A) Tetralogy of Fallot. The direction of shunting across the ventricular septal defect (VSD) depends on the degree of the subpulmonic stenosis; when severe, a right-to-left shunt results (arrow). (B) Dextro-transposition of the great arteries with and without VSD. Ao, Aorta; LA, left atrium; LV, left ventricle; PT, pulmonary trunk; RA, right atrium; RV, right ventricle.

(3) an aorta that overrides the VSD, and (4) right ventricular hypertrophy (see Fig. 12.5A). The first three features result embryologically from anterosuperior displacement of the infundibular septum, and the right ventricular hypertrophy is a secondary consequence of the pressure overload.

MORPHOLOGY

The heart is typically enlarged and is classically boot-shaped due to marked right ventricular hypertrophy. The VSD is usually large with the aortic valve at the superior border, overriding the defect and both ventricular chambers. The obstruction to right ventricular outflow is most often due to narrowing of the infundibulum (**subpulmonic stenosis**) but can be accompanied by pulmonary valvular stenosis. Sometimes there is complete atresia of the pulmonary valve and variable portions of the pulmonary arteries, such that blood flow through a PDA, dilated bronchial arteries, or both, is necessary for survival. Aortic valve insufficiency or an ASD may also be present; a right aortic arch is present in about 25% of cases.

Clinical Features

Depending on the severity of the subpulmonic stenosis, untreated TOF can be tolerated into adulthood; 10% of unoperated individuals are alive at 20 years of age, and 3% survive for 40 years. If the subpulmonic stenosis is mild, the abnormality resembles an isolated VSD, and the shunt may be left-to-right, without cyanosis (so-called *pink tetralogy*). With more severe right ventricular outflow obstruction, right-sided pressures approach or exceed left-sided pressures, and right-to-left shunting develops, producing cyanosis (classic TOF). Most infants with TOF are cyanotic at birth or soon thereafter. As the child grows and the heart increases in size, the pulmonic orifice does not expand proportionally, worsening the obstruction. The subpulmonic stenosis, however, protects the pulmonary vasculature from pressure overload, and right ventricular failure is rare because the right ventricle is decompressed by the shunting of blood into the left ventricle and aorta. Complete surgical repair is possible but is particularly complicated for individuals with pulmonary atresia and dilated bronchial arteries.

Transposition of the Great Arteries

Transposition of the great arteries (TGA) produces ventriculoarterial discordance (the ventricle outflow going to the wrong outflow vessel). In the more common variant (dextro-TGA or d-TGA), the aorta arises from the right ventricle, and the pulmonary artery emanates from the left ventricle (ventriculoarterial discordance; Fig. 12.5B and Fig. 12.6). The atrium-to-ventricle connections are normal (concordant), with the right atrium joining the right ventricle and the left atrium emptying into the left ventricle. The embryologic defect in complete TGA stems from abnormal formation of the spiraling truncal and aortopulmonary septa. The result is separation of the systemic and pulmonary circulations, a condition incompatible with postnatal life unless a shunt exists for adequate mixing of blood.

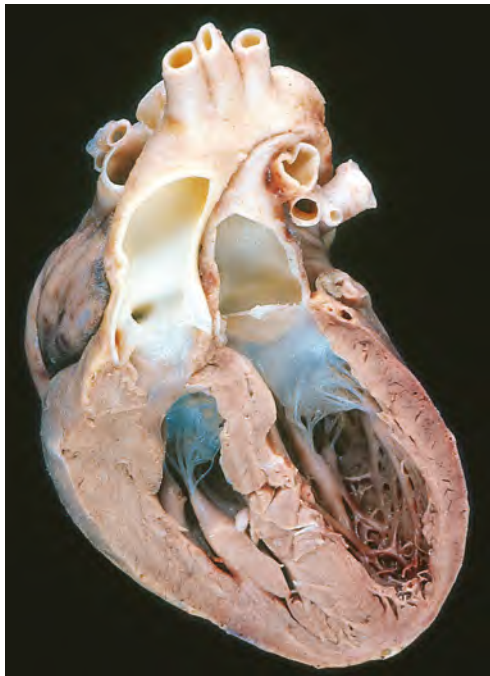


Fig. 12.6 Dextro-transposition of the great arteries. (Courtesy William D. Edwards, MD, Mayo Clinic, Rochester, MN.)

The outlook for infants with d-TGA depends on the degree of blood “mixing,” the severity of tissue hypoxia, and the ability of the right ventricle to maintain the systemic circulation. Patients with d-TGA and a VSD (approximately 35%) often have a stable shunt. However, dependence on a patent foramen ovale or ductus arteriosus for blood mixing (present in approximately 65% of patients) is problematic. These systemic-to-pulmonary connections tend to close early and thus require intervention to create a new shunt within the first few days of life (e.g., balloon atrial septostomy). With time, right ventricular hypertrophy becomes prominent because this chamber is functioning as the systemic ventricle (right ventricle supporting systemic circulation). Concurrently, the left ventricle becomes thin-walled (atrophic) as it supports only the low-resistance pulmonary circulation. Without surgery, most patients die within months. However, improved early surgical intervention, typically repositioning of the pulmonic artery and aorta, allows many patients with d-TGA to survive into adulthood.

In the less common levo-TGA (l-TGA, also commonly referred to as *congenitally corrected TGA*), the right atrium connects to a ventricle with the internal morphology of a left ventricle (atrioventricular discordance), which in turn empties into the pulmonary arteries (ventriculoarterial discordance); at the same time, the left atrium connects to a morphologic right ventricle, which empties into the aorta (Fig. 12.7). l-TGA does not lead to cyanosis and indeed can be entirely asymptomatic, being diagnosed in adulthood only during a workup for other cardiac issues. Nevertheless, l-TGA does result in hypertrophy of the morphologic right ventricle and eventually it can cause heart failure; it is also often associated with other CHD such as VSD, ASD, and PFO.

Tricuspid Atresia

Tricuspid atresia represents complete occlusion of the tricuspid valve orifice. It results embryologically from unequal division of the AV canal; thus, the mitral valve is larger than normal, and there is right

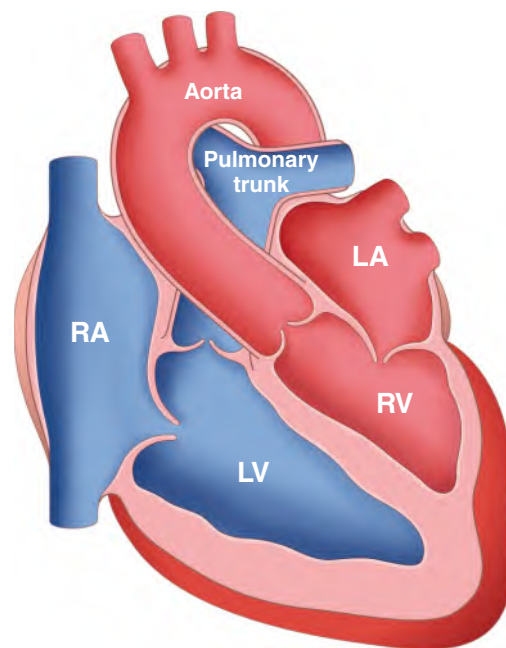


Fig. 12.7 “Congenitally corrected” levo-transposition of the great arteries. The right atrium (RA) flows to the morphologic left ventricle (LV), which connects to the pulmonary arteries; the left atrium (LA) connects to the morphologic right ventricle (RV), which empties into the aorta, which is anomalously anterior and to the left.

ventricular underdevelopment (hypoplasia). The circulation can be maintained by right-to-left shunting through an interatrial communication (ASD or PFO), in addition to a VSD that connects the left ventricle and the pulmonary artery arising from the hypoplastic right ventricle. Cyanosis is present virtually from birth, and there is a high early mortality.

Obstructive Lesions

Congenital obstruction to blood flow can occur at the level of the heart valves or within a great vessel. Common examples include aortic or pulmonary valve stenosis or atresia, and coarctation of the aorta. Obstruction can also occur within a chamber, as with subpulmonic stenosis in TOF.

Coarctation of the Aorta

Coarctation (narrowing, constriction) of the aorta ranks high in frequency among the common structural anomalies. It is twice as common in males as in females; interestingly, females with Turner syndrome are also frequently affected (Chapter 5). There are two classic forms: (1) an infantile form—often symptomatic in early childhood—with tubular hypoplasia of the aortic arch proximal to a PDA (preductal coarctation), and (2) an adult (postductal) form with a discrete ridge-like infolding of the aorta just opposite the closed ductus arteriosus (ligamentum arteriosum) distal to the arch vessels (Fig. 12.8). Encroachment on the aortic lumen is variable, sometimes leaving only a small channel and at other times producing only minimal narrowing. Although coarctation of the aorta may occur as a solitary defect, in 50% of cases it is accompanied by a bicuspid aortic valve; less commonly, it is associated with congenital aortic stenosis, ASD, VSD, mitral regurgitation, or berry aneurysms of the circle of Willis.

MORPHOLOGY

Preductal coarctation is characterized by circumferential narrowing of the aortic segment between the left subclavian artery and the ductus arteriosus; the ductus typically is patent and is the main source of (unoxygenated) blood delivered to the distal aorta. The pulmonary trunk is dilated to accommodate the increased blood flow; because the right side of the heart now perfuses the body, the right ventricle typically is hypertrophied.

In the more common **adult postductal coarctation**, the aorta is sharply constricted by a tissue ridge adjacent to the nonpatent ligamentum arteriosum (see Fig. 12.8B). The constricted segment is made up of smooth muscle and elastic fibers derived from the aortic media. Proximal to the coarctation, the aortic arch and its branch vessels are dilated, and the left ventricle is hypertrophied.

Clinical Features

Clinical manifestations depend on the severity of the narrowing and the patency of the ductus arteriosus. Coarctation of the aorta with a PDA usually manifests early in life; indeed, it may cause signs and symptoms immediately after birth. In such cases, the delivery of unsaturated blood through the PDA produces cyanosis localized to the lower half of the body. Many such infants do not survive the neonatal period without surgical or catheter-based intervention to occlude the PDA.

The outlook is different with coarctation of the aorta without a PDA, unless the aortic constriction is severe. Most children are asymptomatic, and the disease may go unrecognized until well into adult life. Typically, there is hypertension in the upper extremities with weak pulses and hypotension in the lower extremities, associated with

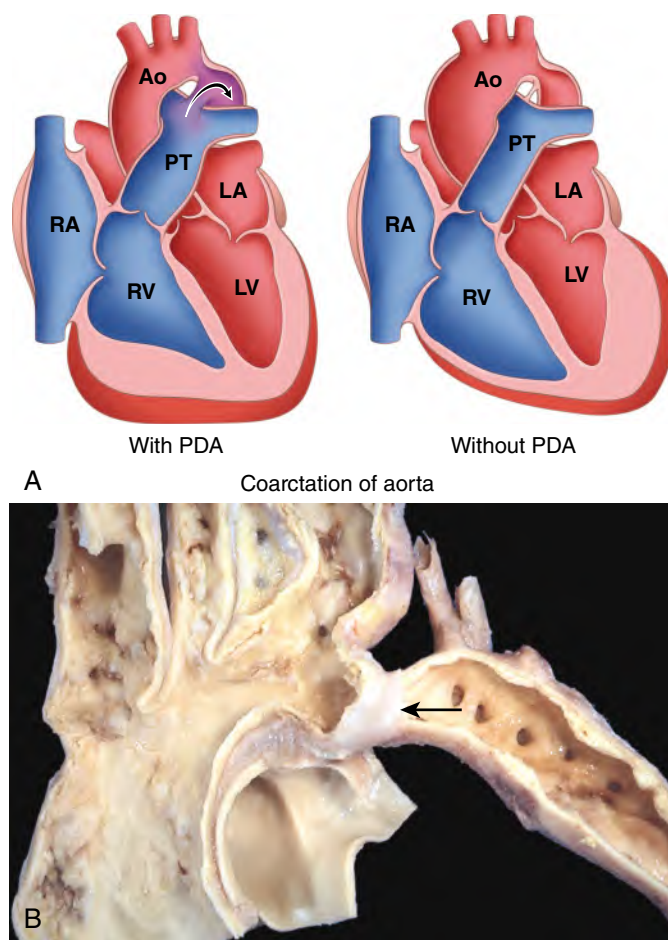


Fig. 12.8 (A) Aortic coarctation with and without patent ductus arteriosus (PDA). (B) Coarctation of the aorta, postductal type. The coarctation is a segmental narrowing of the aorta (arrow). The dilated ascending aorta and major branch vessels are to the left of the coarctation. Ao, Aorta; LA, left atrium; LV, left ventricle; PT, pulmonary trunk; RA, right atrium; RV, right ventricle. (A, Courtesy William D. Edwards, MD, Mayo Clinic, Rochester, MN. B, Courtesy Sid Murphree, MD, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

manifestations of arterial insufficiency (i.e., claudication and coldness). Particularly characteristic is the development of collateral circulation between the pre-coarctation and post-coarctation arteries through enlarged intercostal and internal mammary arteries, often producing radiographically visible erosions (notching) of the undersurfaces of the ribs.

With significant coarctations, murmurs are present throughout systole; sometimes a vibratory thrill is also present. The longstanding pressure overload leads to concentric left ventricular hypertrophy. With uncomplicated coarctation of the aorta, surgical resection and end-to-end anastomosis or replacement of the affected aortic segment by a prosthetic graft yields excellent results.

Pulmonary Stenosis and Atresia

Pulmonary stenosis or atresia is a relatively frequent malformation leading to obstruction at the level of the pulmonary valve. This can be mild to severe; the lesion can also be isolated or part of a more complex anomaly—either TOF or TGA. Right ventricular hypertrophy typically develops, and there is sometimes poststenotic dilation of the pulmonary artery due to injury of the wall by high-pressure jets

through the stenotic valve. With coexistent subpulmonic stenosis (as in TOF), the pulmonary trunk is not dilated and may in fact be hypoplastic. When the valve is entirely atretic, there is no communication between the right ventricle and lungs. In such cases, the anomaly is associated with a hypoplastic right ventricle and an ASD; blood reaches the lungs through a PDA. Mild stenosis may be asymptomatic and compatible with long life, whereas symptomatic cases require surgical correction.

Aortic Stenosis and Atresia

Congenital narrowing and obstruction of the aortic valve can occur at three locations: valvular, subvalvular, and supra-ventricular. Congenital aortic valve stenosis is an isolated lesion in 80% of cases. With valvular aortic stenosis, the cusps may be hypoplastic (small), dysplastic (thickened, nodular), or abnormal in number (usually with one or no commissures). In severe congenital aortic stenosis or atresia, obstruction of the left ventricular outflow tract leads to hypoplasia of the left ventricle and ascending aorta, sometimes accompanied by dense, porcelain-like left ventricular endocardial fibroelastosis; the ductus arteriosus must be patent to allow blood flow to the aorta and coronary arteries. The constellation of findings is called hypoplastic left heart syndrome, and unless PDA patency is preserved, duct closure in the first week of life is generally lethal.

Subaortic stenosis is caused by a thickened ring or collar of dense endocardial fibrous tissue below the level of the cusps. Supravalvular aortic stenosis is a congenital aortic dysplasia with thickening and constriction of the ascending aortic wall. Elastin gene mutations can cause supravalvular stenosis by disrupting elastin-smooth muscle cell interactions during aortic morphogenesis. Subaortic stenosis is usually associated with a prominent systolic murmur and sometimes a thrill. Pressure hypertrophy of the left ventricle develops because of the obstruction to blood flow; unless very severe, the stenosis is generally well tolerated, although left ventricular hypertrophy still carries a risk of sudden cardiac death.

KEY CONCEPTS

Congenital Heart Disease

- CHD represents defects of cardiac chambers or the great vessels; these typically either result in shunting of blood between the right and left circulation or cause outflow obstruction. Lesions range from relatively asymptomatic to rapidly fatal. Environmental and genetic causes both contribute, and the manifestations depend on the timing of the environmental insult and which step in cardiac development is affected.
- Left-to-right shunts are most common and are typically associated with ASD, VSD, or PDA. These lesions result in chronic right-sided pressure and volume overloads that eventually cause pulmonary hypertension with reversal of flow and right-to-left shunts with cyanosis (Eisenmenger syndrome).
- Right-to-left shunts are most often caused by TOF or TGA. These lead to cyanosis and are associated with polycythemia and paradoxical emboli.
- Obstructive lesions include valve stenoses and aortic coarctation; the clinical severity of the lesion depends on the degree of stenosis and the patency of the ductus arteriosus.

ISCHEMIC HEART DISEASE

IHD encompasses a group of related entities resulting from myocardial ischemia—an imbalance between myocardial supply (perfusion) and cardiac demand for oxygenated blood. Ischemia not only limits tissue oxygenation (and thus ATP generation), but also reduces

the availability of nutrients and the removal of metabolic wastes (Chapter 2). Thus, cardiac ischemia is generally less well tolerated than hypoxemia per se, such as may occur with severe anemia, cyanotic heart disease, or advanced lung disease.

In more than 90% of cases, myocardial ischemia results from reduced blood flow due to obstructive atherosclerotic lesions in the epicardial coronary arteries; consequently, IHD is frequently referred to as coronary artery disease (CAD). In most cases, there is a long period (up to decades) of silent, slow progression of coronary lesions before the sudden onset of symptoms. Thus, IHD is often the late manifestation of coronary atherosclerosis that began during childhood or adolescence (Chapter 11).

IHD can declare itself through one or more clinical presentations:

- *Myocardial infarction (MI)*, in which ischemia causes cardiac necrosis
- *Angina pectoris* (literally “chest pain”), in which ischemia is not severe enough to cause infarction, but the symptoms nevertheless portend infarction risk
- *Chronic IHD* with heart failure
- *Sudden cardiac death (SCD)*

In addition to coronary atherosclerosis, myocardial ischemia can be caused by coronary emboli, myocardial vessel inflammation, or vascular spasm. Moreover, otherwise modest coronary artery occlusions may become consequential in the setting of increased cardiac energy demand (e.g., myocardial hypertrophy or increased heart rate), hypoxemia, or systemic hypotension (e.g., shock). Some conditions can have multiple deleterious effects. Thus, tachycardia increases oxygen demand (because of more contractions per unit time) while decreasing blood supply (by decreasing the relative time spent in diastole, when cardiac perfusion occurs).

Epidemiology

IHD is the single largest cause of mortality worldwide, accounting for over 15% of global deaths; in high-resource countries, this amounts to almost 9 million casualties each year. In low-resource countries, advances in the treatment and prevention of infectious diseases and the increasing adoption of westernized diets are accelerating IHD incidence; it is predicted soon to become the leading cause of mortality there as well.

But there is cause for optimism; since peaking in the mid-1960s, the overall death rate from IHD has fallen in the United States by over 50%. This remarkable improvement can be attributed to:

- *Prevention*, achieved by modifying important risk factors, such as smoking, cholesterol level, and hypertension. Additional risk reduction can occur through weight loss, exercise, and maintaining good glycemic control in diabetic patients.
- *Diagnostic and therapeutic advances*, allowing earlier and more effective treatments. The latter include cholesterol-lowering (and inflammation-lowering) drugs such as statins, thrombolysis for acute coronary occlusions, better medical management after MI, coronary angioplasty and stenting, coronary artery bypass graft (CABG) surgery, and improved therapies for heart failure and arrhythmias using ventricular assist devices (VADs), implantable defibrillators, and cardiac resynchronization approaches. Even a simple daily prophylactic aspirin has been shown to have therapeutic benefit.

Continuing this encouraging trend will be challenging, particularly in view of the increased longevity of baby boomers (which will lead to a doubling of individuals older than 65 years of age by 2050), and the obesity epidemic. New therapeutic advances will depend on understanding the genetic determinants of coronary atherosclerosis and IHD.

Thus, the observation that MIs occur in only a fraction of individuals with coronary disease suggests that simple control of atherosclerotic risk factors is only part of the story.

Pathogenesis

The dominant cause of IHD syndromes is insufficient coronary perfusion relative to myocardial demand; in most cases, this is due to either:

- Chronic, progressive atherosclerotic narrowing of the epicardial coronary arteries, or
- Variable degrees of superimposed *acute* plaque change, thrombosis, and vasospasm

Chronic Arterial Occlusion

More than 90% of patients with IHD have atherosclerosis involving one or more of the epicardial coronary arteries (Chapter 11). A fixed lesion obstructing greater than 70% of vascular cross-sectional area (so-called *critical stenosis*, as shown in Fig. 12.9) is typically cited as the threshold for symptomatic ischemia precipitated by increased physical activity (characteristically manifesting as exertional angina). With this degree of obstruction, compensatory coronary arterial vasodilation is no longer sufficient to meet even moderate increases in myocardial demand. Obstruction of 90% of the cross-sectional area of the lumen generally leads to inadequate coronary blood flow, even at rest. Slowly developing obstructions induce the formation of collateral circulation that can mitigate the effects of even high-grade stenoses by allowing alternate channels to perfuse at-risk myocardium. Consequently, rather than just assessing the extent of local vascular stenosis, measurements of coronary flow across the stenosis (flow reserves) provide a better assessment of the consequences of fixed coronary artery occlusions, particularly when multifocal.

Although only a single major coronary epicardial vessel may be affected, two or all three—the left anterior descending (LAD), left circumflex (LCX), and right coronary artery (RCA)—are often involved simultaneously by atherosclerosis. Clinically significant plaques can be located anywhere along the course of the vessels, although they tend to predominate within the first several centimeters of the LAD and LCX. Sometimes the major epicardial branches are also involved (i.e., LAD diagonal branches, LCX obtuse marginal branches, or posterior descending branch of the RCA), but atherosclerosis of the intramyocardial (penetrating) branches is rare. Thus, most atherosclerotic stenoses can be accessed by coronary catheterization.

Acute Plaque Change

In most patients, unstable angina, infarction, and sudden cardiac death occur because of abrupt plaque change followed by thrombosis—hence the term *acute coronary syndrome* (see Fig. 12.9).

The initiating event is typically a sudden disruption (rupture or erosion) of a partially occlusive plaque. Rupture, fissuring, ulceration, or erosion of plaques exposes highly thrombogenic constituents or underlying subendothelial basement membrane, leading to rapid thrombosis. In addition, hemorrhage into the core of plaques can expand plaque volume, thereby acutely exacerbating the degree of luminal occlusion. Even partial luminal occlusion by a thrombus can compromise blood flow sufficiently to cause an infarction of the innermost zone of the myocardium (*subendocardial infarct*). Mural thrombi in a coronary artery can also embolize; indeed, small emboli may be found in the distal intramyocardial circulation (along with associated microinfarcts) at autopsy of patients with unstable angina. Most seriously, completely obstructive thrombus over a disrupted plaque typically results in MI.

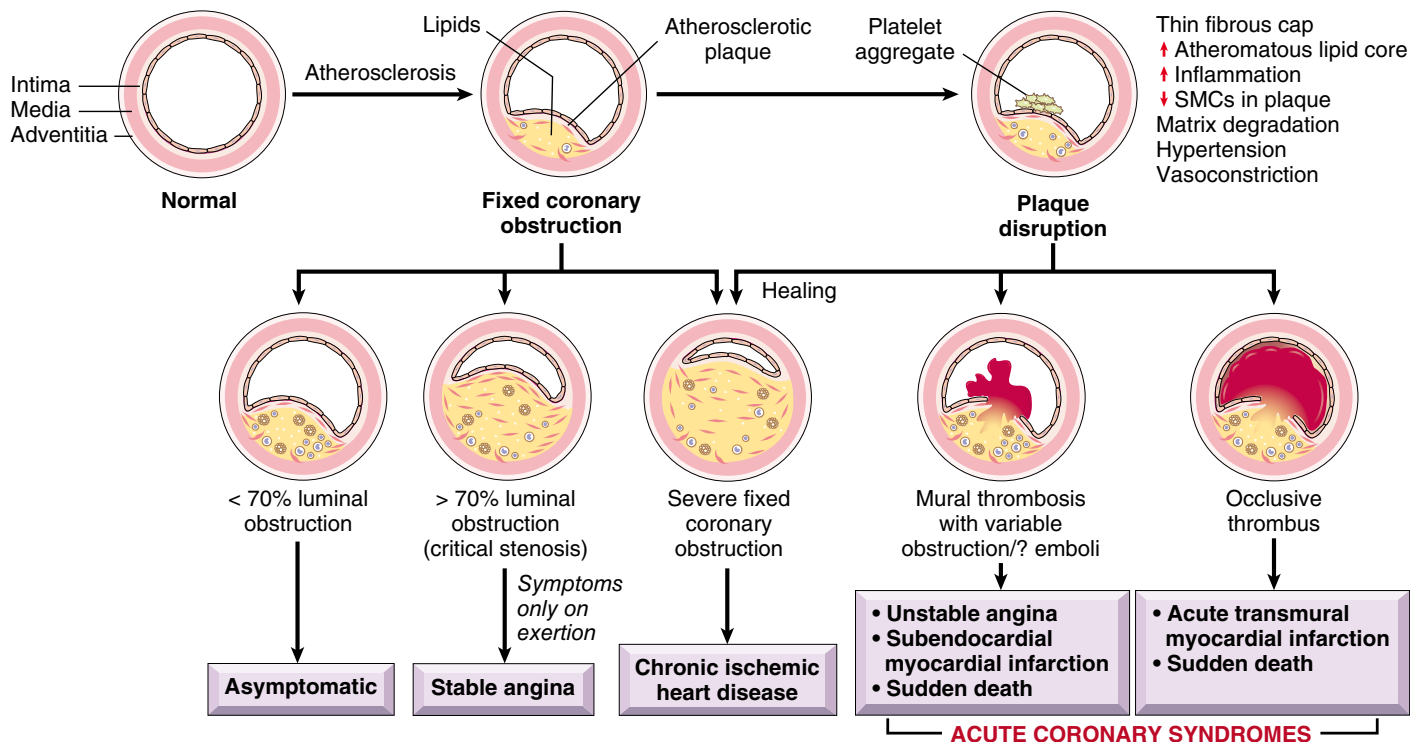


Fig. 12.9 Diagram of sequential progression of coronary artery lesions leading to various acute coronary syndromes. SMCs, Smooth muscle cells. (Modified from Schoen FJ. *Interventional and Surgical Cardiovascular Pathology: Clinical Correlations and Basic Principles*. Saunders; 1989:63.)

Factors that trigger loss of endothelial cells without plaque rupture (*plaque erosion*) include endothelial injury and apoptosis, likely attributable to some combination of inflammatory and toxic exposures. Acute *plaque rupture*, on the other hand, involves factors that influence plaque susceptibility to disruption by mechanical stress. These include intrinsic aspects of plaque composition and structure (Chapter 11) and extrinsic factors, such as blood pressure and platelet reactivity.

Plaques that contain large atheromatous cores or have thin overlying fibrous caps are more likely to rupture and are therefore termed vulnerable. Fissures frequently occur at the junction of the fibrous cap and the adjacent normal (plaque-free) arterial segment, where the mechanical stresses are greatest and the fibrous cap is thinnest. Fibrous caps are also continuously remodeling; the overall balance of collagen synthesis and degradation within the plaque determines its mechanical strength and stability. Collagen is produced by smooth muscle cells and degraded by the action of metalloproteases elaborated by macrophages. Consequently, atherosclerotic lesions with a paucity of smooth muscle cells or large numbers of inflammatory cells are particularly vulnerable to rupture. Of interest, statins may have a beneficial effect not only by reducing circulating cholesterol levels but also by reversing endothelial dysfunction and stabilizing plaques through a reduction in plaque inflammation.

In a majority of cases, the culprit lesion in patients who experience an MI is not critically stenotic or even symptomatic before its rupture. Thus, it is sobering to note that many *asymptomatic* adults are at significant risk for a catastrophic coronary event. At present, it is impossible to predict plaque rupture in any given patient. Plaque disruption and ensuing nonocclusive thrombosis are additional common, repetitive, and often clinically silent complications of atheromas. The healing of subclinical disrupted plaques is an important mechanism by which atherosclerotic lesions progressively enlarge (see Fig. 12.9).

Angina Pectoris

Angina pectoris is characterized by paroxysmal, recurrent attacks of substernal or precordial chest discomfort caused by transient (15 seconds to 15 minutes) myocardial ischemia that is insufficient to induce myocyte necrosis. Anginal pain is a consequence of ischemia-induced release of adenosine, bradykinin, and other molecules that stimulate sympathetic and vagal afferent nerves. Three overlapping patterns of angina pectoris are recognized, caused by varying combinations of inadequate perfusion, increased demand, and coronary arterial pathology. Importantly, not all ischemic events are perceived as pain by patients; so-called *silent ischemia* is particularly common in the geriatric population and in the setting of diabetic neuropathy, and can manifest as dyspnea, nausea and vomiting, palpitations, diaphoresis, or fatigue—or no symptoms at all.

- *Stable (typical) angina* is the most common form of angina; it is caused by an imbalance in coronary perfusion (due to chronic stenosing coronary atherosclerosis) relative to myocardial demand. Stable angina does not occur at rest, but in a given patient can be reliably induced by activities that increase the energy requirements of the heart, including physical activity, emotional excitement, or psychological stress. Angina pectoris is classically described as a crushing or squeezing substernal sensation that can radiate down the left arm or to the left jaw (*referred pain*). The pain is usually relieved by rest (decreasing demand) or by administering vasodilators, such as nitroglycerin and calcium channel blockers (thereby increasing perfusion).
- *Prinzmetal variant angina* is an uncommon form of episodic myocardial ischemia caused by coronary artery spasm. Although

individuals with Prinzmetal variant angina can also have significant coronary atherosclerosis, the anginal attacks are unrelated to physical activity, heart rate, or blood pressure, and can occur at rest. Prinzmetal angina generally responds promptly to vasodilators.

- *Unstable or crescendo angina* refers to a pattern of increasingly frequent, prolonged (>20 minutes), or severe angina, precipitated by progressively lower levels of physical activity or even occurring at rest. Unstable angina is associated with plaque disruption and superimposed thrombosis, distal embolization of the thrombus, and/or vasospasm; it is a harbinger of MI, as it portends a high risk of complete vascular occlusion.

Myocardial Infarction

MI, commonly referred to as a heart attack, is the death of cardiac muscle due to prolonged ischemia. Roughly 800,000 individuals in the United States experience an MI each year (nearly one MI every 40 seconds), causing almost 400,000 deaths annually. The major underlying cause is atherosclerosis. Although MIs can occur at virtually any age, 10% occur in people younger than 40 years of age, and 45% occur in people younger than 65 years. Nevertheless, the frequency rises progressively with age and with increasing atherosclerotic risk factors (Chapter 11). Through middle age, male sex increases the relative risk of MI; indeed, females are generally protected against MI during their reproductive years. However, a postmenopausal decline in estrogen production (with associated increases in inflammatory markers, cholesterol, and blood pressure) is often associated with worsening CAD; consequently, IHD is the most common cause of death in older females. Unfortunately, postmenopausal hormonal replacement therapy has not been shown to be protective, and in fact, in some cases—perhaps attributable to a prothrombotic effect—may be detrimental.

Pathogenesis

Coronary Arterial Occlusion. The following sequence of events likely underlies most MIs (see Chapter 11 for additional details):

- An atheromatous plaque is eroded or suddenly disrupted by endothelial injury, intraplaque hemorrhage, or mechanical forces, exposing subendothelial collagen and necrotic plaque contents to the blood.
- Platelets adhere, aggregate, and are activated, releasing thromboxane A₂, adenosine diphosphate (ADP), and serotonin—causing further platelet aggregation and vasospasm (Chapter 4).
- Activation of coagulation by tissue factor and other mechanisms adds to the growing thrombus.
- Within minutes, the thrombus can evolve to completely occlude the coronary artery lumen.

The evidence for this scenario derives from autopsy studies of patients dying of acute MI, as well as imaging studies demonstrating a high frequency of thrombotic occlusion early after MI; as noted previously, these thrombi are usually at a site that did not previously have a critical (>70%) fixed stenosis. When angiography is performed within 4 hours of the onset of MI, it demonstrates coronary thrombosis in almost 90% of cases. However, when angiography is performed 12 to 24 hours after onset of symptoms, evidence of thrombosis is seen in only 60% of patients, even without intervention. Thus, at least some occlusions clear spontaneously through lysis of the thrombus or relaxation of spasm. This sequence of events in a typical MI has therapeutic implications: early thrombolysis and/or angioplasty can be highly successful in limiting the extent of myocardial necrosis.

In approximately 10% of cases, MI occurs in the absence of the typical coronary atherothrombosis. In such situations, other mechanisms may be responsible for the reduced coronary blood flow:

- **Vasospasm** with or without coronary atherosclerosis, perhaps in association with platelet aggregation or due to drug ingestion (e.g., cocaine or ephedrine).
- **Embolism.** Emboli can originate from a mural thrombus in the left atrium in association with atrial fibrillation; vegetations of infective endocarditis; intracardiac prosthetic material; or from the right side of the heart or the peripheral veins in patients with a patent foramen ovale.
- **Uncommon causes** of MI include disorders of small intramural coronary vessels (e.g., vasculitis), hematologic abnormalities (e.g., sickle cell disease), amyloid deposition in vascular walls, vascular dissection, marked myocardial hypertrophy (e.g., due to aortic stenosis), lowered systemic blood pressure (e.g., shock), or inadequate myocardial protection during cardiac surgery.

Myocardial Response. Coronary arterial obstruction diminishes blood flow to a region of myocardium, causing ischemia, rapid myocardial dysfunction, and eventually—with prolonged vascular compromise—myocyte death. The anatomic region supplied by that artery is referred to as the area at risk. The outcome depends predominantly on the severity and duration of flow deprivation.

The first biochemical consequence of myocardial ischemia is the cessation of aerobic metabolism within seconds, leading to inadequate production of high-energy phosphates (e.g., creatine phosphate and adenosine triphosphate) and accumulation of potentially noxious metabolites (e.g., lactic acid). Because of the exquisite dependence of myocardial function on oxygen and nutrients, myocardial contractility ceases within a minute or so of the onset of severe ischemia. Such loss of function contributes to decreased systolic function long before myocyte death occurs.

As detailed in Chapter 2, ultrastructural changes (including myofibrillar relaxation, glycogen depletion, and cell and mitochondrial swelling) also develop within a few minutes of the onset of ischemia. Nevertheless, these early manifestations of ischemic injury are reversible. Indeed, experimental and clinical evidence shows that only severe ischemia (blood flow 10% or less of normal) lasting 20 to 30 minutes or longer leads to irreversible damage (necrosis) of cardiac myocytes. Thereafter, progressive loss of viability occurs, becoming complete by 6 to 12 hours. The benefits of reperfusion are greatest when it is achieved quickly. This provides the rationale for rapid diagnosis in acute MI—to permit early coronary intervention to establish reperfusion and salvage as much at-risk myocardium as possible.

The earliest detectable feature of myocyte necrosis is disruption of the integrity of the sarcolemmal membrane, allowing intracellular macromolecules to leak out of necrotic cells into the cardiac interstitium and ultimately into the microvasculature and lymphatics. This escape of intracellular myocardial proteins into the circulation forms the basis for blood tests that can sensitively detect irreversible myocyte damage and are important for managing MI (see later). With prolonged severe ischemia, injury to the microvasculature follows injury to the cardiac myocytes. The temporal progression of these events is summarized in Table 12.4.

The progression of ischemic necrosis in the myocardium is summarized in Fig. 12.10. Irreversible injury of ischemic myocytes first occurs in the subendocardial zone. This region is especially susceptible to ischemia because it is the last area to receive blood delivered by the epicardial vessels, and because it is exposed to relatively high intramural pressures, which act to impede the inflow of blood. With more prolonged ischemia, a wavefront of cell death moves centripetally in the

TABLE 12.4 Approximate Time of Onset of Key Events in Ischemic Cardiac Myocytes

Feature	Time
Onset of ATP depletion	Seconds
Loss of contractility	<2 minutes
ATP reduced to 50% of normal	10 minutes
ATP reduced to 10% of normal	40 minutes
Irreversible cell injury	20–40 minutes
Microvascular injury	>1 hour

ATP, Adenosine triphosphate.

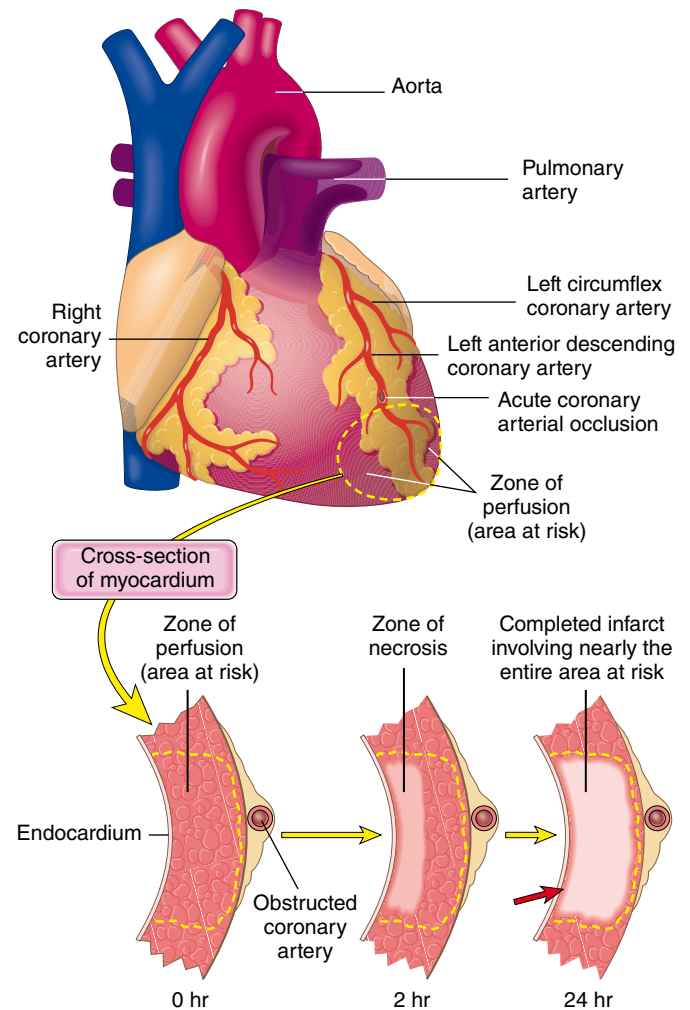


Fig. 12.10 Progression of myocardial necrosis after coronary artery occlusion. The area that depends on the occluded vessel for perfusion is the at-risk myocardium (outline in yellow). Necrosis begins in a small zone of the myocardium beneath the endocardial surface in the center of the ischemic zone. A very narrow zone of myocardium (red arrow) immediately beneath the endocardium is spared from necrosis because oxygen and nutrition are provided by diffusion from the ventricle.

myocardium, driven by progressive tissue edema and myocardial-derived reactive oxygen species and inflammatory mediators.

The location, size, and specific morphologic features of an acute MI depend on:

- The location, severity, and rate of development of coronary obstructions due to atherosclerosis and thromboses

- The *size of the vascular bed* perfused by the obstructed vessels
- The *duration* of the occlusion
- The *metabolic and oxygen needs* of the myocardium at risk
- The extent of *vascular collateralization*
- The presence, site, and severity of *coronary arterial spasm*
- *Other factors*, such as heart rate, cardiac rhythm, and blood oxygenation

An infarct usually achieves its full extent within 6 to 12 hours; in the absence of intervention, an infarct caused by occlusion of an epicardial vessel can involve the entire wall thickness (*transmural infarct*). Clinical intervention within this critical window of time can lessen the size of the infarct within the territory at risk.

Patterns of Infarction. The distribution of myocardial necrosis correlates with the location and cause of the decreased perfusion (Fig. 12.11).

Knowledge of the areas of myocardium perfused by the major coronary arteries allows correlation of specific vascular obstructions with their corresponding areas of MI. Typically, the LAD branch of the left coronary artery supplies most of the apex of the heart, the anterior wall of the left ventricle, and the anterior two-thirds of the ventricular septum. By convention, the coronary artery—either RCA or LCX—that perfuses the posterior third of the septum is called “*dominant*” (even though the LAD and LCX collectively perfuse the majority of the left ventricular myocardium). In a right dominant circulation (present in approximately 80% of individuals), the RCA supplies the entire right ventricular free wall, the posterobasal wall of the left ventricle, and the posterior third of the ventricular septum, while the

LCX generally perfuses only the lateral wall of the left ventricle. Thus, RCA occlusions can potentially lead to left ventricular damage.

Although most hearts have numerous intercoronary anastomoses (collateral circulation), relatively little blood normally courses through these. However, when a coronary artery is progressively narrowed over time, blood flows via the collaterals from the high- to the low-pressure circulation, causing the channels to enlarge. Progressive dilation of collaterals, stimulated by ischemia, may spare areas of myocardium that would otherwise be deprived of adequate perfusion.

Transmural infarctions occur when there is occlusion of an epicardial vessel (in the absence of any therapeutic intervention)—the necrosis involves virtually the full thickness of the ventricular wall in the distribution of the affected coronary. This pattern of infarction is usually associated with a combination of chronic coronary atherosclerosis, acute plaque change, and superimposed thrombosis (discussed earlier).

Subendocardial (nontransmural) infarctions can occur because of a plaque disruption followed by a coronary thrombus that becomes lysed (therapeutically or spontaneously) before myocardial necrosis extends across the full thickness of the wall. Subendocardial infarcts can also result from prolonged reduction in systemic blood pressure, as in shock superimposed on chronic, otherwise noncritical, coronary stenoses. In subendocardial infarcts that occur because of global hypotension, myocardial damage is often circumferential, rather than being limited to the distribution of a single major coronary artery.

Multifocal microinfarction refers to a pattern that is seen when there is pathology involving only smaller intramural vessels. It can occur in the setting of microembolization, vasculitis, or vascular spasm, for example, due to endogenous catecholamines (epinephrine)

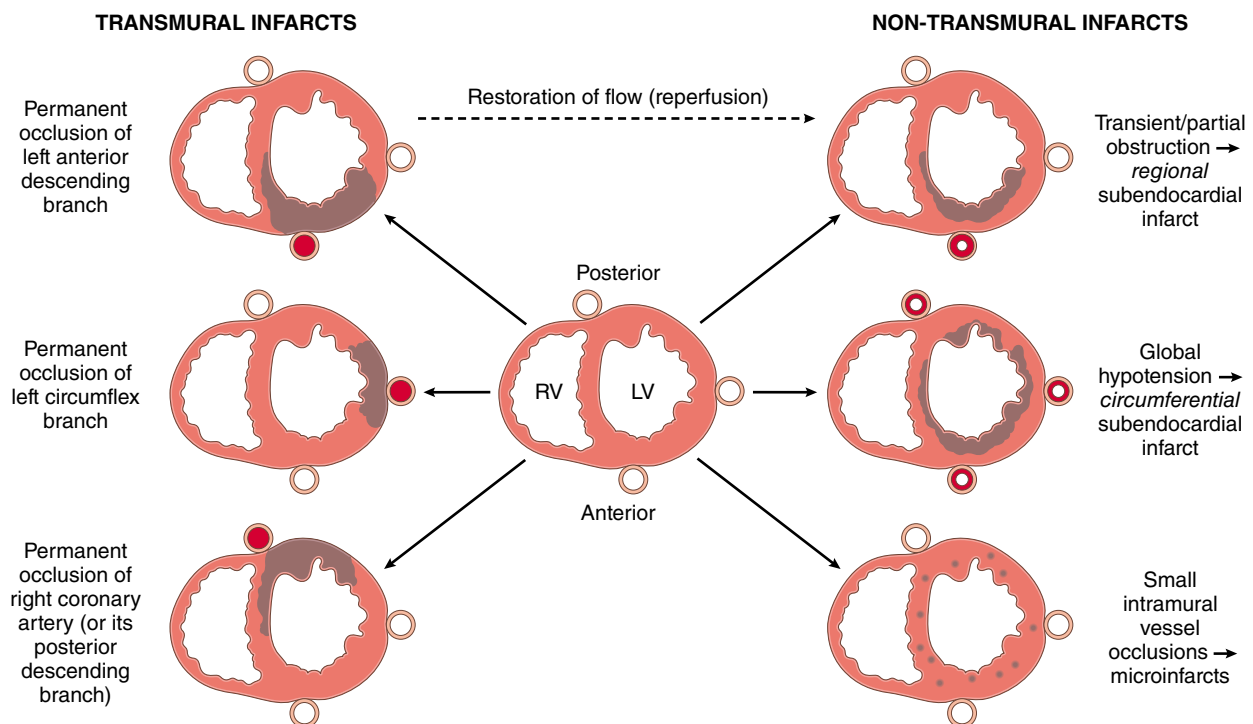


Fig. 12.11 Distribution of myocardial ischemic necrosis correlates with the location and nature of decreased perfusion. *Left*, the positions of transmural acute infarcts resulting from occlusions of the major coronary arteries; *top to bottom*, left anterior descending, left circumflex, and right coronary arteries. *Right*, the types of infarcts that result from a partial or transient occlusion, global hypotension, or intramural small-vessel occlusions. As shown at top, reperfusion may only spare a portion of the myocardium that is at risk. *LV*, Left ventricle; *RV*, right ventricle.

or drugs (cocaine or ephedrine). Elevated levels of catecholamines also increase heart rate and myocardial contractility, exacerbating ischemia caused by the vasospasm. The outcome of such vasospasm can be sudden cardiac death (usually caused by a fatal arrhythmia) or—with recurrent episodes—an ischemic dilated cardiomyopathy (discussed later).

MORPHOLOGY

The temporal evolution of the morphologic changes in acute MI and subsequent healing are summarized in [Table 12.5](#).

Nearly all transmural infarcts involve at least a portion of the left ventricle (comprising the free wall and ventricular septum); they encompass nearly the entire perfusion zone of the occluded coronary artery save for a narrow rim (approximately 0.1 mm) of viable subendocardial myocardium that is preserved by diffusion of oxygen and nutrients from the ventricular lumen.

The frequencies of involvement of each of the three main arterial trunks and the corresponding sites of myocardial lesions resulting in infarction (in the typical right dominant heart) are as follows ([Fig. 12.11](#)):

- Left anterior descending coronary artery (40% to 50%): infarcts involving the anterior wall of left ventricle near the apex; the anterior portion of ventricular septum; and the apex circumferentially
- Right coronary artery (30% to 40%): infarcts involving the inferior/posterior wall of the left ventricle; posterior portion of ventricular septum; and the inferior/posterior right ventricular free wall in some cases
- Left circumflex coronary artery (15% to 20%): infarcts involving the lateral wall of the left ventricle except at the apex

Other locations of critical coronary arterial lesions causing infarcts are sometimes encountered, such as the left main coronary artery, the secondary (diagonal) branches of the left anterior descending coronary artery, or the marginal branches of the left circumflex coronary artery. Of MIs caused by a right coronary obstruction,

15% to 30% extend from the posterior free wall of the septal portion of the left ventricle into the adjacent right ventricular wall. Isolated infarction of the right ventricle is unusual (only 1% to 3% of cases), as is infarction of the atria.

The gross and microscopic appearance of an infarct depends on the interval of time between the MI and death. Damaged myocardium undergoes a progressive sequence of morphologic changes involving ischemic coagulative necrosis, the predominant mechanism of cell death in MI (although apoptosis can also occur); this is followed by stereotypical inflammation and repair that parallels responses to injury in other tissues (Chapter 2).

Early morphologic recognition of acute MI can be difficult, particularly when death occurs within only a few hours of symptom onset. MIs less than 12 hours old are usually not apparent on gross examination alone; however, if the infarct preceded death by at least 2 to 3 hours, it is possible to highlight the area of necrosis by immersion of tissue slices in a solution of **triphenyl tetrazolium chloride**. This histochemical stain imparts a brick-red color to intact, noninfarcted myocardium with preserved lactate dehydrogenase activity. Because dehydrogenases leak out through the damaged membranes of dead cells, an infarct appears as an unstained pale zone ([Fig. 12.12](#)). By 12 to 24 hours after infarction, an MI can usually be identified grossly as a reddish-blue area of discoloration caused by congestion and extravasated blood. By 3 to 7 days, it is rimmed by a hyperemic zone of highly vascularized early wound healing (**granulation tissue**). Thereafter, the infarct becomes progressively more sharply defined, yellow-tan, and soft. Over the succeeding weeks, the injured region evolves to a fibrous scar.

The histopathologic changes also proceed in a fairly predictable sequence ([Fig. 12.13](#)). The typical changes of coagulative necrosis become detectable in the first 4 to 12 hours; so-called *wavy fibers* result from the contractions of viable myocardium that cause the stretching and buckling of adjacent, noncontractile dead fibers. At even earlier timepoints, there can be subtle changes in nuclear staining and cytoplasmic hypereosinophilia associated with the loss of

TABLE 12.5 Evolution of Morphologic Changes in Myocardial Infarction

Time	Gross Features	Light Microscope	Electron Microscope
Reversible Injury			
0–0.5 hr	None	None	Relaxation of myofibrils; glycogen loss; mitochondrial swelling
Irreversible Injury			
0.5–4 hr	None	Usually none; variable waviness of fibers at border	Sarcolemmal disruption; mitochondrial amorphous densities
4–12 hr	Dark mottling (occasional)	Early coagulative necrosis; edema; hemorrhage	
12–24 hr	Dark mottling	Ongoing coagulative necrosis; pyknosis of nuclei; myocyte hypereosinophilia; marginal contraction band necrosis; early neutrophilic infiltrate	
1–3 days	Mottling with yellow-tan infarct center	Coagulative necrosis, with loss of nuclei and striations; brisk interstitial infiltrate of neutrophils	
3–7 days	Hyperemic border; central yellow-tan softening	Beginning disintegration of dead myofibers, with dying neutrophils; early phagocytosis of dead cells by macrophages at infarct border; hyperemic border represents early granulation tissue	
7–10 days	Maximally yellow-tan and soft, with depressed red-tan margins	Well-developed phagocytosis of dead cells; granulation tissue at margins	
10–14 days	Red-gray depressed infarct borders	Well-established granulation tissue with new blood vessels and collagen deposition	
2–8 wk	Gray-white scar, progressive from border toward core of infarct	Increased collagen deposition, with decreased cellularity	
>2mo	Scarring complete	Dense collagenous scar	

negatively charged nucleic acids. **Myocyte vacuolization** is a *sublethal ischemic change* at the margins of infarcts caused by intracellular accumulations of salt and water within the sarcoplasmic reticulum; such myocytes are viable but poorly contractile.

Necrotic myocardium elicits acute inflammation (typically most prominent 1 to 3 days after MI), followed by a wave of macrophages that remove necrotic myocytes and neutrophil fragments (most pronounced 3 to 7 days after MI). The infarcted zone is progressively replaced by granulation tissue (most prominent 7 to 10 days after MI), which in turn forms the provisional scaffolding on which dense collagenous scar forms; because cardiac myocytes are terminally differentiated cells, no cardiomyocyte proliferation is seen. In most instances, scarring is well advanced by the end of the sixth week, but the efficiency of repair depends on the size of the original lesion and the ability of the host tissues to heal. Healing requires the migration of inflammatory cells and ingrowth of new vessels from the infarct margins. Thus, an MI heals from its borders toward the center, and a large infarct may not heal as fast or as completely as a small one. Moreover, malnutrition, poor vasculature, or exogenous antiinflammatory steroids can impede effective infarct scarring (Chapter 3). Once an MI is completely healed, it is impossible to distinguish its age: whether present for 8 weeks or 10 years, fibrous scars look the same.

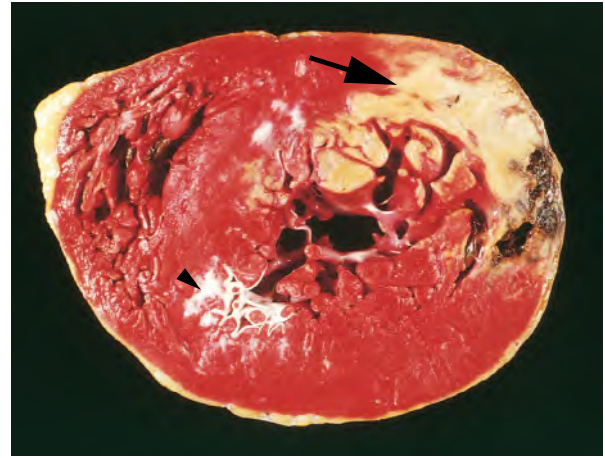


Fig. 12.12 Acute myocardial infarct, predominantly of the posterolateral left ventricle, demonstrated grossly by a lack of staining by triphenyl tetrazolium chloride (a substrate for the lactate dehydrogenase enzyme in viable myocardium) in areas of necrosis (*arrow*). Note the myocardial hemorrhage at one edge of the infarct that was associated with cardiac rupture, and the anterior scar (*arrowhead*), indicative of an old infarct. The specimen is oriented with the posterior wall at the top.

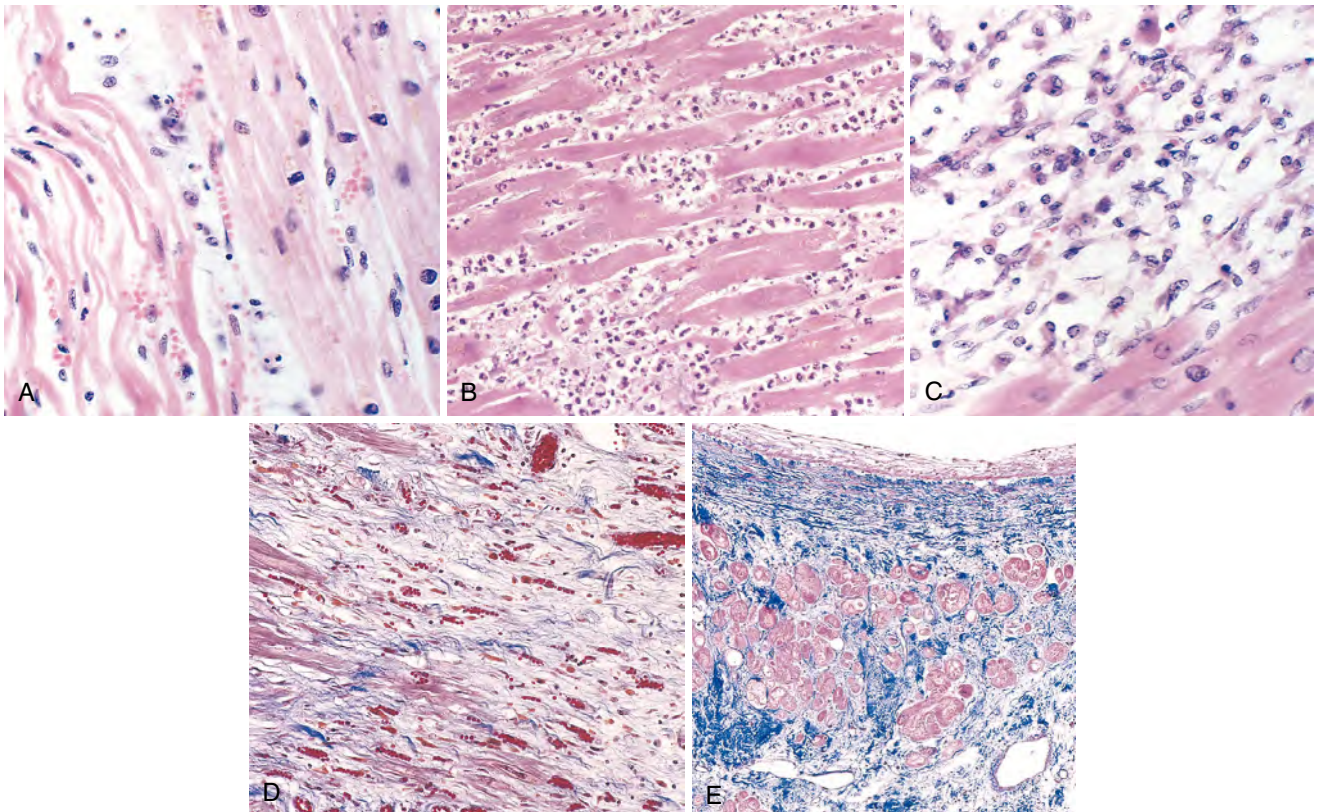


Fig. 12.13 Microscopic features of myocardial infarction and its repair. (A) A 1-day-old infarct showing coagulative necrosis and wavy fibers (elongated and narrow, as compared with adjacent normal fibers at *right*). Widened spaces between the dead fibers contain edema fluid and scattered neutrophils. (B) Dense infiltrate of neutrophils in an acute myocardial infarction that is 3 to 4 days old. (C) Removal of necrotic myocytes by phagocytosis (approximately 7 to 10 days). (D) Granulation tissue characterized by loose collagen and abundant capillaries. (E) Healed myocardial infarct in which the necrotic tissue has been replaced by a dense collagenous scar. The residual cardiac muscle cells show evidence of compensatory hypertrophy. D and E are stained with Masson trichrome, rendering collagenous connective tissue a deep blue color; note the accumulation of collagen, which is transformed into scar.

The following discussion considers the changes that result from interventions that can limit infarct size by salvaging myocardium that is not yet necrotic.

Infarct Modification by Reperfusion. The therapeutic goal in acute MI is to salvage the maximal amount of ischemic myocardium; this is accomplished by restoration of tissue perfusion as quickly as possible (hence the adage “time is myocardium”). Such reperfusion is achieved by thrombolysis (dissolution of thrombus by tissue plasminogen activator), angioplasty, or coronary arterial bypass graft. Although preservation of viable (but at-risk) heart improves both short- and long-term outcomes, reperfusion is not an unalloyed blessing. Late restoration of blood flow into ischemic tissues may be associated with arrhythmias and can incite greater local damage than might otherwise have occurred—so-called *reperfusion injury*.

The effects of reperfusion on myocardial viability and function are summarized in Fig. 12.14. Although the clinical significance of myocardial reperfusion injury is debated, it has been estimated that up to 50% (or more) of the ultimate infarct size can be attributed to its effects. To date, clinical trials to prevent reperfusion injury have not been fruitful, but this remains an active area of investigation. Factors that contribute to reperfusion injury include:

- **Mitochondrial dysfunction.** Ischemia alters mitochondrial membrane permeability, which leads to swelling and rupture of the outer membrane, releasing mitochondrial contents that promote apoptosis.
- **Myocyte hypercontraction.** During periods of ischemia, the intracellular levels of calcium are increased because of impaired calcium cycling and sarcolemmal damage. After reperfusion, the contraction of myofibrils is augmented and uncontrolled, causing cytoskeletal damage and cell death.
- **Free radicals** including superoxide anion (O_2^-), hydrogen peroxide (H_2O_2), hypochlorous acid ($HOCl$), nitric oxide–derived peroxynitrite, and hydroxyl radicals ($\bullet OH$) are produced within minutes of reperfusion and cause damage to the myocytes by altering membrane proteins and phospholipids.
- **Leukocyte aggregation**, which may occlude the microvasculature and contribute to the “no-reflow” phenomenon. Leukocytes also elaborate proteases and elastases that cause cell death. No reflow phenomenon is the failure of blood to perfuse an ischemic area after the physical

obstruction has been removed or bypassed. The underlying mechanism is related to arterial microvasculature damage.

- **Platelet and complement activation** also contribute to microvascular injury. Complement activation is thought to play a role in the no-reflow phenomenon by injuring the endothelium (Chapter 2).

The typical appearance of reperfused myocardium in the setting of an acute MI is shown in Fig. 12.15. As a consequence of vascular injury and leakiness, such infarcts are typically hemorrhagic. Microscopically, irreversibly damaged myocytes after reperfusion develop contraction band necrosis, marked by intense eosinophilic bands of hypercontracted sarcomeres created by an influx of calcium across plasma membranes that heightens actin-myosin interactions. In the absence of ATP, the sarcomeres cannot relax and get stuck in an agonal tetanic state. Thus, reperfusion alters the morphology of irreversibly injured cells.

The biochemical abnormalities (and their functional consequences) may also persist for days to weeks in reperfused myocytes. Such changes are thought to underlie a phenomenon referred to as *stunned myocardium*, a state of prolonged contractile dysfunction induced by short-term ischemia that usually recovers after several days with almost complete restoration of metabolic function. Myocardium that is subjected to chronic, sublethal ischemia can also enter a state of lowered metabolism and function called *hibernation*. Subsequent revascularization (e.g., by CABG surgery, angioplasty, or stenting) often restores normal function to hibernating myocardium.

Clinical Features

MI is diagnosed by clinical symptoms, laboratory tests for the presence of myocardial proteins in the plasma, and characteristic electrocardiographic changes. Patients with MI classically present with prolonged (more than 30 minutes) chest pain described as crushing, stabbing, or squeezing, associated with a rapid, weak pulse. Profuse sweating (diaphoresis), and nausea and vomiting are common and can suggest involvement of the posterior-inferior ventricle with secondary vagal stimulation. Dyspnea is a frequent symptom due to impaired contractility of the ischemic myocardium and the resultant pulmonary congestion and edema. However, in as many as 25% of patients the onset is entirely asymptomatic (e.g., in the setting of diabetic neuropathy), and the disease is discovered only by electrocardiographic changes or laboratory tests that show evidence of myocardial damage (see later).

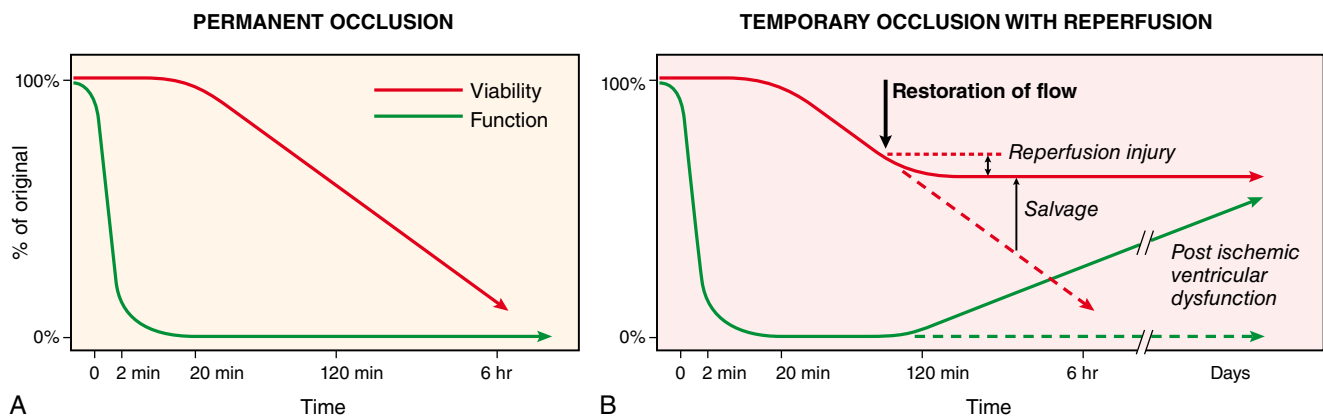


Fig. 12.14 Effects of reperfusion on myocardial viability and function. After coronary occlusion, contractile function is lost within 2 minutes, and viability begins to diminish after approximately 20 minutes. (A) If perfusion is not restored, nearly all myocardium in the affected region dies. (B) If flow is restored, some necrosis is prevented, and at least some function can return. However, reperfusion itself may induce damage (reperfusion injury) and return of function of salvaged myocardium may be delayed for hours to days (postischemic ventricular dysfunction or stunning).

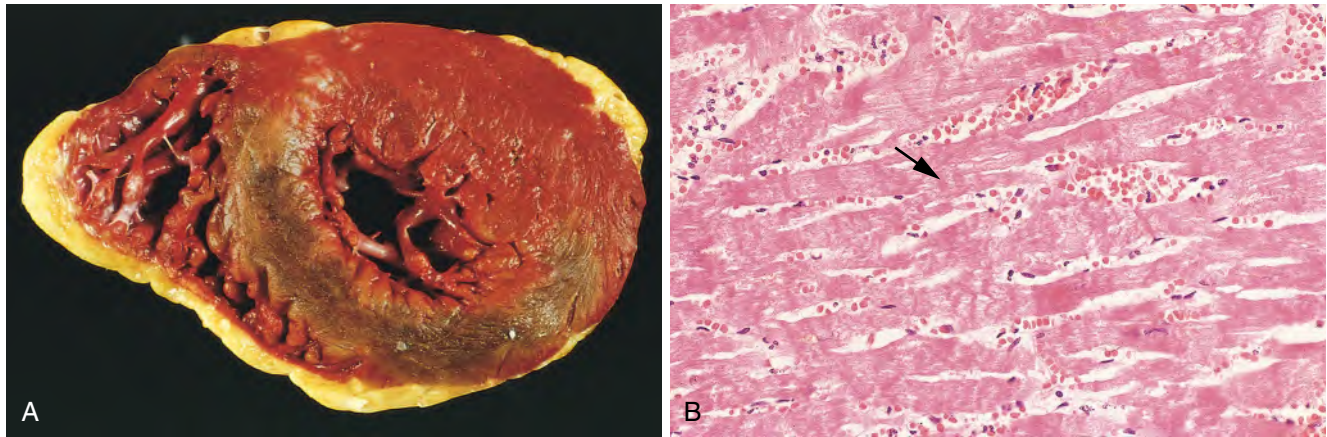


Fig. 12.15 Consequences of myocardial ischemia followed by reperfusion. Gross (A) and microscopic (B) appearance of myocardium modified by reperfusion. (A) Large, densely hemorrhagic, anterior wall acute myocardial infarction in a patient with left anterior descending artery thrombus treated with streptokinase, a fibrinolytic agent (slice is stained with triphenyl tetrazolium chloride). Specimen oriented with posterior wall at top. (B) Myocardial necrosis with hemorrhage and contraction bands, visible as dark bands spanning some myofibers (arrow).

Owing to the characteristic electrocardiographic changes resulting from myocardial ischemia or necrosis in various distributions, a transmural infarct is sometimes referred to as an ST-elevation myocardial infarct (STEMI) and a subendocardial infarct as a non-ST-elevation infarct (NSTEMI). Depending on the extent and location of the vascular involvement, smaller infarctions show nonspecific changes or can even be electrocardiographically silent.

The laboratory evaluation of MI is based on measuring the blood levels of proteins that leak out of irreversibly damaged myocytes. The most clinically useful biomarkers of myocardial damage are cardiac-specific troponins T and I (cTnT and cTnI), proteins that normally regulate calcium-mediated contraction of cardiac muscle (Fig. 12.16). The diagnosis of myocardial injury is established when blood levels of troponin are elevated above threshold levels; the tempo and magnitude of appearance of these serum markers after MI depends on

several factors, including the volume of damaged myocardium, blood flow and lymphatic drainage in the area of the infarct, and the rate of elimination of the marker from the blood. Cardiac troponins begin to rise in 2 to 4 hours, peak at 24 to 48 hours, and remain elevated for 7 to 10 days after an acute infarct. With reperfusion, troponin levels may be higher and peak earlier owing to more rapid washout of the marker from the necrotic tissue.

Significant acute elevation in serum troponin is also seen in other conditions that cause death of cardiac myocytes, such as myocarditis and myocardial trauma. On the other hand, low-level serum troponin elevation (“troponin leak”) can occur in a host of other conditions, including CHF, pulmonary embolus, renal failure, and sepsis. These elevations do not usually follow the same time course as those following MI, so serial measurements may be helpful in distinguishing different etiologies.

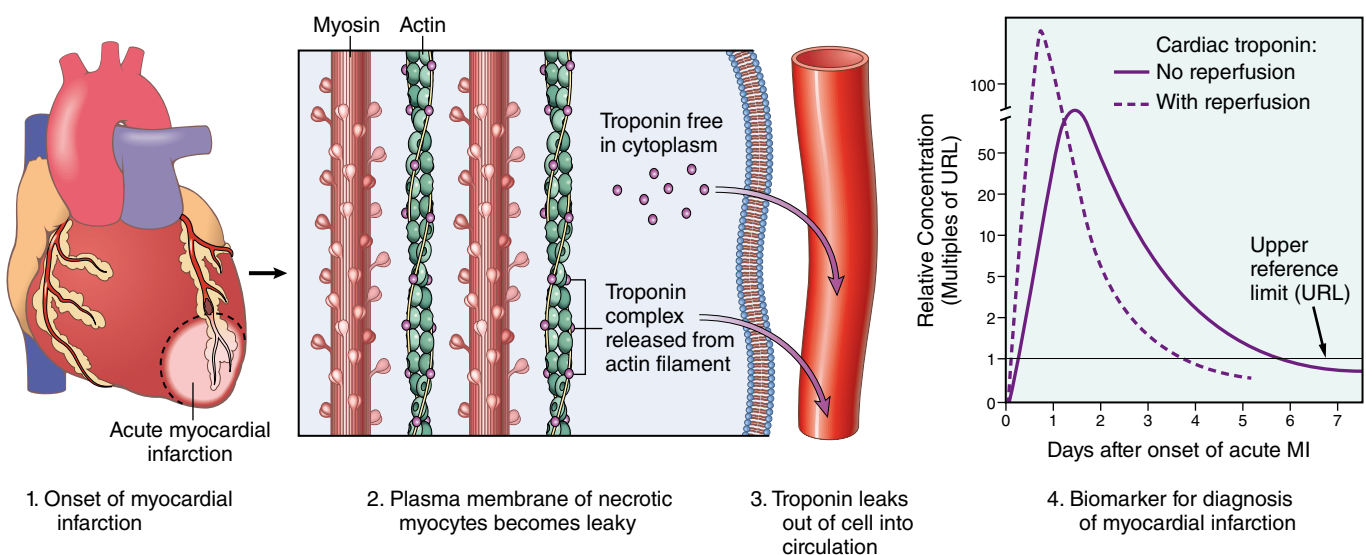


Fig. 12.16 Release of myocyte proteins in myocardial infarction (MI). Troponin I or troponin T are routinely used as diagnostic biomarkers of myocyte injury.

Recommended initial therapies for acute MI are based on the pathophysiologic features discussed earlier and include:

- *Oxygen supplementation* to improve blood oxygen saturation for patients with hypoxia or respiratory distress
- *Nitrates* to induce vasodilation and reverse vasospasm
- *Antiplatelet agents* such as aspirin, ADP receptor inhibitors, and GPIIb/IIIa inhibitors
- *Anticoagulant therapy* with unfractionated heparin, low-molecular-weight heparin, direct thrombin inhibitors, and/or factor Xa inhibitors to prevent coronary artery thrombus propagation
- *Beta blockers* to decrease myocardial oxygen demand and to reduce the risk of arrhythmias, unless contraindicated, such as in heart failure
- *Prompt reperfusion* to salvage myocardium, by either fibrinolytic medications or catheter-based interventions
- *Improve myocardial oxygen supply versus demand* by management of other factors, such as anxiety, ischemic pain, abnormal hemodynamics, anemia, and respiratory disorders
- *Early arrhythmia monitoring* and management

Consequences and Complications of Myocardial Infarction

Extraordinary progress has been made in improving patient outcomes after acute MI. The overall in-hospital death rate for MI is less than 7%,

with STEMI patients experiencing slightly higher mortality (9%) than NSTEMI patients (6%). Unfortunately, mortality for out-of-hospital MIs is substantially worse: one-third of patients who suffer a STEMI at home will die, usually of an arrhythmia within 1 hour of symptom onset and before they can receive medical attention. Such statistics make the rising rate of CAD in countries with limited hospital facilities all the more worrisome.

Nearly three-quarters of patients experience one or more complications after an acute MI (Fig. 12.17):

- *Contractile dysfunction.* In general, MIs affect left ventricular pump function in proportion to the volume of damage. In most cases, there is some degree of left ventricular failure manifested as hypotension, pulmonary congestion, and pulmonary edema. Severe pump failure (*cardiogenic shock*) occurs in roughly 10% of patients with transmural MIs and typically is associated with infarcts that damage 40% or more of the left ventricle.
- *Papillary muscle dysfunction.* Although papillary muscles rupture infrequently after MI, they are often dysfunctional and can be poorly contractile because of ischemia, leading to postinfarct mitral regurgitation. Much later, papillary muscle fibrosis and shortening, or global ventricular dilation also can cause mitral regurgitation.
- *Right ventricular infarction.* Although isolated right ventricular infarction occurs in only 1% to 3% of MIs, the right ventricle is affected

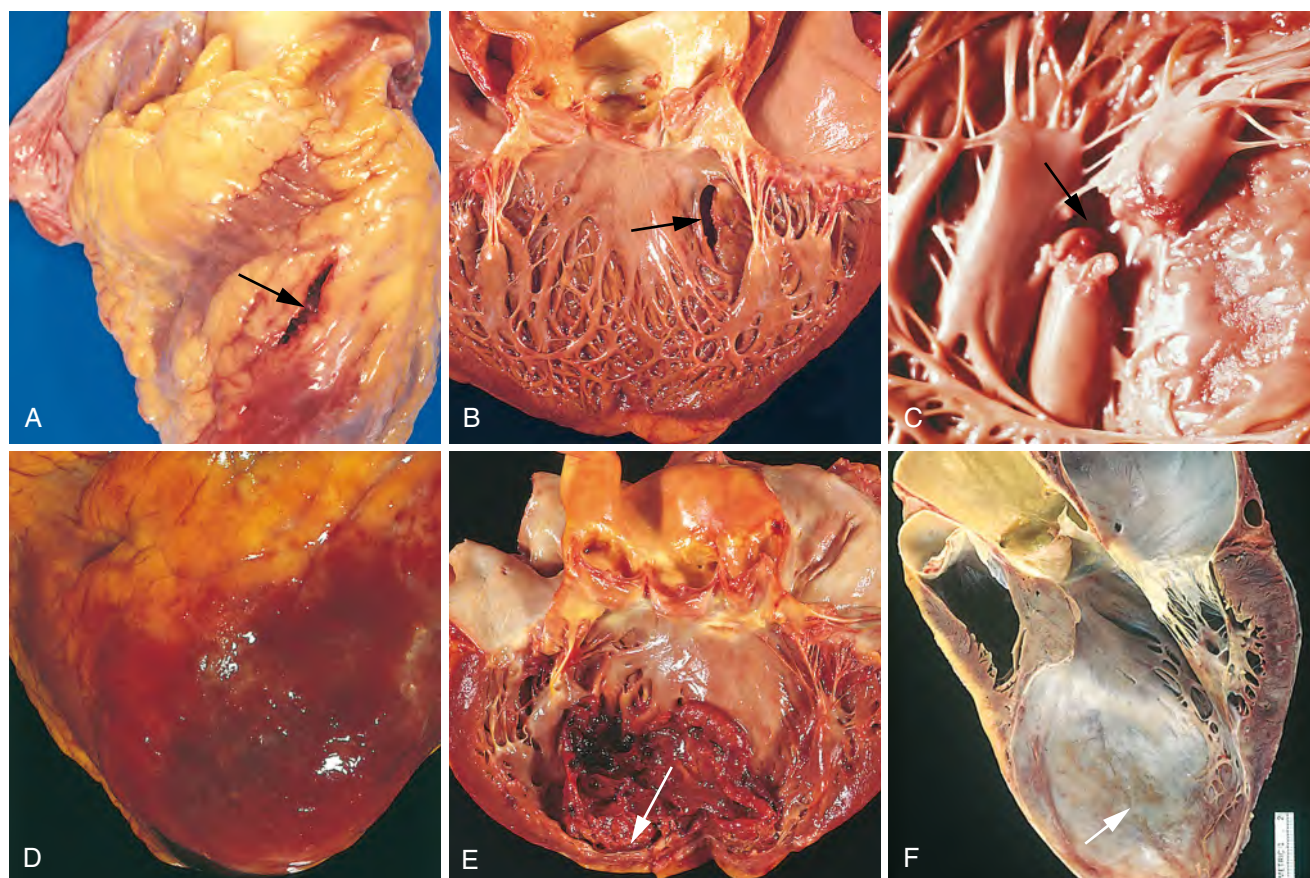


Fig. 12.17 Complications of myocardial infarction. (A) Anterior myocardial rupture in an acute infarct (arrow). (B) Rupture of the ventricular septum (arrow). (C) Complete rupture of a necrotic papillary muscle (arrow). (D) Fibrinous pericarditis, showing a dark, roughened epicardial surface overlying an acute infarct. (E) Early expansion of anteroapical infarct with wall thinning (arrow) and mural thrombus. (F) Large apical left ventricular aneurysm (arrow). The left ventricle is on the right in this apical four-chamber view of the heart. (A–E, From Schoen FJ. *Interventional and Surgical Cardiovascular Pathology: Clinical Correlations and Basic Principles*. WB Saunders; 1989. F, Courtesy William D. Edwards, MD, Mayo Clinic, Rochester, MN.)

by RCA occlusions leading to posterior septal or left ventricular infarction. In either case, right-sided heart failure is a common outcome, leading to pooling of blood in the venous circulation and systemic hypotension.

- **Myocardial rupture.** Rupture complicates only 1% to 3% of MIs but is frequently fatal. Left ventricular free wall rupture is most common, usually resulting in rapidly fatal hemopericardium and cardiac tamponade (see Fig. 12.17A). Ventricular septal rupture creates a VSD with left-to-right shunting (see Fig. 12.17B), and papillary muscle rupture leads to severe mitral regurgitation (see Fig. 12.17C). Rupture occurs most commonly within 3 to 7 days after infarction—the time in the healing process when lysis of necrotic myocardium is maximal and when much of the infarct has been converted to soft, friable granulation tissue. Risk factors for free wall rupture include age older than 60 years, anterior or lateral wall infarctions, female sex, lack of left ventricular hypertrophy, and first MI (because scarring associated with prior MIs tends to limit the risk for myocardial tearing).
- **Arrhythmias.** MIs lead to myocardial irritability and conduction disturbances that can cause sudden death. Approximately 90% of patients develop some form of rhythm disturbance, with the incidence being higher in STEMIs versus NSTEMIs. MI-associated arrhythmias include heart block of variable degree (including asystole), bradycardia, supraventricular tachyarrhythmias, ventricular premature contractions or ventricular tachycardia, and ventricular fibrillation. The risk for serious arrhythmias (e.g., ventricular fibrillation) is greatest in the first hour and declines thereafter.
- **Pericarditis.** Transmural MIs can elicit a fibrinohemorrhagic pericarditis; this is an epicardial manifestation of the underlying myocardial inflammation (see Fig. 12.17D). Heralded by anterior chest pain and a pericardial friction rub, pericarditis typically appears 2 to 3 days after infarction and then gradually resolves over the next few days. Extensive infarcts or severe pericardial inflammation occasionally lead to large effusions or organize to form dense adhesions that eventually manifest as a constrictive lesion. Rarely patients develop intense pericarditis weeks after MI (*Dressler syndrome*) due to antibodies directed against damaged pericardium and myocardium.
- **Chamber dilation.** Because of the weakening of necrotic muscle, there may be disproportionate stretching, thinning, and dilation of the infarcted region (especially with anteroseptal infarcts).
- **Mural thrombus.** With any infarct, the combination of attenuated myocardial contractility (causing stasis), chamber dilation, and endocardial damage (causing a thrombogenic surface) can foster mural thrombosis (see Fig. 12.17E), and potentially lead to embolism.
- **Ventricular aneurysm.** A late complication, aneurysms of the ventricle most commonly result from a large transmural anteroseptal infarct that heals with the formation of a thinned wall of scar tissue (see Fig. 12.17F). Ventricular aneurysms frequently give rise to formation of mural thrombi, arrhythmias, and heart failure, but do not rupture.
- **Progressive heart failure.** This is discussed later in this chapter.

The risk of postinfarct complications and the prognosis of the patient depend primarily on the infarct size, location, and fraction of the wall thickness involved (subendocardial or transmural). Thus, large transmural infarcts yield a higher probability of cardiogenic shock, arrhythmias, and late CHF. Patients with anterior transmural infarcts are at greatest risk for free-wall rupture, dilation, mural thrombi, and aneurysm. In contrast, posterior transmural infarcts are more likely to be complicated by conduction blocks, right ventricular

involvement, or both; when acute VSDs occur in this area they are more difficult to manage. Moreover, female sex, age older than 70 years, diabetes mellitus, and previous MI are poor prognostic factors in patients with STEMIs. With subendocardial infarcts, pericarditis, rupture, and aneurysm only rarely occur.

In addition to the sequence of repair in the infarcted tissues described earlier, the noninfarcted segments of the ventricle undergo hypertrophy and dilation; collectively, these changes are termed ventricular remodeling. The compensatory hypertrophy of noninfarcted myocardium is initially hemodynamically beneficial. However, this adaptive effect may be overwhelmed by ventricular dilation (with or without ventricular aneurysm) and increased oxygen demand, which can exacerbate ischemia and depress cardiac function. There may also be changes in ventricular shape and stiffening of the ventricle due to scar formation and hypertrophy that further diminish cardiac output. Some of these deleterious effects appear to be reduced by ACE inhibitors, which lessen the ventricular remodeling that occurs after infarction.

Long-term prognosis after MI depends on many factors, the most important of which are the residual left ventricular function and the extent of any vascular obstructions in vessels that perfuse the remaining viable myocardium. The overall mortality within the first year is roughly 20% (with most deaths in that time span occurring within 30 days); thereafter, each passing year is associated with an additional 1% to 2% mortality among survivors. Infarct prevention (through control of risk factors) in individuals who have never experienced MI (primary prevention) and prevention of reinfarction in MI survivors (secondary prevention) are important strategies that have achieved considerable success.

Chronic Ischemic Heart Disease

The designation chronic IHD (often called “ischemic cardiomyopathy” by clinicians) is used here to describe progressive CHF as a consequence of accumulated ischemic myocardial damage and/or inadequate compensatory responses. In most instances, there has been a prior MI and sometimes previous coronary arterial interventions and/or bypass surgery. Chronic IHD usually appears postinfarction due to the functional decompensation of hypertrophied noninfarcted myocardium (see earlier discussion of cardiac hypertrophy). In other cases severe obstructive CAD may present as chronic CHF in the absence of prior infarction. Patients with chronic IHD account for roughly 40% of cardiac transplant recipients.

MORPHOLOGY

Patients with chronic IHD have cardiomegaly, with left ventricular hypertrophy and dilation. Invariably there is some degree of stenotic coronary atherosclerosis. Discrete scars representing healed infarcts are usually present. The mural endocardium often has patchy fibrous thickenings (due to abnormal wall shear forces), and mural thrombi may be present. Microscopic findings include myocardial hypertrophy, diffuse subendocardial myocyte vacuolization, and interstitial and replacement fibrosis.

The relationship of the causes, pathophysiology, and consequences of MI are summarized in Fig. 12.18, including the possible outcomes of chronic IHD and sudden death, discussed later.

KEY CONCEPTS

Ischemic Heart Disease

- The vast majority of IHD is due to coronary artery atherosclerosis. Vaso-spasm, vasculitis, or embolism are less common causes.

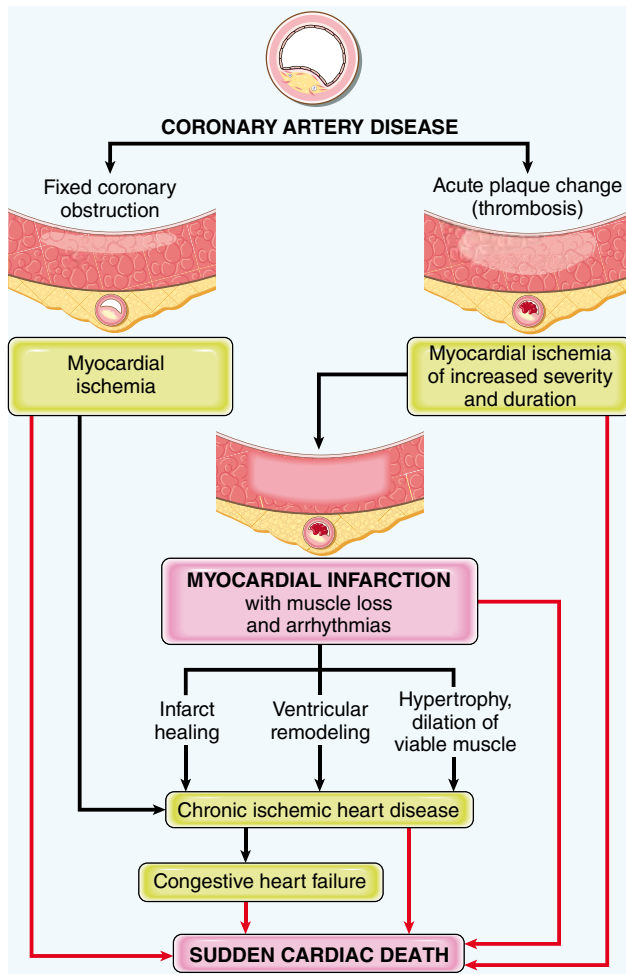


Fig. 12.18 Schematic for the causes and outcomes of ischemic heart disease, showing the interrelationships among coronary artery disease and its major sequelae.

- Cardiac ischemia results from a mismatch in coronary supply and myocardial demand, and it presents as several, somewhat discrete syndromes:
 - Angina pectoris is chest pain on exertion due to inadequate perfusion and is typically due to atherosclerotic disease with greater than 70% fixed stenosis (so-called critical stenosis).
 - Unstable angina results from a small fissure or rupture of atherosclerotic plaque triggering platelet aggregation, vasoconstriction, and formation of a mural thrombus that needs not necessarily be occlusive. Chest pain may occur, even at rest or with minimal exertion.
 - Acute MI typically results from acute thrombosis after plaque disruption, mostly of plaques that did not previously produce critical stenosis.
 - Chronic IHD is progressive heart failure due to ischemic injury, either from prior infarctions or chronic low-grade ischemia.
- Severe myocardial ischemia leads to loss of function within 1 to 2 minutes and necrosis after 20 to 30 minutes. MI is diagnosed based on symptoms, electrocardiographic changes, and measurement of serum troponins. Gross and histologic changes of infarction require hours to days to develop.
- Infarction can be modified by therapeutic intervention (e.g., thrombolysis or stenting), which salvages myocardium at risk, but may induce reperfusion-related injury.
- Complications of infarction include ventricular rupture, papillary muscle dysfunction, aneurysm formation, mural thrombus, arrhythmia, pericarditis, and CHF.

ARRHYTHMIAS

Aberrant cardiac rhythms can cause sudden cardiac death. Abnormalities in myocardial conduction can be sustained or sporadic (paroxysmal). Aberrant rhythms can be initiated anywhere in the conduction system, from SA node down to the level of an individual myocyte; they are typically designated as originating from the atrium (supraventricular) or within the ventricular myocardium. Arrhythmias can manifest as *tachycardia* (fast heart rate), *bradycardia* (slow heart rate), an irregular rhythm with normal ventricular contraction, chaotic depolarization without functional ventricular contraction (*ventricular fibrillation*), or no electrical activity at all (*asystole*). Patients may be unaware of a rhythm disorder or may note a “racing heart” or palpitations (irregular rhythm); loss of adequate cardiac output due to sustained arrhythmia can produce lightheadedness (near syncope), loss of consciousness (syncope), or sudden cardiac death (see later).

Ischemic injury is a common cause of rhythm disorders, either through direct damage or through the dilation of heart chambers that alters conduction system firing. Listed below are some of the common mechanisms of arrhythmias:

- *Sick sinus syndrome*. If the SA node is damaged, other fibers or even the AV node can take over pacemaker function, albeit at a much slower intrinsic rate (causing bradycardia).
- *Atrial fibrillation*. If the atrial myocytes become irritable and depolarize independently and sporadically (as occurs with atrial dilation), the signals are variably transmitted through the AV node, leading to an “irregularly irregular” heart rate.
- *Heart block*. If the AV node is dysfunctional, varying degrees of heart block occur, ranging from simple prolongation of the P-R interval on electrocardiogram (first-degree heart block), to intermittent transmission of the signal (second-degree heart block), to complete failure (third-degree heart block).

As discussed earlier, coordinated cardiac contraction depends on the orderly transmission of electrical currents from myocyte to myocyte via gap junctions. Thus, abnormalities in the structure or spatial distribution of gap junctions, which are seen in a variety of disorders (e.g., IHD and dilated cardiomyopathies), can cause arrhythmias. Ischemia, myocyte hypertrophy, and inflammation (e.g., myocarditis or sarcoidosis) also promote increased “irritability” that leads to spontaneous aberrant myocyte depolarization; because of the electrical interconnection of myocytes, such random events can cause inappropriate firing of adjacent cells and create abnormal electrical pathways (so-called *reentry circuits*) that lead to ventricular tachycardia, and can progress to fatal ventricular fibrillation. Likewise, deposition of nonconducting material (e.g., amyloid) and even small areas of fibrosis can disrupt myocyte-to-myocyte signaling, again sowing the seeds for development of reentry circuits that can give rise to potentially fatal arrhythmias.

Heritable conditions associated with arrhythmias are important to recognize because they may alert physicians to the need for intervention to prevent sudden cardiac death (discussed later). Some of these disorders are associated with recognizable anatomic abnormalities (e.g., congenital anomalies, hypertrophic cardiomyopathy, mitral valve prolapse). However, other heritable disorders precipitate arrhythmias and sudden death in the absence of structural cardiac pathology (so-called primary electrical disorders). These syndromes can only be reliably diagnosed by genetic testing, which is performed in those with a positive family history or an unexplained nonlethal arrhythmia.

Channelopathies are the most important of the primary electrical abnormalities of the heart that predispose to arrhythmias (see Table 12.6). Channelopathies are caused by mutations in genes that are

TABLE 12.6 Selected Examples of Causal Genes in Inherited Arrhythmogenic Diseases^a

Disorder	Gene	Function
Long QT syndrome ^b	<i>KCNQ1</i>	K ⁺ channel (LOF)
	<i>KCNH2</i>	K ⁺ channel (LOF)
	<i>SCN5A</i>	Na ⁺ channel (GOF)
	<i>CAV3</i>	Caveolin, Na ⁺ current (GOF)
Short QT syndrome ^b	<i>KCNQ1</i>	K ⁺ channel (GOF)
	<i>KCNH2</i>	K ⁺ channel (GOF)
Brugada syndrome ^b	<i>SCN5A</i>	Na ⁺ channel (LOF)
	<i>CACNB2b</i>	Ca ⁺⁺ channel (LOF)
	<i>SCN1b</i>	Na ⁺ channel (LOF) ^a
CPVT syndrome ^b	<i>RYR2</i>	Diastolic Ca ⁺⁺ release (GOF)
	<i>CASQ2</i>	Diastolic Ca ⁺⁺ release (LOF)

CPVT, Catecholaminergic polymorphic ventricular tachycardia; GOF, gain-of-function mutations; LOF, loss-of-function mutations.

^aDifferent mutations can cause the same general syndrome, and mutations in some genes can cause multiple different phenotypes; thus, loss-of-function (LOF) mutations may cause long QT intervals, whereas gain-of-function (GOF) mutations result in short repolarization intervals.

^bLong QT syndrome manifests as arrhythmias associated with excessive prolongation of the cardiac repolarization; patients often present with stress-induced syncope or sudden cardiac death (SCD), and some forms are associated with swimming. Short QT syndrome patients have arrhythmias associated with abbreviated repolarization intervals; they can present with palpitations, syncope, and SCD. Brugada syndrome manifests as ECG abnormalities (ST segment elevations and right bundle branch block) in the absence of structural heart disease; patients classically present with syncope or SCD during rest or sleep, or after large meals. CPVT does not have characteristic ECG changes; patients often present in childhood with life-threatening arrhythmias due to adrenergic stimulation (stress-related). Modified from Cerrone M, Priori SG. Genetics of sudden death: focus on inherited channelopathies. *Eur Heart J*. 2011;32(17):2109-2118.

required for normal ion channel function. These disorders (mostly with autosomal dominant inheritance) either involve genes that encode the structural components of ion channels (including Na⁺, K⁺, and Ca⁺⁺ channels) or accessory proteins that are essential for normal channel function. Ion channels are responsible for conducting the electrical currents that mediate contraction of the heart, and it is thus not surprising that defects in these channels may provoke arrhythmias. The prototype channelopathy is the *long QT syndrome*, characterized by prolongation of the QT segment in ECGs with susceptibility to serious ventricular arrhythmias. Ion channels are needed for the normal function of many tissues, and certain channelopathies are also associated with skeletal muscle disorders and diabetes. Nevertheless, the most common channelopathies are isolated disorders of the heart, and their most feared consequence is sudden cardiac death (SCD; discussed next).

Sudden Cardiac Death

SCD is most commonly defined as unexpected death from cardiac causes within 1 hour of symptom onset or within 24 hours of having been last seen well; this happens in some 325,000 individuals each year in the United States and 4 to 5 million people worldwide. **The mechanism of SCD is most often a lethal arrhythmia (e.g., asystole or ventricular fibrillation).** Although ischemic injury (and other pathologies) can directly affect the major components of the conduction system, most cases of fatal arrhythmia are triggered by electrical irritability of the myocardium distant from the major elements of the conduction system.

CAD is the leading cause of SCD, responsible for most cases; unfortunately, SCD can be the first manifestation of IHD. In such cases, there is typically only chronic severe atherosclerotic disease with fixed critical stenoses; acute plaque disruption is found in only 10% to 20%

of cases, and 80% to 90% of patients with SCD who are successfully resuscitated do not show any enzymatic or ECG evidence of myocardial necrosis. Healed remote MIs are present in about 40%, and subendocardial myocyte vacuolization indicative of severe chronic ischemia is a common feature. With the decrease of prevalence of IHD in high-resource countries, SCD is increasingly being seen in individuals with hearts that are hypertrophic and fibrotic (from hypertension, obesity, use of cocaine, etc.) without IHD.

With younger patients, nonatherosclerotic causes are more common etiologies for SCD. These include:

- *Cardiac conduction abnormalities*, hereditary or acquired
- *Dilated or hypertrophic cardiomyopathy*
- *Congenital coronary arterial abnormalities*
- *Myocarditis*
- *Mitral valve prolapse*
- *Pulmonary hypertension*
- *Other miscellaneous causes*, such as pericardial tamponade, pulmonary embolism, systemic metabolic and hemodynamic alterations, catecholamines, and non-prescription drug use, particularly of cocaine and methamphetamine.

The prognosis of patients vulnerable to SCD is markedly improved by pharmaceutical intervention and by implantation of automatic cardioverter defibrillators that can sense and electrically counteract episodes of ventricular fibrillation.

KEY CONCEPTS

Arrhythmias

- Arrhythmias can be caused by ischemic or structural changes in the conduction system or by intrinsic myocyte electrical instability. In structurally normal hearts, arrhythmias are often due to mutations in ion channels that cause aberrant repolarization or depolarization.
- SCD typically results from ventricular fibrillation and is most frequently a consequence of CAD. Myocardial irritability usually results from nonlethal ischemia or from preexisting fibrosis from previous myocardial injury. SCD is less often due to acute plaque rupture with thrombosis that induces a rapidly fatal arrhythmia.
- Multiple re-entry circuits generated within the ventricular myocardium may progress through ventricular tachycardia to life-threatening ventricular fibrillation.

HYPERTENSIVE HEART DISEASE

Hypertensive heart disease (HHD) is a consequence of the increased demands placed on the heart by hypertension, causing pressure overload and ventricular hypertrophy. Although most commonly seen in the left heart as the result of systemic hypertension, pulmonary hypertension can cause right-sided HHD, or cor pulmonale.

Systemic (Left-Sided) Hypertensive Heart Disease

Hypertrophy of the heart is an adaptive response to the pressure overload of chronic hypertension. However, such compensatory changes may be ultimately maladaptive and can lead to myocardial dysfunction, myocardial “demand” ischemia, cardiac dilation, CHF, and, in some cases, sudden death.

The minimal pathologic criteria for the diagnosis of systemic HHD are (1) left ventricular hypertrophy (usually concentric) in the absence of other cardiovascular pathology and (2) a clinical history or pathologic evidence of hypertension in other organs (e.g., kidney).

The Framingham Study established unequivocally that even mild hypertension (levels only slightly above 140/90 mm Hg)—if sufficiently prolonged—induces left ventricular hypertrophy; roughly one-third of the United States population has hypertension of at least that degree. More recently, the criteria for diagnosing hypertension were changed; according to the American Heart Association diastolic pressures above 80 mm Hg, or systolic pressures above 130 mm Hg constitute clinically significant elevations in blood pressure, whereas European Society of Hypertension considers elevation above 140/90 as threshold for defining hypertension. Using the AHA criteria, almost one-half of individuals in the general population are hypertensive. The pathogenesis of hypertension is discussed in Chapter 11.

MORPHOLOGY

Hypertension induces left ventricular pressure overload hypertrophy, initially without ventricular dilation. As a result, the left ventricular wall thickening increases the weight of the heart disproportionately to the increase in overall cardiac size (Fig. 12.19A). The heart weight may exceed 500 g, and the left ventricular wall thickness may exceed 2.0 cm. Over time, the increased thickness of the left ventricular wall, associated with increased interstitial connective tissue, imparts a stiffness that impairs diastolic filling, often leading to left atrial enlargement.

Microscopically, the earliest change of systemic HHD is an increase in the transverse diameter of myocytes, which may be difficult to appreciate on routine microscopy. At a more advanced stage, variable degrees of cellular and nuclear enlargement become apparent, often accompanied by perivascular and interstitial fibrosis.

Compensated systemic HHD may be asymptomatic, producing only electrocardiographic or echocardiographic evidence of left ventricular enlargement. In many patients, systemic HHD comes to attention because of new atrial fibrillation induced by left atrial enlargement, or by progressive CHF. Depending on the severity, duration, and underlying basis of the hypertension, and on the adequacy of therapeutic control, the patient may (1) enjoy normal longevity and die of unrelated causes, (2) develop IHD due to both the potentiating effects of hypertension on coronary atherosclerosis and the ischemia induced by increased oxygen

demand from the hypertrophic muscle, (3) experience renal damage or cerebrovascular stroke as direct effects of hypertension, or (4) experience progressive heart failure or SCD. Effective control of hypertension can prevent cardiac hypertrophy and can even lead to its regression; with normalization of blood pressure, the associated risks of HHD are diminished.

Pulmonary (Right-Sided) Hypertensive Heart Disease (Cor Pulmonale)

Normally, because the pulmonary vasculature is the low-pressure side of the circulation, the right ventricle has a thinner and more compliant wall than the left ventricle. Isolated pulmonary HHD, or cor pulmonale, stems from right ventricular pressure overload. Chronic cor pulmonale is characterized by right ventricular hypertrophy, dilation, and potentially right-sided failure. Typical causes of chronic cor pulmonale are disorders of the lungs, especially chronic parenchymal diseases such as emphysema, and primary pulmonary hypertension (see Table 12.7; see also Chapter 15). Acute cor pulmonale can follow massive pulmonary embolism. It should also be remembered that pulmonary hypertension most commonly occurs as a complication of left-sided heart disease.

MORPHOLOGY

In **acute cor pulmonale** there is marked dilation of the right ventricle without hypertrophy. On cross-section, the normal crescent shape of the right ventricle chamber is transformed to a dilated ovoid. In **chronic cor pulmonale**, the right ventricular wall thickens, sometimes up to 1.0 cm or more (Fig. 12.19B). More subtle right ventricular hypertrophy may take the form of thickening of the muscle bundles in the outflow tract, immediately below the pulmonary valve, or thickening of the moderator band, the muscle bundle that connects the ventricular septum to the anterior right ventricular papillary muscle. Sometimes, the hypertrophied right ventricle compresses the left ventricular chamber or leads to regurgitation and fibrous thickening of the tricuspid valve.

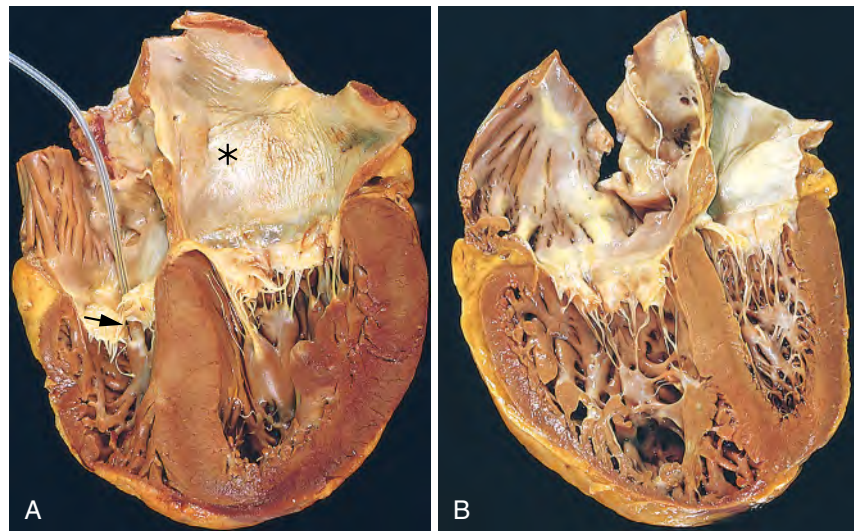


Fig. 12.19 Hypertensive heart disease, systemic and pulmonary. (A) Systemic (left-sided) hypertensive heart disease. There is marked concentric thickening of the left ventricular wall causing reduction in lumen size. The left ventricle and left atrium (*asterisk*) are on the *right* in this apical four-chamber view of the heart. A pacemaker is present in the right ventricle (*arrow*). (B) Pulmonary (right-sided) hypertensive heart disease (cor pulmonale). The right ventricle is markedly dilated and has a thickened free wall and hypertrophied trabeculae (apical four-chamber view of heart, right ventricle at *left*). The shape of the left ventricle (*to the right*) has been distorted by the enlarged right ventricle.

TABLE 12.7 Disorders Predisposing to Cor Pulmonale

Diseases of the Pulmonary Parenchyma
Chronic obstructive pulmonary disease
Diffuse pulmonary interstitial fibrosis
Pneumoconioses
Cystic fibrosis
Bronchiectasis
Diseases of the Pulmonary Vessels
Recurrent pulmonary thromboembolism
Primary pulmonary hypertension
Extensive pulmonary arteritis (e.g., granulomatosis with polyangiitis)
Drug-, toxin-, or radiation-induced vascular obstruction
Extensive pulmonary tumor microembolism
Disorders Affecting Chest Movement
Kyphoscoliosis
Marked obesity (sleep apnea, Pickwickian syndrome)
Neuromuscular diseases
Disorders Inducing Pulmonary Arterial Constriction
Metabolic acidosis
Hypoxemia
Chronic altitude sickness
Obstruction of major airways
Idiopathic alveolar hypoventilation

KEY CONCEPTS

Hypertensive Heart Disease

- HHD can affect either the left ventricle or the right ventricle; the latter is called cor pulmonale. Elevated pressures induce myocyte hypertrophy and interstitial fibrosis that increases wall thickness and myocardial stiffness.
- The chronic pressure overload of systemic hypertension causes left ventricular concentric hypertrophy, often associated with left atrial dilation due to impaired diastolic filling of the ventricle. Persistently elevated pressure overload can cause ventricular failure with dilation.
- Cor pulmonale results from pulmonary hypertension due to primary lung parenchymal or vascular disorders. There is commonly right ventricular and right atrial hypertrophy; right ventricular and atrial dilation can also occur.

VALVULAR HEART DISEASE

Valvular disease can come to clinical attention due to stenosis, insufficiency (synonyms: regurgitation or incompetence), or both.

- **Stenosis** is the failure of a valve to open completely, obstructing forward flow. Acquired valvular stenosis is almost always due to a primary leaflet abnormality and is virtually always a chronic process (e.g., calcification or valve scarring).
- **Insufficiency** results from failure of a valve to close completely, thereby allowing regurgitation (backflow) of blood. Valvular insufficiency can result from either intrinsic disease of valve leaflets (e.g., endocarditis) or disruption of the supporting structures (e.g., the aorta, mitral annulus, tendinous cords, papillary muscles, or ventricular free wall) without primary leaflet injury. It can appear abruptly, as with chordal rupture, or insidiously, as a consequence of leaflet scarring and retraction.

Stenosis or insufficiency can occur alone or together in the same valve. Valvular disease can involve only one valve (the mitral valve being the most common target) or more than one valve. Abnormal flow through diseased valves typically produces abnormal heart sounds

called murmurs; severe lesions can even be externally palpated as thrills. Depending on the valve involved, murmurs are best heard at different locations on the chest wall; moreover, the nature (regurgitation versus stenosis) and severity of the valvular disease determines the quality and timing of the murmur (e.g., harsh systolic or soft diastolic murmurs).

The clinical consequences of valve dysfunction vary depending on the valve involved, the degree of impairment, the tempo of disease onset, and the rate and quality of compensatory mechanisms. For example, sudden destruction of an aortic valve cusp by infection (infectious endocarditis; see later) can cause acute, massive, and rapidly fatal aortic regurgitation. In contrast, rheumatic mitral stenosis typically develops indolently over years, and its clinical effects can be well tolerated for extended periods. Certain conditions can complicate valvular heart disease by increasing the demands on the heart; for example, the increased output demands of pregnancy can exacerbate valve disease and lead to unfavorable maternal or fetal outcomes. Valvular stenosis or insufficiency often produces secondary changes, both proximal and distal to the affected valve, particularly in the myocardium. Generally, valvular stenosis leads to pressure overload cardiac hypertrophy, whereas valvular insufficiency leads to volume overload; both situations can culminate in heart failure. In addition, the ejection of blood through narrowed stenotic valves can produce high speed “jets” of blood that can injure the endothelium or endocardium where they impact or can result in significant hemolysis.

Valvular abnormalities can be congenital (discussed earlier) or acquired. Acquired valvular stenosis is almost always a consequence of a remote or chronic injury of the valve leaflets that declares itself clinically only after many years. In contrast, acquired valvular insufficiency can result from intrinsic disease of the valve leaflets or damage to or distortion of the supporting structures (e.g., the aorta, mitral annulus, tendinous cords, papillary muscles, ventricular free wall).

The causes of acquired heart valve diseases are summarized in Table 12.8. The most frequent causes of the major valvular lesions are:

- **Aortic stenosis:** calcification and sclerosis of anatomically normal or congenitally bicuspid aortic valves
- **Aortic insufficiency:** dilation of the ascending aorta, often secondary to hypertension and/or aging
- **Mitral stenosis:** rheumatic heart disease
- **Mitral insufficiency:** myxomatous degeneration in mitral valve prolapse or left ventricular dilation due to ischemic or nonischemic heart failure

Calcific Valvular Degeneration

Heart valves are subjected to high levels of repetitive mechanical stress, particularly at the hinge points of the leaflets; this is a consequence of (1) millions of cardiac contractions per year, (2) substantial tissue deformations during each contraction, and (3) transvalvular pressure gradients in the closed phase of each contraction of approximately 120 mm Hg for the mitral and 80 mm Hg for the aortic valve. It is therefore not surprising that these delicate structures can sustain cumulative damage and calcification that lead to clinically important dysfunction. There is also a strong association between atherosclerotic risk factors and calcific valvular degeneration, due to shared pathogenic aspects of injury.

Calcific Aortic Stenosis

The most common of all valvular abnormalities, calcific aortic stenosis is usually the consequence of age-associated “wear and tear” of either anatomically normal valves or congenitally bicuspid valves (found in approximately 1% of the population). The prevalence of calcific aortic stenosis is estimated to be 5% for those age 65 and older and is increasing as the general population ages. Aortic stenosis of

TABLE 12.8 Etiology of Acquired Heart Valve Disease

Mitral Valve Disease	Aortic Valve Disease
Mitral Stenosis	Aortic Stenosis
Postinflammatory scarring (rheumatic heart disease)	Postinflammatory scarring (rheumatic heart disease) Senile calcific aortic stenosis Calcification of congenitally deformed valve
Mitral Regurgitation	Aortic Regurgitation
Abnormalities of leaflets and commissures: • Postinflammatory scarring • Infective endocarditis • Mitral valve prolapse • “Fen-phen”-induced valvular fibrosis Abnormalities of tensor apparatus: • Rupture of papillary muscle • Papillary muscle dysfunction (fibrosis) • Rupture of chordae tendineae Abnormalities of left ventricular cavity and/or annulus: • Left ventricular enlargement (myocarditis, dilated cardiomyopathy) • Calcification of mitral ring	Intrinsic valvular disease: • Postinflammatory scarring (rheumatic heart disease) • Infective endocarditis Aortic disease: • Degenerative aortic dilation • Syphilitic aortitis • Ankylosing spondylitis • Rheumatoid arthritis • Marfan syndrome

Fen-phen, Fenfluramine-phenentermine.

Data from Schoen FJ. Surgical pathology of removed natural and prosthetic valves. *Hum Pathol*. 1987;18(6):558-567.

previously normal valves (termed degenerative calcific aortic stenosis) usually comes to clinical attention in the seventh to ninth decades of life, whereas stenotic bicuspid valves tend to become clinically significant one to two decades earlier.

Aortic valve calcification is likely a consequence of recurrent chronic injury due to hyperlipidemia, hypertension, inflammation, and other factors similar to those implicated in atherosclerosis. The chronic progressive injury leads to valvular degeneration and incites

the deposition of hydroxyapatite (the same calcium salt found in bone). Although this model provides a good starting point for understanding calcific degeneration, it is increasingly clear that the valve injury of calcific aortic stenosis differs in some important respects from atherosclerosis. Most notably, the abnormal valves contain cells resembling osteoblasts that synthesize bone matrix proteins and promote the deposition of calcium salts. Moreover, interventions that improve atherosclerotic risk (e.g., statins) do not appear to significantly prevent valvular calcific degeneration. Bicuspid valves incur greater mechanical stress than normal tricuspid valves, which may explain their accelerated stenosis.

MORPHOLOGY

The gross morphologic hallmark of nonrheumatic, calcific aortic stenosis is heaped up calcified masses on the outflow surfaces of the cusps that ultimately prevent cuspal opening. The free edges of the cusps are usually not involved (Fig. 12.20A). Microscopically, the layered architecture of the valve is largely preserved. The calcific process begins in the fibrosa on the outflow surface of the valve, at the points of maximal cusp flexion (near the margins of attachment). Inflammation is variable, and metaplastic bone can be seen. In aortic stenosis, the functional valve area is decreased by large nodular calcific deposits that can eventually cause measurable outflow obstruction; this subjects the left ventricular myocardium to progressively increasing pressure overload. In contrast with rheumatic (and congenital) aortic stenosis (see Fig. 12.22E), commissural fusion is not usually seen.

Clinical Features

In calcific aortic stenosis (superimposed on a previously normal or bicuspid aortic valve), the obstruction to left ventricular outflow leads to gradual narrowing of the valve orifice (valve area approximately 0.5 to 1 cm² in severe aortic stenosis; normal is approximately 4 cm²) and an increasing pressure gradient across the calcified valve. Left ventricular pressures rise to 200 mm Hg or more in such instances, producing concentric left ventricular (pressure overload) hypertrophy. The hypertrophied myocardium tends to be ischemic (as a result of diminished microcirculatory perfusion, often complicated by coronary atherosclerosis), and angina pectoris may occur.

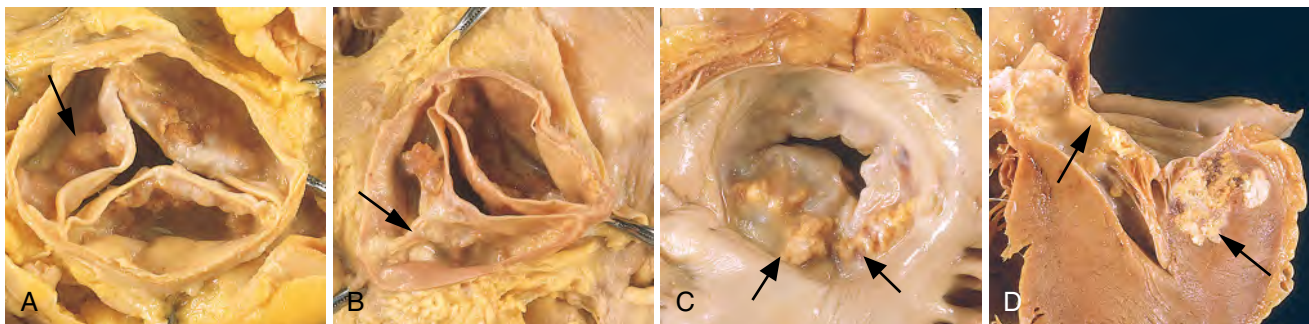


Fig. 12.20 Calcific valvular degeneration. (A) Calcific aortic stenosis of a previously healthy valve (viewed from aortic aspect). Nodular masses of calcium are heaped up within the sinuses of Valsalva (arrow). Note that the commissures are not fused, as occurs with post-rheumatic aortic valve stenosis (see Fig. 12.22E). (B) Calcific aortic stenosis of a congenitally bicuspid valve. One cusp has a partial fusion at its center, called a raphe (arrow). (C–D) Mitral annular calcification, with calcific nodules at the base (attachment margin) of the anterior mitral leaflet (arrows). (C) Left atrial view. (D) Cut section of myocardium showing the lateral wall with dense calcification that extends into the underlying myocardium (arrow).

Both systolic and diastolic myocardial function may be impaired; eventually, cardiac decompensation and CHF can ensue. The onset of symptoms (angina, CHF, or syncope) in aortic stenosis heralds cardiac decompensation and carries an extremely poor prognosis. If untreated, most patients with aortic stenosis will die within 5 years of developing angina, within 3 years of developing syncope, and within 2 years of CHF onset. Treatment requires surgical valve replacement, as medical therapy is ineffective in severe symptomatic aortic stenosis.

Calcific Stenosis of Congenitally Bicuspid Aortic Valve

Bicuspid aortic valve (BAV) is a developmental abnormality with prevalence in the population of approximately 1%; calcified BAV comprises approximately 50% of cases of aortic stenosis in adults. Some cases of BAV show familial clustering, often with associated aortic or left ventricular outflow tract malformations. Loss-of-function mutations in the *NOTCH1* receptor gene have been specifically associated with BAV in a few families.

In a congenitally bicuspid aortic valve, there are only two functional cusps, usually of unequal size, with the larger cusp having a midline raphe, resulting from incomplete commissural separation during development; less frequently, the cusps are of equal size and the raphe is absent. The raphe is frequently a major site of calcific deposits (see Fig. 12.20B).

Although BAV is usually asymptomatic early in life, the onset of calcific degeneration heralds a clinical course similar to that described earlier for calcific aortic stenosis. Other bicuspid valve complications include aortic regurgitation and infective endocarditis (IE.) Interestingly, structural abnormalities of the aortic wall also commonly accompany BAV, even when the valve is hemodynamically normal, and this may potentiate aortic dilation or aortic dissection.

Mitral Annular Calcification

As opposed to the predominantly cuspal involvement in aortic valve calcification, degenerative calcific deposits in the mitral valve typically develop in the fibrous annulus. Grossly, these appear as irregular, stony hard, occasionally ulcerated nodules (2 to 5 mm in thickness) at the base of the leaflets (see Fig. 12.20C,D). Mitral annular calcification usually does not affect valvular function. However, in exceptional cases it can lead to:

- *Regurgitation* by interfering with physiologic contraction of the valve ring
- *Stenosis* by impairing opening of the mitral leaflets
- *Arrhythmias* and occasionally sudden death by penetration of calcium deposits to a depth sufficient to impinge on the atrioventricular conduction system

Because calcific nodules may also provide a site for thrombus formation, patients with mitral annular calcification have a mildly increased risk of embolic stroke, and the calcific nodules can become a nidus for IE. Mitral annular calcification increases with age and is more common in females and individuals with mitral valve prolapse (see later).

Mitral Valve Prolapse (Myxomatous Degeneration of the Mitral Valve)

In mitral valve prolapse (MVP), one or both mitral valve leaflets are “floppy” and protrude into the left atrium during systole. MVP affects approximately 2% to 3% of adults in the United States and is more common in females; it is most often an incidental finding on physical examination but may lead to serious complications in a minority of affected individuals.

Pathogenesis

The etiologic basis for the changes that weaken the valve leaflets and associated structures is unknown in most cases. Uncommonly, MVP is associated with heritable disorders of connective tissue including Marfan syndrome, caused by fibrillin-1 (*FBN1*) mutations that alter cell-matrix interactions and dysregulate TGF- β signaling (Chapter 5). Interestingly, mice with mutated *FBN1* develop a form of MVP that is prevented by TGF- β inhibitors, indicating that excess TGF- β activity can cause the characteristic structural laxity and myxomatous changes.

MORPHOLOGY

The characteristic anatomic change in MVP is ballooning (hooding) of the mitral leaflets (Fig. 12.21A–C). The affected leaflets are often enlarged, redundant, thick, and rubbery. The associated tendinous cords may be elongated, thinned, or even ruptured, and the annulus may be dilated. The tricuspid, aortic, or pulmonary valves may also be affected. The key histologic change in the tissue is marked **myxomatous degeneration** of the spongiosa layer, reflected by increased deposition of a highly sulfated hydrophilic matrix (Fig. 12.21E); the collagenous fibrosa layer of the valve is also attenuated, affecting the structural integrity of the leaflet. Secondary changes reflect the stresses and tissue injury incident to the billowing leaflets: (1) fibrous thickening of the valve leaflets, particularly where they rub against each other; (2) linear fibrous thickening of the left ventricular endocardial surface where the abnormally long cords snap or rub against it; (3) thickening of the mural endocardium of the left ventricle or atrium as a consequence of friction-induced injury caused by the prolapsing, hypermobile leaflets; (4) thrombi on the atrial surfaces of the leaflets or the atrial walls (Fig. 12.21B); (5) focal calcifications at the base of the posterior mitral leaflet (Fig. 12.21C); and (6) left atrial dilation, often with associated atrial fibrillation (Fig. 12.21A). Notably, mitral valve myxomatous degeneration can also occur as a secondary consequence of regurgitation of other etiologies (e.g., ischemic dysfunction).

Clinical Features

Most individuals diagnosed with MVP are asymptomatic; in such cases, the condition is discovered incidentally by auscultation of mid-systolic clicks, caused by abrupt tension on the redundant valve leaflets and chordae tendineae as the valve attempts to close; there may or may not be an associated regurgitant murmur. The diagnosis is confirmed by echocardiography. A minority of patients have chest pain mimicking angina (although not exertional in nature), and a subset has dyspnea, presumably related to valvular insufficiency. Although the great majority of persons with MVP have no untoward effects, approximately 3% develop one of four serious complications: (1) infective endocarditis (IE); (2) mitral insufficiency, sometimes with chordal rupture; (3) stroke or other systemic infarct, resulting from embolism of leaflet thrombi; or (4) arrhythmias, both ventricular and atrial. Rarely, MVP is the only finding in sudden cardiac death (SCD), presumed secondary to ischemic injury with chronic increased tension on the papillary muscles. Valve repair or replacement surgery can be done for symptomatic patients or those with increased risk for significant complications.

Rheumatic Fever and Rheumatic Heart Disease

Rheumatic fever (RF) is an acute, immunologically mediated, multisystem inflammatory disease classically occurring a few weeks after group A streptococcal pharyngitis; occasionally, RF can follow streptococcal infections at other sites, such as the skin. Acute rheumatic carditis is a common manifestation of active RF and may progress over time to chronic rheumatic heart disease (RHD).

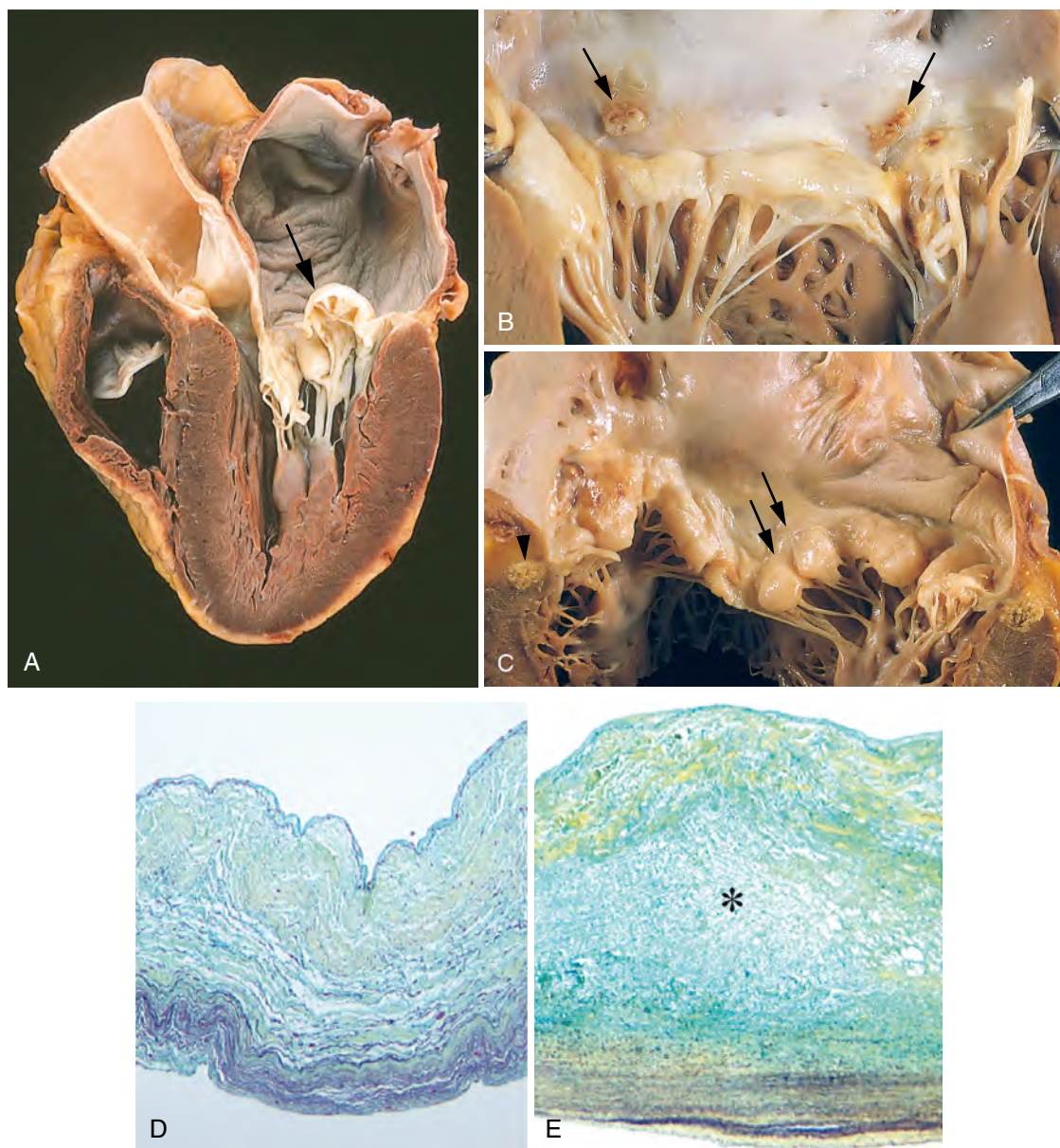


Fig. 12.21 Myxomatous degeneration of the mitral valve. (A) Long axis view (left ventricle is on the *right*) demonstrating hooding with prolapse of the posterior mitral leaflet into the left atrium (*arrow*). (B) Opened valve, showing pronounced hooding of the posterior mitral leaflet with thrombotic plaques at sites of leaflet-left atrium contact (*arrows*). (C) Opened valve with pronounced hooding (*double arrows*) in a patient who died suddenly. Note also mitral annular calcification on the left side (*arrowhead*). Healthy heart valve (D) and myxomatous mitral valve (E). In myxomatous valves, collagen in the fibrosa layer is loose and disorganized, proteoglycan deposition (*asterisk*) in the central spongiosa layer is markedly expanded, and elastin in the atrialis layer is disorganized. (A, Courtesy William D. Edwards, MD, Mayo Clinic, Rochester, MN. D–E, Movat pentachrome stain, in which collagen is yellow, elastin is black, and proteoglycans are blue; from Rabkin E, Aikawa M, Stone JR, Fukumoto Y, Libby P, Schoen FJ. Activated interstitial myofibroblasts express catabolic enzymes and mediate matrix remodeling in myxomatous heart valves. *Circulation*. 2001;104:2525–2532.)

RHD is characterized principally by deforming fibrotic valvular disease, particularly involving the mitral valve; indeed, RHD is virtually the only cause of mitral stenosis. The incidence and mortality rate of RF and RHD have declined remarkably in many parts of the world over the past century, because of improved sanitation and rapid diagnosis and treatment of streptococcal pharyngitis. Nevertheless, in low-resource countries, and in many crowded, economically depressed urban areas, RHD remains an important public health problem.

Pathogenesis

Acute rheumatic fever results from host immune responses to group A streptococcal antigens that cross-react with host proteins. The characteristic 2- to 3-week delay in symptom onset after infection is explained by the time needed to generate an immune response; In particular, antibodies and CD4⁺ T cells directed against streptococcal M proteins (virulence factor expressed on bacterial surface) recognize cardiac antigens (pericardial, myocardial, or valvular). Antibody binding can

activate complement, as well as recruit Fc-receptor-bearing cells (neutrophils and macrophages); cytokine production by the stimulated T cells leads to macrophage activation (e.g., within *Aschoff bodies*). Damage to heart tissue may thus be caused by a combination of antibody- and T cell-mediated reactions (Chapter 6). In keeping with an immunologic basis of RHD, streptococci are completely absent from the lesions. Because only a small minority of patients who have had streptococcal infection develop rheumatic fever (estimated at 0.3% to 3%), an intrinsic genetic susceptibility is likely to underlie the development of the cross-reactive immune responses. The chronic fibrotic lesions are the predictable consequence of healing and scarring associated with the resolution of the acute inflammation.

MORPHOLOGY

Key pathologic features of acute RF and chronic RHD are shown in Fig. 12.22. During acute RF, focal inflammatory lesions are found in various tissues. Distinctive lesions in the heart—called **Aschoff bodies**—are composed of foci of T lymphocytes, occasional plasma cells, and plump activated macrophages called **Anitschkow cells**. These macrophages have abundant cytoplasm and

central round-to-ovoid nuclei (occasionally binucleate) in which the chromatin condenses into a central, slender, wavy ribbon (“**caterpillar cells**”).

During acute RF, diffuse inflammation and Aschoff bodies may be found in any of the three layers of the heart, resulting in pericarditis, myocarditis, or endocarditis (**pancarditis**).

Inflammation of the endocardium and the left-sided valves typically results in fibrinoid necrosis within the cusps or tendinous cords. Overlying these necrotic foci and along the lines of closure are small (1 to 2 mm) vegetations, called **verrucae**. Thus, RHD is one of the forms of vegetative valve disease, each of which exhibits its own characteristic morphologic features (Fig. 12.23). Subendocardial lesions, perhaps exacerbated by regurgitant jets, can induce irregular thickenings called **MacCallum plaques**, usually in the left atrium.

The cardinal anatomic changes of the mitral valve in chronic RHD are **leaflet thickening, commissural fusion and shortening, and thickening and fusion of the tendinous cords** (see Fig. 12.22D). The mitral valve is virtually always involved in chronic RHD; it is affected in isolation in roughly two-thirds of cases, and along with the aortic valve in another 25%. Tricuspid valve involvement is infrequent, and the pulmonary valve is only rarely affected.

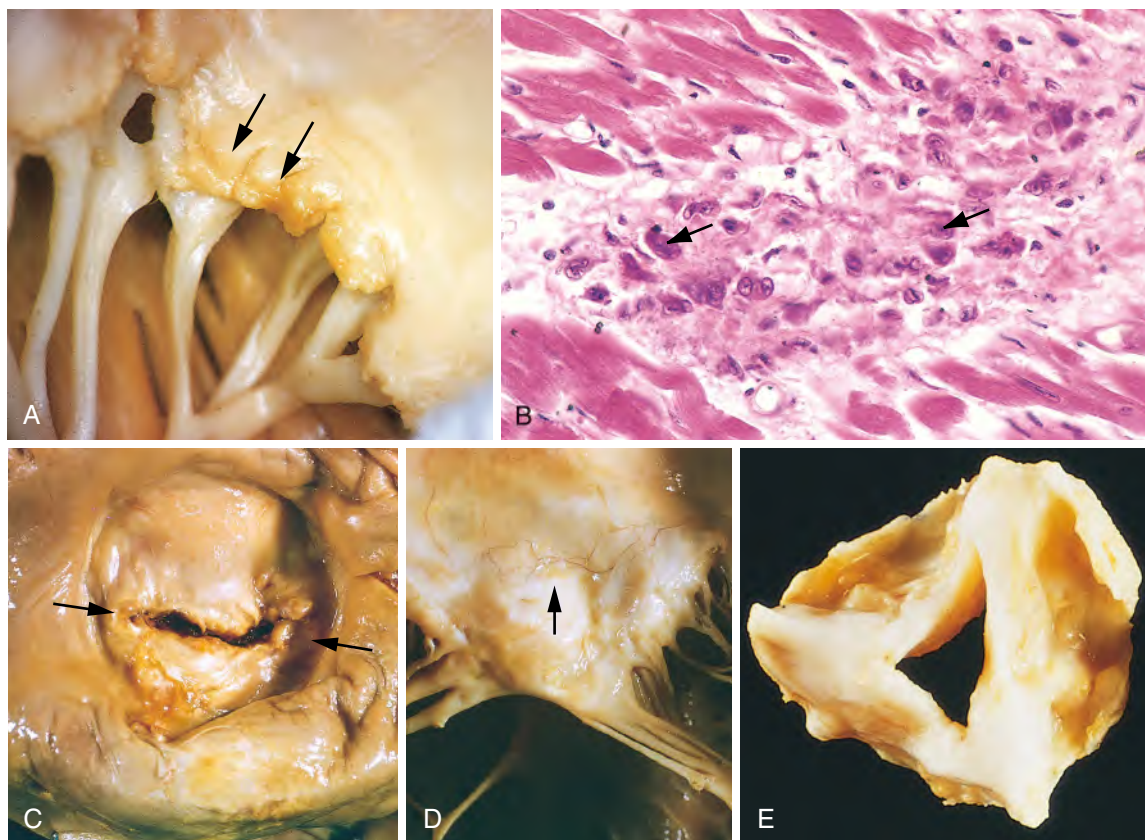


Fig. 12.22 Acute and chronic rheumatic heart disease. (A) Acute rheumatic mitral valvulitis superimposed on chronic rheumatic heart disease. Small vegetations (verrucae) are visible along the line of closure of the mitral valve leaflet (arrows). Previous episodes of rheumatic valvulitis have caused fibrous thickening and fusion of the chordae tendineae. (B) Microscopic appearance of an Aschoff body in a patient with acute rheumatic carditis. The myocardium contains a circumscribed nodule of mixed mononuclear inflammatory cells with associated necrosis; within the inflammation, large activated macrophages show prominent nucleoli, as well as chromatin condensed into long, wavy ribbons (caterpillar cells; arrows). (C–D) Mitral stenosis with diffuse fibrous thickening and distortion of the valve leaflets and commissural fusion (arrows, C), and thickening of the chordae tendineae (D). Note the neovascularization of the anterior mitral leaflet (arrow, D). (E) Surgically resected specimen of rheumatic aortic stenosis, demonstrating thickening and distortion of the cusps with commissural fusion. (E, From Schoen FJ, St. John-Sutton M. Contemporary issues in the pathology of valvular heart disease. *Hum Pathol*. 1967;18:568.)

In rheumatic mitral stenosis, calcification and fibrous bridging across the valvular commissures create “fish mouth” stenoses (see Fig. 12.22C). With tight mitral stenosis, the left atrium progressively dilates and may harbor mural thrombi that can embolize. Long-standing congestive changes in the lungs may induce pulmonary vascular and parenchymal changes; over time, these can lead to right ventricular hypertrophy. The left ventricle is largely unaffected by isolated pure mitral stenosis. Microscopically, valves show organization of the acute inflammation, with postinflammatory neovascularization and transmural fibrosis that obliterate the leaflet architecture. Aschoff bodies are rarely seen in surgical specimens or autopsy tissue from patients with chronic RHD because of the long intervals between the initial insult and the development of the chronic deformity.

Clinical Features

RF is characterized by a constellation of major manifestations:

- *Migratory polyarthritides* of the large joints
- *Pancarditis* (myocarditis, pericarditis, or endocarditis)
- *Subcutaneous nodules* (typically on extensor surfaces of extremities)
- *Erythema marginatum*, an irregular circinate skin rash
- *Sydenham chorea*, a neurologic disorder with involuntary rapid movements

The diagnosis is established in accordance with the revised Jones criteria: evidence of a preceding group A streptococcal infection, and the presence of two major manifestations, or one major and two minor manifestations (nonspecific signs and symptoms that include fever, arthralgia, or elevated blood levels of acute-phase reactants); notably, these criteria are evolving and are applied differently in low- and high-risk settings.

Acute RF typically appears 10 days to 6 weeks after a group A streptococcal infection in about 0.3% to 3% of patients. It occurs most often in children between 5 and 15 years of age, but first attacks can occur in middle to later life. Although pharyngeal cultures for streptococci are negative by the time the illness begins, antibodies to one or more streptococcal enzymes, such as streptolysin O and DNase B, can be detected in the sera of most patients with RF. The predominant clinical manifestations are carditis and arthritis, the latter more

common in adults than in children. Arthritis typically begins with migratory polyarthritides (accompanied by fever) in which one large joint after another becomes painful and swollen for a period of days and then subsides spontaneously, leaving no residual disability. Clinical features related to acute carditis include pericardial friction rubs, tachycardia, and arrhythmias. Myocarditis can cause cardiac dilation that may culminate in functional mitral valve insufficiency or even heart failure. Approximately 1% to 2% of affected individuals die of fulminant RF involvement of the heart.

After an initial attack, there is increased vulnerability to reactivation of the disease with subsequent pharyngeal infections, and the same manifestations are likely to appear with each recurrent attack. Damage to the valves is cumulative. Turbulence induced by ongoing valvular deformities leads to additional fibrosis. This is a prime example of a common theme in valvular heart disease in which the consequences of valve pathology may contribute to progression of that valve pathology in a positive feedback loop.

Clinical manifestations of RHD appear years or even decades after the initial episode of RF and depend on which valves are involved. In addition to cardiac murmurs, cardiac hypertrophy and dilation, and heart failure, individuals with chronic RHD may have arrhythmias (particularly atrial fibrillation in the setting of mitral stenosis), thromboembolic complications, and IE (see later). The long-term prognosis is highly variable. Surgical repair or replacement of diseased valves has greatly improved the outlook for persons with RHD.

Infective Endocarditis

Infective endocarditis (IE) is a microbial infection of the heart valves or the mural endocardium that leads to the formation of vegetations composed of thrombotic debris and organisms, often associated with destruction of the underlying cardiac tissues. The aorta, aneurysms, other blood vessels, and prosthetic devices can also become infected. Although fungi and other classes of microorganisms can be responsible, most infections are bacterial (bacterial endocarditis). Prompt diagnosis, identification of the offending agent, and effective treatment of IE is important in limiting morbidity and mortality.

Traditionally, IE has been classified on clinical grounds into acute and subacute forms based on severity and tempo (reflecting microbial virulence), and whether there is underlying valvular pathology. Thus,

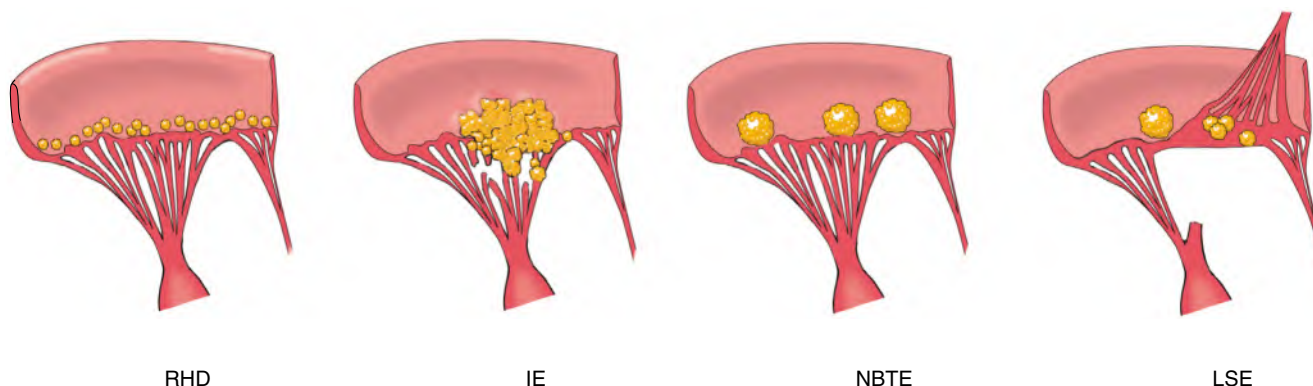


Fig. 12.23 Comparison of the four major forms of vegetative endocarditis. The rheumatic fever phase of rheumatic heart disease (*RHD*) is marked by small, warty vegetations along the lines of closure of the valve leaflets. Infective endocarditis (*IE*) is characterized by large, irregular masses on the valve cusps that can extend onto the chordae (see Fig. 12.24A). Nonbacterial thrombotic endocarditis (*NBTE*) typically exhibits small, bland vegetations, usually attached at the line of closure. One or many may be present (see Fig. 12.25). Libman-Sacks endocarditis (*LSE*) has small- or medium-sized vegetations on either or both sides of the valve leaflets.

acute IE is typically caused by infection of a previously normal heart valve by a highly virulent organism (e.g., *Staphylococcus aureus*) that rapidly produces destructive lesions. These infections may be difficult to cure with antibiotics alone and often require surgery; despite appropriate treatment, there can be substantial morbidity and even mortality. In contrast, subacute IE is characterized by organisms with lower virulence (e.g., viridans streptococci) that cause insidious infections of deformed valves with overall less destruction. In such cases the disease may pursue a protracted course of weeks to months, and cures can often be achieved with antibiotics alone. Of note, a clear delineation between acute and subacute endocarditis does not always exist, and many cases fall somewhere along the spectrum between the two forms.

Pathogenesis

A variety of cardiac abnormalities increase the risk of developing IE. RHD with valvular scarring has historically been the major antecedent disorder; as RHD becomes less common, it has been supplanted by mitral valve prolapse, degenerative calcific valvular stenosis, bicuspid aortic valve (whether calcified or not), artificial (prosthetic) valves, and congenital defects.

Endocarditis of native but previously damaged or otherwise abnormal valves is caused most commonly (50% to 60% of cases) by *Streptococcus viridans*, a normal component of the oral cavity flora. In contrast, more virulent *S. aureus* organisms commonly found on the skin can infect either healthy or deformed valves and are responsible for most cases overall; notably, *S. aureus* is the major offender in IE among intravenous drug users. Other bacterial causes include enterococci and the HACEK group (*Haemophilus*, *Actinobacillus*, *Cardiobacterium*, *Eikenella*, and *Kingella*), all commensals in the oral cavity. More rarely, gram-negative bacilli and fungi can be involved. Prosthetic valve endocarditis occurring in the 1 to 2 months after surgical implantation is typically caused by skin flora (*S. aureus* and *S. epidermidis*); prosthetic valve infections 1 year or more after surgery tend to be caused by streptococci and *S. aureus* (see later discussion of prosthetic valves). In about 10% of all cases of endocarditis, no organism can be isolated from the blood (culture-negative endocarditis); reasons include prior antibiotic therapy, difficulties in isolating the offending agent, or because deeply embedded organisms within the enlarging vegetation are not readily released into the blood.

Foremost among the factors predisposing to endocarditis are those that cause microorganism seeding into the bloodstream (bacteremia or fungemia). The source may be an obvious infection elsewhere, a dental or surgical procedure, a contaminated needle shared by intravenous drug users, or seemingly trivial breaks in the epithelial barriers of the gut, oral cavity, or skin. In patients with valve abnormalities, or with known bacteremia, IE risk can be lowered by antibiotic prophylaxis.

MORPHOLOGY

Vegetations on heart valves are the classic hallmark of IE; these are friable, bulky, potentially destructive lesions containing fibrin, inflammatory cells, and bacteria or other organisms (Figs. 12.23 and 12.24). The aortic and mitral valves are the most common sites of infection, although the valves of the right heart may also be involved, particularly in intravenous drug users. Vegetations can be single or multiple and may involve more than one valve; they can occasionally erode into the underlying myocardium and produce an abscess (**ring abscess**; see Fig. 12.24B). Vegetations are prone to embolization; because the embolic fragments often contain virulent organisms, abscesses frequently develop where they lodge, leading to sequelae such as septic infarcts or mycotic aneurysms.

The vegetations of subacute endocarditis are associated with less valvular destruction than those of acute endocarditis, although the distinction can be subtle. Microscopically, the vegetations of subacute IE typically exhibit granulation tissue at their base indicative of healing. With time, fibrosis, calcification, and a chronic inflammatory infiltrate can develop.

Clinical Features

Acute endocarditis has a stormy beginning with rapid onset of fever, chills, weakness, and lassitude. Although fever is the most consistent sign of IE, it can be slight or absent, particularly in older adults, and the only manifestations may be nonspecific fatigue, weight loss, and a flulike illness. Murmurs are present in the majority of patients with left-sided IE, either from a new valvular defect or from a preexisting abnormality. The modified Duke criteria (Table 12.9) facilitate diagnosis of individuals with suspected IE by taking into account predisposing factors, physical findings, blood culture results, imaging findings, and other laboratory information.

Complications of IE generally begin within the first few weeks of onset and can include glomerular antigen-antibody complex deposition causing glomerulonephritis (Chapter 20). Sepsis, arrhythmias

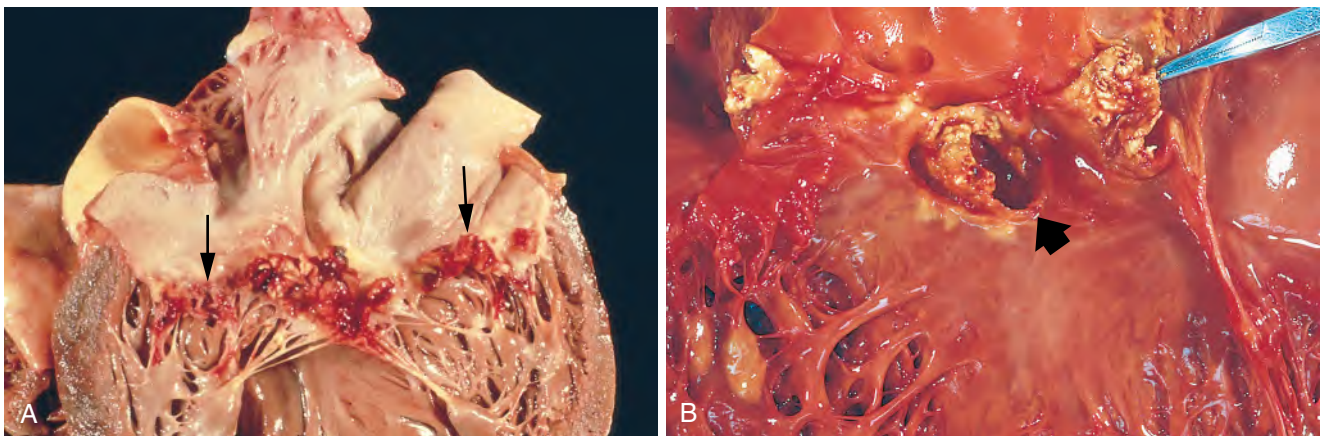


Fig. 12.24 Infective (bacterial) endocarditis. (A) Endocarditis of mitral valve (subacute, caused by *Streptococcus viridans*). The large, friable vegetations are denoted by arrows. (B) Acute endocarditis of congenitally bicuspid aortic valve (caused by *Staphylococcus aureus*) with extensive cuspal destruction and ring abscess (arrow).

TABLE 12.9 Diagnostic Criteria for Infective Endocarditis

Pathologic Criteria
Microorganisms, demonstrated by culture or histologic examination, in a vegetation, embolus from a vegetation, explanted prosthetic valve or sewing ring, or cardiac implantable electronic device (CIED)
Clinical Criteria (2 Major, or 1 Major and 3 Minor, or 5 Minor Criteria)
Major
- Blood culture, nucleic acid–based technique, or immunofluorescence assay positive for a characteristic organism or persistently positive for an unusual organism - Echocardiographic, computed tomography, or positron-emission tomography imaging of a valve-related or implant-related oscillating mass or abscess, or partial separation of artificial valve - New valvular regurgitation on imaging - Direct inspection during heart surgery
Minor
- Predisposing heart lesion, prosthetic valve, CIED, or intravenous drug use - Fever - Vascular lesions, including major arterial emboli, septic pulmonary infarcts, mycotic aneurysm, intracranial hemorrhage, conjunctival hemorrhages, and Janeway lesions^a - Immunological phenomena, including glomerulonephritis, Osler nodes,^b Roth spots,^c and rheumatoid factor - Microbiologic evidence, including a single culture positive for an unusual organism

^aJaneway lesions are small erythematous or hemorrhagic, macular, nontender lesions on the palms and soles and are the consequence of septic embolic events.

^bOsler nodes are small, tender subcutaneous nodules that develop in the pulp of the digits or occasionally more proximally in the fingers and persist for hours to several days.

^cRoth spots are oval retinal hemorrhages with pale centers.

Modified from Fowler VG Jr, Durack DT, Seltman-Suty C, et al. The 2023 Duke-International Society for Cardiovascular Infectious Diseases criteria for infective endocarditis: updating the Modified Duke Criteria. *Clin Infect Dis*. 2023;77(4):518-526.

(suggesting invasion into underlying myocardium and conduction system), and systemic embolization bore particularly ill for the patient. Left untreated, IE is generally fatal. However, with appropriate long-term (6 weeks or more) antibiotic therapy and/or valve replacement, mortality is reduced. For infections involving low-virulence organisms (e.g., *S. viridans*), the cure rate is 98%, and for enterococci and *S. aureus* infections, cure rates range from 60% to 90%; however, with infections due to gram-negative bacilli or fungi, one-half of the patients ultimately succumb. The cure rate for endocarditis arising on prosthetic valves is uniformly worse, and valve re-replacement is commonly required.

Earlier diagnosis and effective treatment have nearly eliminated some previously common clinical manifestations of long-standing IE—for example, small thromboemboli (manifest as splinter or subungual hemorrhages), erythematous or hemorrhagic nontender lesions on the palms or soles (Janeway lesions), subcutaneous nodules in the pulp of the digits (Osler nodes), and retinal hemorrhages in the eyes (Roth spots).

Noninfected Vegetations

Noninfected (sterile) vegetations occur in nonbacterial thrombotic endocarditis (NBTE) and the endocarditis of systemic lupus erythematosus (SLE).

Nonbacterial Thrombotic Endocarditis

Nonbacterial thrombotic endocarditis (NBTE) is characterized by the deposition of small (1 to 5 mm) sterile thrombi on the leaflets of the cardiac valves (Figs. 12.23 and 12.25). Histologically, these are bland thrombi, loosely attached to the underlying valve; the vegetations are nondestructive and do not elicit any inflammatory reaction. Although the local effect of the vegetations is usually trivial, they can be a source of systemic emboli that produce significant infarcts in the brain, heart, or elsewhere.

NBTE is often encountered in debilitated patients, such as those with cancer or sepsis—hence the previous term *marantic endocarditis* (root word *marasmus*, relating to malnutrition). Valvular damage is not a prerequisite for NBTE; indeed, the condition usually occurs on previously normal valves. Rather, hypercoagulable states are the usual precursor to NBTE; such conditions include chronic disseminated intravascular coagulation, hyperestrogenic states, and those associated with underlying malignancy, particularly mucinous adenocarcinomas. The latter association is likely related to the procoagulant effects of tumor-derived mucin or tissue factor that can also cause migratory thrombophlebitis (*Trousseau syndrome*, Chapter 4). Endocardial trauma, as from an indwelling catheter, is another predisposing condition.

Endocarditis of Systemic Lupus Erythematosus

Small (1 to 4 mm), sterile vegetations in the setting of systemic lupus erythematosus (Chapter 6) are termed *Libman-Sacks endocarditis*. The lesions develop as a result of immune complex deposition, with activation of complement and recruitment of Fc-receptor-bearing cells; histologically, there is an intense valvulitis and fibrinoid necrosis of the valve substance. The vegetations can occur anywhere on the valve surface, on the chordae, or even on the atrial or ventricular endocardium (see Fig. 12.23). Persistent injury can eventually result in valvular scarring and leaflet fusion, analogous to that seen in RHD. Similar lesions can occur in the antiphospholipid antibody syndrome (Chapter 4).

Carcinoid Heart Disease

Carcinoid syndrome is a systemic disorder marked by flushing, diarrhea, dermatitis, and bronchoconstriction caused by bioactive compounds such as serotonin released by carcinoid tumors (Chapter 17). Carcinoid heart disease refers to the cardiac manifestations caused by the tumor-derived mediators bioactive compounds and occurs in roughly one-half of the patients in whom the systemic syndrome develops. Cardiac lesions do not typically occur until there are multiple liver metastases, because the liver normally catabolizes the circulating mediators before they can affect the heart. Classically, the endocardium and valves of the right heart are primarily affected because they are the first cardiac tissues bathed by the mediators released by gastrointestinal carcinoid tumors. The left side of the heart is afforded some measure of protection because the pulmonary vascular bed degrades the mediators. However, left heart carcinoid lesions can occur in the setting of atrial or septal defects and right-to-left flow, or they can be induced by primary pulmonary carcinoid tumors.

Pathogenesis

The mediators elaborated by carcinoid tumors include serotonin (5-hydroxytryptamine), kallikrein, bradykinin, histamine, prostaglandins, and tachykinins. Although it is not clear which of these is causal, plasma levels of serotonin and urinary excretion of the serotonin metabolite 5-hydroxyindoleacetic acid correlate with the severity of the cardiac lesions. The valvular plaques in carcinoid syndrome are also similar to lesions that occurred in patients taking fenfluramine (an appetite suppressant, no longer marketed) or ergot alkaloids (used

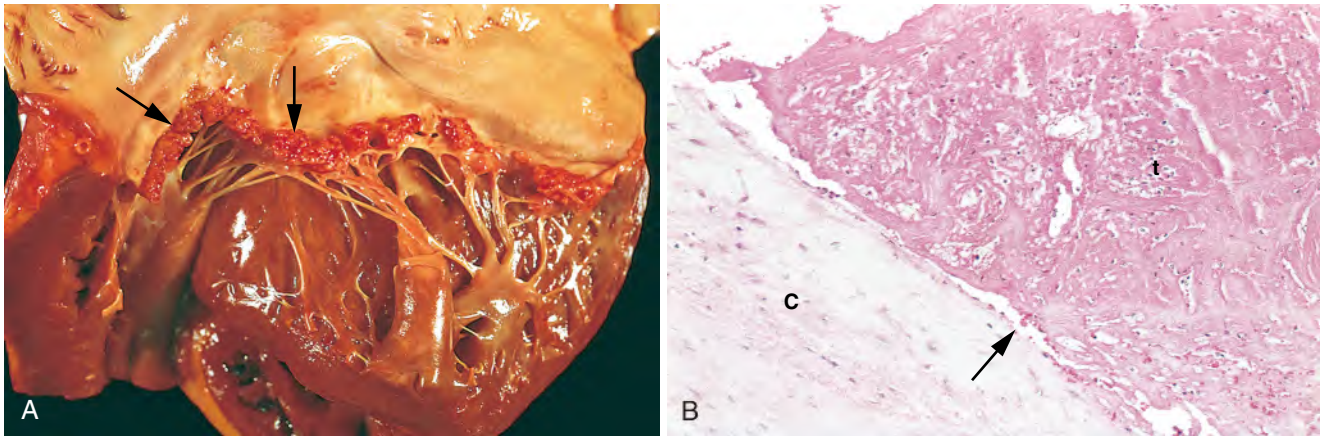


Fig. 12.25 Nonbacterial thrombotic endocarditis (NBTE). (A) Nearly complete row of thrombotic vegetations along the line of closure of the mitral valve leaflets (*arrows*). (B) Photomicrograph of NBTE, showing bland thrombus, with virtually no inflammation in the valve cusp (*c*) or the thrombotic deposit (*t*). The thrombus is only loosely attached to the cusp (*arrow*).

previously for migraine headaches); interestingly, these agents affect systemic serotonin metabolism. Similarly, left-sided plaques have been reported after methysergide or ergotamine therapy for migraines; notably, these drugs are metabolized to serotonin as they pass through the pulmonary vasculature.

MORPHOLOGY

The cardiovascular lesions associated with the **carcinoid syndrome** are distinctive, glistening white intimal plaque-like thickenings of the endocardial surfaces of the cardiac chambers and valve leaflets (Fig. 12.26). The lesions are composed of smooth muscle cells and sparse collagen fibers embedded in an acid mucopolysaccharide-rich matrix material. Underlying structures are intact. With right-sided involvement, typical findings are tricuspid insufficiency and pulmonary stenosis.

Complications of Prosthetic Valves

Although prosthetic heart valves are less-than-perfect substitutes for the native tissues, their introduction has radically altered the prognosis for patients with valve disease. Two types of prosthetic valves are currently used, each with its own advantages and disadvantages:

- **Mechanical valves.** These consist of rigid nonphysiologic materials, currently almost exclusively configured as hinged semicircular flaps (bileaflet tilting disk valves).
- **Tissue valves (bioprostheses).** Cryopreserved human cadaveric valves, or devascularized animal valve leaflets are mounted on a prosthetic frame and have the advantage of flexibility and function similar to native valves. Increasingly—especially in patients who are not good candidates for open surgery—bioprosthetic valves are deployed by catheter-based approaches (for example, transcatheter aortic valve replacement) within the original diseased valve annulus.

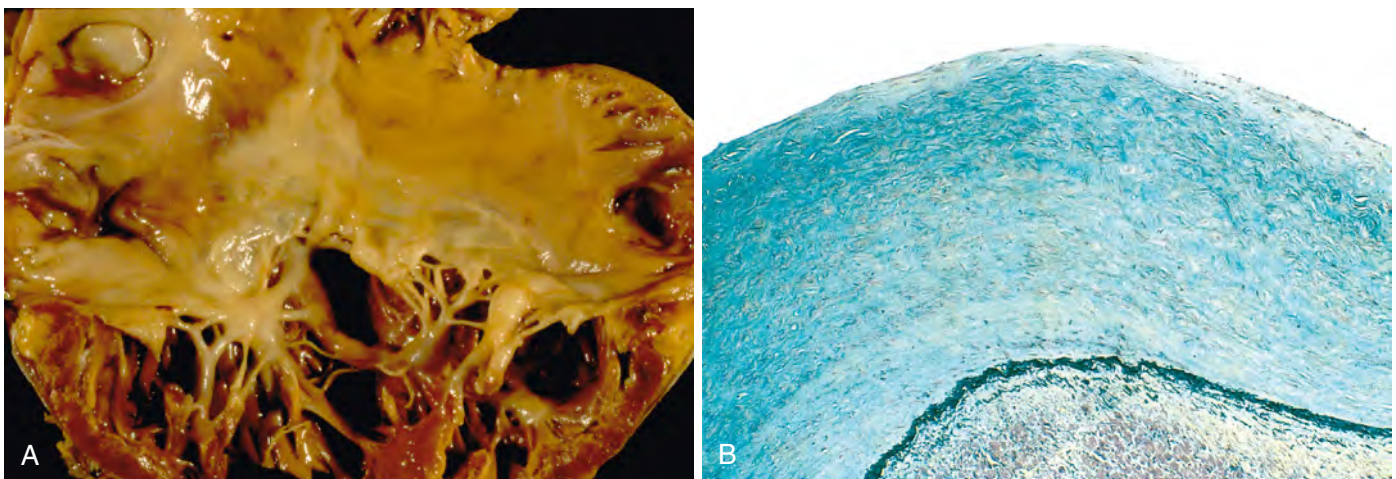


Fig. 12.26 Carcinoid heart disease. (A) Characteristic endocardial fibrotic lesion involving the right ventricle and tricuspid valve. (B) Microscopic appearance of carcinoid heart disease with endocardial thickening. Movat stain shows myocardial elastic tissue (*black*) underlying the acid mucopolysaccharide-rich lesion (*blue-green*). The underlying myocardium is unaffected.

Approximately 60% of substitute valve recipients develop a serious prosthesis-related problem within 10 years after the surgery. The complications that occur depend on which type of valve has been implanted (Fig. 12.27).

- **Thromboembolism** is the major consideration with mechanical valves (see Fig. 12.27A); this may take the form of either thrombotic occlusion of the prosthesis or emboli released from thrombi formed on the valve. Because blood flow in all mechanical devices is non-laminar, foci of turbulence and stasis are produced by prostheses that predispose to thrombus formation. The risk of such complications necessitates long-term anticoagulation in all individuals with mechanical valves, with the attendant risk of hemorrhagic stroke or other forms of serious bleeding.
- **Structural deterioration** rarely causes failure of any of the mechanical valves in current use. However, virtually all bioprostheses eventually become incompetent due to calcification and/or tearing (see Fig. 12.27B).
- **Infective endocarditis** is a potentially serious complication of any valve replacement. The vegetations of prosthetic valve endocarditis are usually located at the prosthesis-tissue interface, and they often cause the formation of a ring abscess, which can eventually lead to a paravalvular regurgitant blood leak. In addition, vegetations may directly involve the tissue of bioprosthetic valvular cusps.
- **Other complications** include paravalvular leak due to inadequate healing, obstruction due to overgrowth of fibrous tissue during healing, valve-orifice disproportion—where the effective valve area is too small for the needs of the patient, leading to a relative stenosis—or intravascular hemolysis due to high shear forces.

KEY CONCEPTS

Valvular Heart Disease

- Valve pathology can lead to occlusion (stenosis) and/or regurgitation (insufficiency); acquired aortic and mitral valve stenoses account for approximately two-thirds of all valve disease; much of the remainder is mitral regurgitation.

- Valve calcification is a degenerative process that typically results in stenosis.
- Abnormal matrix synthesis and turnover result in myxomatous degeneration and insufficiency.
- Inflammatory valve diseases lead to postinflammatory scarring. Rheumatic heart disease (RHD) results from antistreptococcal antibodies and T cells that cross-react with cardiac tissues; it most commonly affects the mitral valve and is responsible for 99% of acquired mitral stenosis.
- Infective endocarditis (IE) can be aggressive and rapidly destroy normal valves (acute IE) or can be indolent and minimally destructive of previously abnormal valves (subacute IE). Systemic embolization can produce septic infarcts.
- Nonbacterial thrombotic endocarditis (NBTE) occurs on previously normal valves due to hypercoagulable states or immune complex-related injury embolization is an important complication.
- Mechanical prosthetic valves have thrombotic or hemorrhagic complications related to the nonlaminar flow of blood and the need for chronic anticoagulation. Bioprosthetic valves are nonviable and are therefore susceptible to long-term calcification and/or degeneration with tearing. Both types of valves have an increased risk of developing endocarditis relative to native valves.

CARDIOMYOPATHIES

Cardiac diseases due to intrinsic myocardial dysfunction are termed **cardiomyopathies** (literally, “heart muscle diseases”); these can be **primary cardiomyopathies** (that is, principally confined to the myocardium) or **secondary cardiomyopathies**, presenting as the cardiac manifestation of a systemic disorder (Table 12.10). This definition excludes myocardial dysfunction secondary to CAD, hypertension, valvular disease, and congenital heart diseases. Cardiomyopathies are a diverse group that includes inflammatory disorders (e.g., myocarditis), immunologic diseases (e.g., sarcoidosis), systemic metabolic disorders (e.g., hemochromatosis), muscular dystrophies, and other genetic disorders that selectively affect myocardial fibers. In many cases, the cardiomyopathy is of unknown etiology and is termed *idiopathic*; however, an

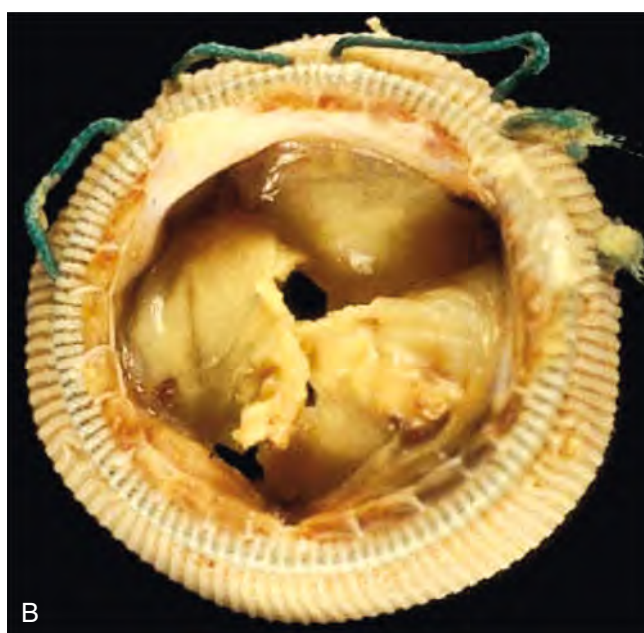


Fig. 12.27 Complications of artificial heart valves. (A) Thrombosis of a mechanical prosthetic valve. (B) Calcification with secondary tearing of a porcine bioprosthetic heart valve, viewed from the inflow aspect.

TABLE 12.10 Cardiomyopathies: Functional Patterns and Causes

Functional Pattern	Left Ventricular Ejection Fraction ^a	Mechanisms of Heart Failure	Causes	Secondary Myocardial Dysfunction (Mimicking Cardiomyopathy)
Dilated	<40%	Impairment of contractility (systolic dysfunction)	Genetic; alcohol; peripartum; myocarditis; hemochromatosis; chronic anemia; anthracycline medications; sarcoidosis; idiopathic	Ischemic heart disease; valvular heart disease; hypertensive heart disease; congenital heart disease
Hypertrophic	50%–80%	Impairment of compliance (diastolic dysfunction)	Genetic; Friedreich ataxia; storage diseases; infants of diabetic mothers	Hypertensive heart disease; aortic stenosis
Restrictive	35%–55%	Impairment of compliance (diastolic dysfunction)	Amyloidosis; radiation-induced fibrosis; idiopathic	Pericardial constriction

^aRange of normal values is approximately 50% to 65%.

increasing number of previously “idiopathic” cardiomyopathies are now known to have a genetic basis.

Cardiomyopathies can be classified according to a variety of criteria, including the underlying genetic basis of dysfunction; we have already discussed several of the arrhythmia-inducing channelopathies. Here, we will organize the discussion of the various disorders based on three distinct pathologic patterns (Fig. 12.28 and Table 12.10):

- *Dilated cardiomyopathy (DCM)* (including arrhythmogenic cardiomyopathy)
- *Hypertrophic cardiomyopathy*
- *Restrictive cardiomyopathy*

Among the three major patterns, DCM is most common (90% of cases), and restrictive cardiomyopathy is the least frequent. Within each pattern, there is a spectrum of clinical severity, and in some cases clinical features overlap among the groups. Regardless of pattern, most cases can be linked to a specific cause (Tables 12.10 and 12.11).

Dilated Cardiomyopathy

Dilated cardiomyopathy (DCM) is characterized morphologically and functionally by progressive cardiac dilation and contractile (systolic) dysfunction, usually with concomitant hypertrophy. Many cases are familial, but the DCM phenotype can result from diverse causes, both primary and secondary.

Pathogenesis

Several different pathways can lead to DCM (Fig. 12.29). Identifying which is causal in any given case may be challenging because diagnosis is only made after the patient has developed end-stage heart failure marked by the nonspecific findings of dilation and poor contractility. If ischemic, valvular, hypertensive, or congenital causes can be excluded, a diagnosis of primary dilated cardiomyopathy can be proposed. Although an exhaustive evaluation may not pinpoint a specific etiology (idiopathic DCM), familial (genetic) forms of DCM are increasingly recognized, and the same pathology can also result from a host of environmental exposures:

- *Genetic factors.* DCM is probably familial in up to 50% of cases, caused by mutations in a diverse group of more than 20 genes encoding proteins that are components of the cytoskeleton, sarcolemma, or nuclear envelope (e.g., lamin A/C) (Fig. 12.29). The diverse functions of these proteins suggest there are many types of intracellular aberrations that can contribute to myocyte dysfunction leading to DCM. In particular, truncation mutations in

TTN, a gene that encodes titin (so-called because it is the largest protein expressed in humans), account for approximately 10% to 20% of all cases of DCM. In the genetic forms of DCM, autosomal dominant inheritance is predominant, but there is often variable penetrance even within families, suggesting that the progression to DCM is multifactorial.

- X-linked, autosomal recessive, and mitochondrial inheritance of DCM are less common. In some families, mitochondrial gene deletions are present that affect oxidative phosphorylation; in others, there are mutations in genes encoding enzymes involved in fatty acid β -oxidation. Mitochondrial defects typically manifest in the pediatric population, whereas X-linked DCM typically presents after puberty and into early adulthood. X-linked cardiomyopathy can also be associated with mutations affecting dystrophin, a protein that couples the cytoskeleton to the extracellular matrix that is mutated in the most common skeletal myopathies, Duchenne and Becker muscular dystrophies (Chapter 27); some dystrophin gene mutations have DCM as the primary clinical feature. Probably due to the common developmental origin of contractile myocytes and conduction elements, congenital abnormalities of conduction may also be associated with DCM.
- *Myocarditis.* Sequential endomyocardial biopsies have documented progression from myocarditis to DCM. In other studies, the detection of the genetic fingerprints of Coxsackie B and other viruses within the myocardium of patients with DCM suggests that viral myocarditis may be causal (see later discussion).
- *Alcohol and other toxins.* Alcohol use disorder is strongly associated with the development of DCM, raising the possibility that ethanol toxicity (Chapter 9) or a secondary nutritional disturbance can underlie myocardial injury. Alcohol or its metabolites (especially acetaldehyde) have a direct toxic effect on the myocardium. Moreover, chronic alcohol use may be associated with thiamine deficiency, which can lead to beriberi heart disease (a form of DCM). Nevertheless, no morphologic features serve to distinguish alcoholic cardiomyopathy from DCM of other causes. Cobalt is an example of a heavy metal with cardiotoxicity and has caused DCM in the setting of inadvertent tainting (e.g., in beer production). Cardiotoxic drugs used for chemotherapy (discussed later) are also important causes of DCM.
- *Peripartum cardiomyopathy* can occur late in pregnancy or up to 5 months postpartum; the mechanism is probably multifactorial, including contributions from genetic susceptibility, pregnancy-associated hypertension, volume overload, nutritional deficiency, and/or other subtle metabolic derangements. In mouse models, DCM can also be induced by increased levels of circulating

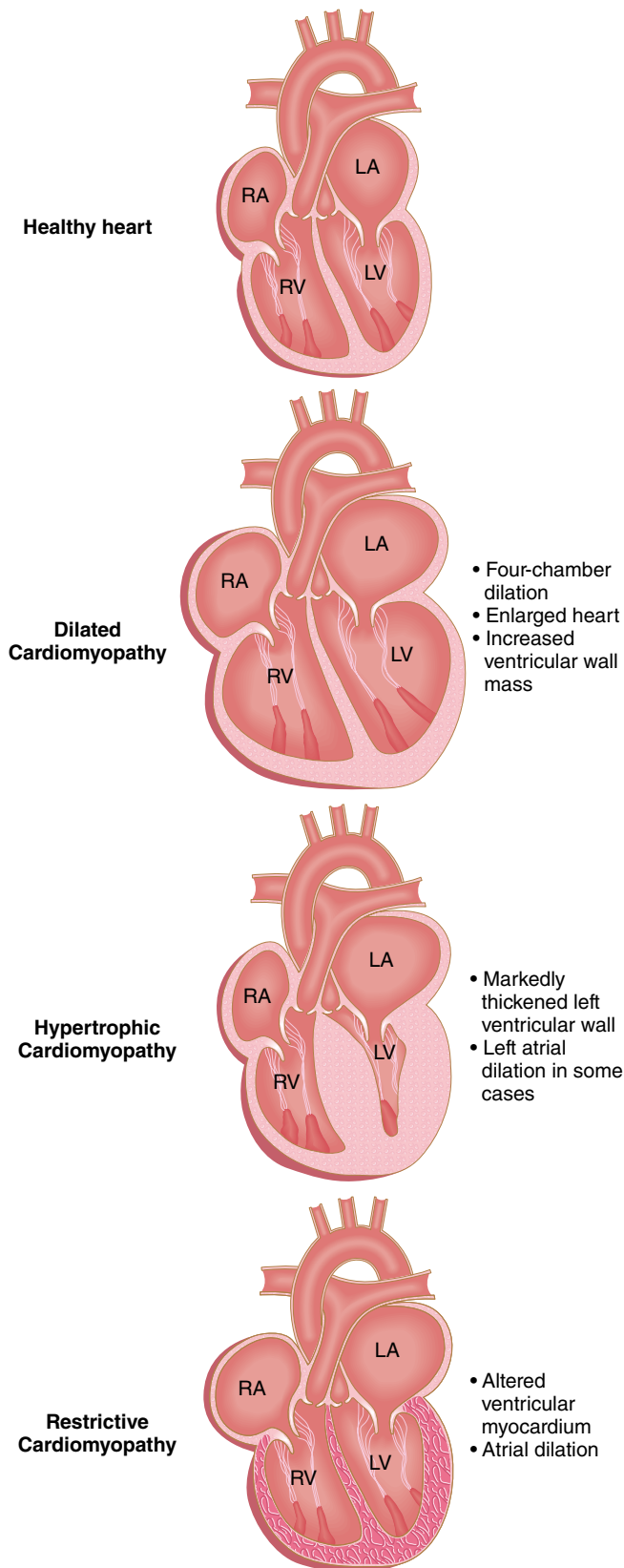


Fig. 12.28 The three major morphologic patterns of cardiomyopathy. Dilated cardiomyopathy leads primarily to systolic dysfunction, whereas restrictive and hypertrophic cardiomyopathies result in diastolic dysfunction. Note the changes in atrial and/or ventricular wall thickness. Ao, Aorta; LA, left atrium; LV, left ventricle; RA, right atrium; RV, right ventricle.

TABLE 12.11 Conditions Associated With Heart Muscle Diseases

Cardiac Infections
• Viruses • Chlamydia • Rickettsia • Bacteria • Fungi • Protozoa
Toxins
• Alcohol • Cobalt • Catecholamines • Carbon monoxide • Lithium • Hydrocarbons • Arsenic • Cyclophosphamide • Doxorubicin (Adriamycin) and daunorubicin
Metabolic Conditions
• Hyperthyroidism • Hypothyroidism • Hyperkalemia • Hypokalemia • Nutritional deficiency (protein, thiamine, other avitaminoses) • Hemochromatosis
Neuromuscular Disease
• Friedreich ataxia • Muscular dystrophy • Congenital atrophies
Storage Disorders and Other Depositions
• Hunter-Hurler syndrome • Glycogen storage disease • Fabry disease • Amyloidosis
Infiltrative Diseases
• Carcinomatosis • Sarcoidosis • Radiation-induced fibrosis
Immunologic Conditions
• Myocarditis (several forms) • Posttransplant rejection

antiangiogenic mediators including vascular endothelial growth factor inhibitors (e.g., sFLT1, as occurs with preeclampsia) or antiangiogenic cleavage products of the hormone prolactin (which rises late in pregnancy). Thus, in pregnant females with a particular genetic predisposition, elevated antiangiogenic molecules can potentially lead to a microvascular angiogenic imbalance (microvascular loss in excess of angiogenesis), ultimately resulting in functional ischemic injury.

- *Iron overload* in the heart can result from either hereditary hemochromatosis (Chapter 18) or from excessive red cell transfusions. DCM is the most common manifestation of such iron excess, and it may be caused by interference with metal-dependent enzyme systems or by injury from iron-mediated production of reactive oxygen species.
- *Supraphysiologic stress* can also result in DCM. This can happen with persistent tachycardia, hyperthyroidism, or even in the fetuses of insulin-dependent diabetic mothers. It can be seen with excess

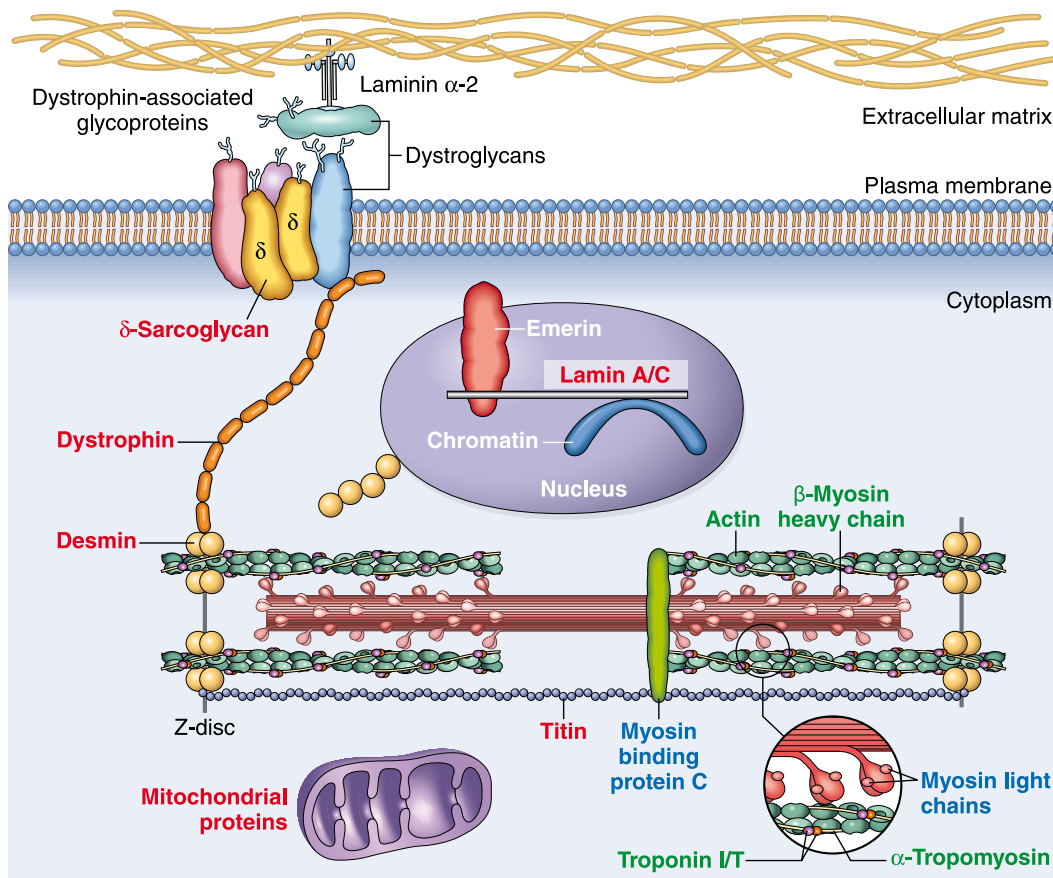


Fig. 12.29 Schematic of a myocyte, showing key proteins mutated in dilated cardiomyopathy (red labels), hypertrophic cardiomyopathy (blue labels), or both (green labels). Mutation of titin is the most common genetic cause of DCM. Titin spans the sarcomere and connects the Z and M bands, thereby limiting the passive range of motion of the sarcomere as it is stretched, and also functions like a molecular spring, thereby affecting the passive elasticity of striated muscle.

circulating catecholamines in individuals with pheochromocytomas (tumors that elaborate epinephrine, Chapter 24); use of cocaine or vasopressor agents such as dopamine can have similar consequences. This catecholamine effect also occurs in the setting of intense autonomic stimulation of the heart, for example, secondary to intracranial lesions or emotional duress. *Takotsubo cardiomyopathy* is an entity characterized by left ventricular contractile dysfunction after extreme psychological stress (thus also called “*broken heart syndrome*”). For unclear reasons, the left ventricular apex is most often affected, leading to apical ballooning that resembles a *takotsubo*, Japanese for “fishing pot for trapping octopus” (hence the name). The mechanism of catecholamine cardiotoxicity is uncertain but likely relates either to direct myocyte toxicity due to calcium overload or to focal vasoconstriction in the face of an increased heart rate.

MORPHOLOGY

In the DCM phenotype, the heart is typically enlarged, heavy (often weighing two to three times normal), and flabby, due to dilation of all chambers (Fig. 12.30). Mural thrombi can result from relative stasis of the blood in poorly contractile chambers and may be a source of thromboemboli. To be considered DCM, the heart should have no primary valvular alterations; if mitral (or tricuspid) regurgitation is present, it results from left (or right) ventricular chamber

dilation (functional regurgitation). Similarly, the coronary arteries must be free from significant narrowing or any obstructions present must be insufficient to explain the degree of cardiac dysfunction.

The histologic abnormalities in DCM are nonspecific (see Fig. 12.30B). Moreover, the severity of morphologic changes may not reflect either the degree of dysfunction or the patient’s prognosis. Interstitial and endocardial fibrosis of variable degree is present, and small subendocardial scars may replace individual cells or groups of cells, probably reflecting healing of previous ischemic necrosis of myocytes caused by hypertrophy-induced imbalance between perfusion and demand. Most muscle cells are hypertrophied with enlarged nuclei, but some are attenuated, stretched, and irregular. In DCM caused by mutations in titin, myocytes may exhibit hyperchromatic, highly distorted “*ninja star*” nuclei (see Fig. 12.30C); identifying them in greater than or equal to 5% of myocytes is highly suggestive of a titin-truncation mutation.

Clinical Features

The fundamental defect in DCM is ineffective contraction. Thus, in end-stage DCM, the cardiac ejection fraction typically is less than 25% (normal is 50% to 70%). DCM can occur at any age, including in childhood, but it most commonly affects individuals between 20 and 50 years of age. It presents with slowly progressive signs and symptoms

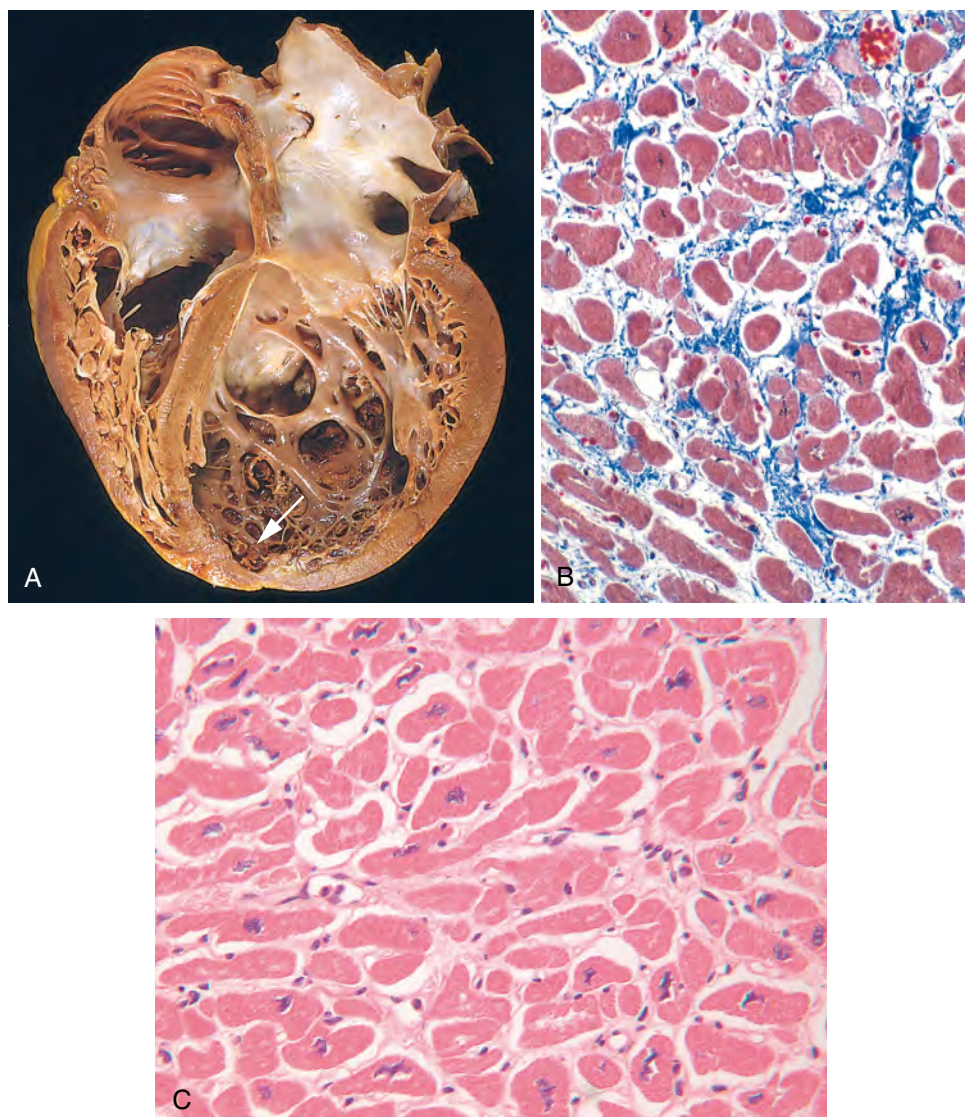


Fig. 12.30 Dilated cardiomyopathy. (A) Four-chamber dilation and hypertrophy are evident. There is a mural thrombus (*arrow*) at the apex of the left ventricle (seen on the right in this apical four-chamber view). The coronary arteries were patent. (B) Histologic section demonstrating variable myocyte hypertrophy and interstitial fibrosis (collagen is highlighted as blue in this Masson trichrome stain). (C) Histologic appearance of the enlarged, bizarre, hyperchromatic nuclei (likened to ninja stars) that are seen in greater numbers in patients with dilated cardiomyopathy caused by titin-truncating mutations.

of CHF including dyspnea, easy fatigability, and poor exertional capacity. Secondary mitral regurgitation and abnormal cardiac rhythms are common, and embolism from intracardiac thrombi can occur. Death usually results from progressive cardiac failure or arrhythmia. Although the annual mortality is high (10% to 50%), some severely affected patients respond well to pharmacologic or electrical resynchronization (biventricular pacing) therapies. Cardiac transplantation is also increasingly performed, and long-term ventricular assist can be beneficial. Interestingly, in some patients, relatively short-term mechanical cardiac support can induce durable improvement of cardiac function.

Arrhythmogenic Right Ventricular Cardiomyopathy

Arrhythmogenic right ventricular cardiomyopathy (ARVC) is an autosomal dominant disorder that classically manifests with right-sided heart failure and rhythm disturbances, which can cause sudden cardiac

death. Unlike dilated cardiomyopathy, ARVC predominantly involves right ventricle. Left-sided involvement with left-sided heart failure can also occur. Typically, the right ventricular wall is severely attenuated due to loss of myocytes, accompanied by massive fatty infiltration and focal fibrosis. Cardiac enlargement in ARVC can sometimes lead to a misdiagnosis of dilated cardiomyopathy (Fig. 12.31). Although mononuclear inflammation may be present around degenerating cardiomyocytes, arrhythmogenic cardiomyopathy is not considered a primarily inflammatory cardiomyopathy. Arrhythmogenic cardiomyopathy has usually an autosomal dominant inheritance with a variable penetrance; many of the causal mutations involve genes encoding desmosomal junctional proteins at the intercalated disk (e.g., plakoglobin) or proteins that interact with the desmosome (e.g., the intermediate filament desmin). *Naxos syndrome* is a disorder characterized by arrhythmogenic cardiomyopathy and hyperkeratosis of plantar and palmar skin surfaces that is also associated with plakoglobin mutations.

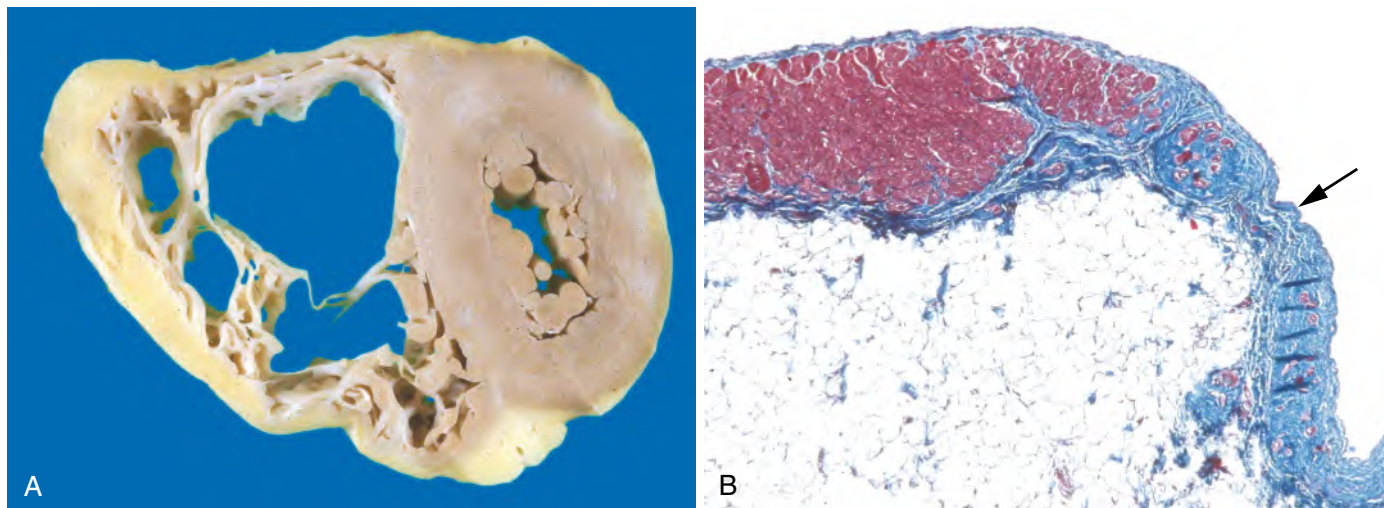


Fig. 12.31 Arrhythmogenic cardiomyopathy. (A) Gross photograph showing dilation of the right ventricle and near-transmural replacement of the right ventricular free-wall by fat and fibrosis. The left ventricle has a virtually normal configuration in this case but can also be involved by the disease process. (B) Histologic section of the right ventricular free wall, demonstrating replacement of myocardium (red) by fibrosis (blue, arrow) and fat (Masson trichrome stain).

Hypertrophic Cardiomyopathy

Hypertrophic cardiomyopathy (HCM) is a common (1 in 500 prevalence), clinically heterogeneous, genetic disorder characterized by myocardial hypertrophy, poorly compliant left ventricular myocardium leading to abnormal diastolic filling, and (in about one-third of cases) intermittent ventricular outflow obstruction. It is the leading cause of left ventricular hypertrophy unexplained by other clinical or pathologic causes. The heart is thick-walled, heavy, and hypercontracting, in striking contrast to the flabby, hypocontracting heart of DCM. HCM causes primarily diastolic dysfunction; systolic function is usually preserved. The two most common diseases that must be distinguished clinically from HCM are deposition diseases (e.g., amyloidosis, Fabry disease) and HHD coupled with age-related subaortic septal hypertrophy (see earlier discussion under Hypertensive Heart Disease in this chapter). Occasionally, valvular or congenital subvalvular aortic stenosis can also mimic HCM.

Pathogenesis

In most cases, the pattern of transmission is autosomal dominant with variable penetrance. **HCM is most commonly caused by mutations in any one of several genes that encode sarcomeric proteins;** there are more than 400 different known mutations in nine different genes, most being missense mutations. A unifying feature is that these are all gain-of-function mutations that affect sarcomeric proteins, causing hypercontractility. This insight has led to novel therapeutics for HCM based on inhibition of myosin activity.

Mutations causing HCM are found most commonly in the genes encoding myosin-binding protein C (*MYBP-C*) or β -myosin heavy chain (*β -MHC/MYH7*), followed by the genes coding for cardiac TnI, TnT, or α -tropomyosin; overall, these account for 70% to 80% of all cases (see Fig. 12.29). Different affected families can have distinct mutations involving the same protein; thus, over 50 different *β -MHC* mutations can cause HCM. The prognosis of HCM varies widely and correlates strongly with specific mutations.

While HCM is a disease associated with hypercontractility caused by sarcomere protein mutations, DCM is mostly associated

with abnormalities of cytoskeletal proteins (Fig. 12.32); the latter can be conceptualized as a disease of abnormal force generation, force transmission, or myocyte signaling. Interestingly, some of the genes mutated in HCM are also mutated in DCM (e.g., *MYH7*), but in DCM the mutations cause loss of function as opposed to the gain-of-function mutations in HCM. Genetic testing in cardiomyopathies is still hindered by the large number of genetic variants of unknown significance, so genetic screening of family members of affected probands must be interpreted within the clinical context.

MORPHOLOGY

The essential feature of HCM is **massive myocardial hypertrophy, usually without ventricular dilation** (Fig. 12.33). The classic pattern (roughly two-thirds of patients) involves disproportionate thickening of the ventricular septum relative to the left ventricle free wall, termed **asymmetric septal hypertrophy**; in others, the hypertrophy is concentric and symmetrical. In the classic pattern, the normally round-to-ovoid left ventricular cavity may be compressed into a banana-like configuration by bulging of the ventricular septum into the lumen (see Fig. 12.33A). Although marked hypertrophy can involve the entire septum, it is usually most prominent in the subaortic region; the left ventricular outflow tract often exhibits a fibrous endocardial plaque and thickening of the anterior mitral leaflet. Both findings result from contact of the anterior mitral leaflet with the septum during ventricular systole; they correlate with the echocardiographic systolic anterior motion of the anterior leaflet, which can lead to functional left ventricular outflow tract obstruction during systole (so-called **hypertrophic obstructive cardiomyopathy**).

The most important histologic features of HCM myocardium are (1) massive cardiomyocyte hypertrophy, with transverse myocyte diameters frequently greater than 40 μm (normal is approximately 15 μm); (2) haphazard disarray of bundles of myocytes, individual myocytes, and contractile elements in sarcomeres within cells (termed **myofiber disarray**); (3) fibrotic narrowing of small intramural arteries; and (4) interstitial and replacement fibrosis (see Fig. 12.33B).

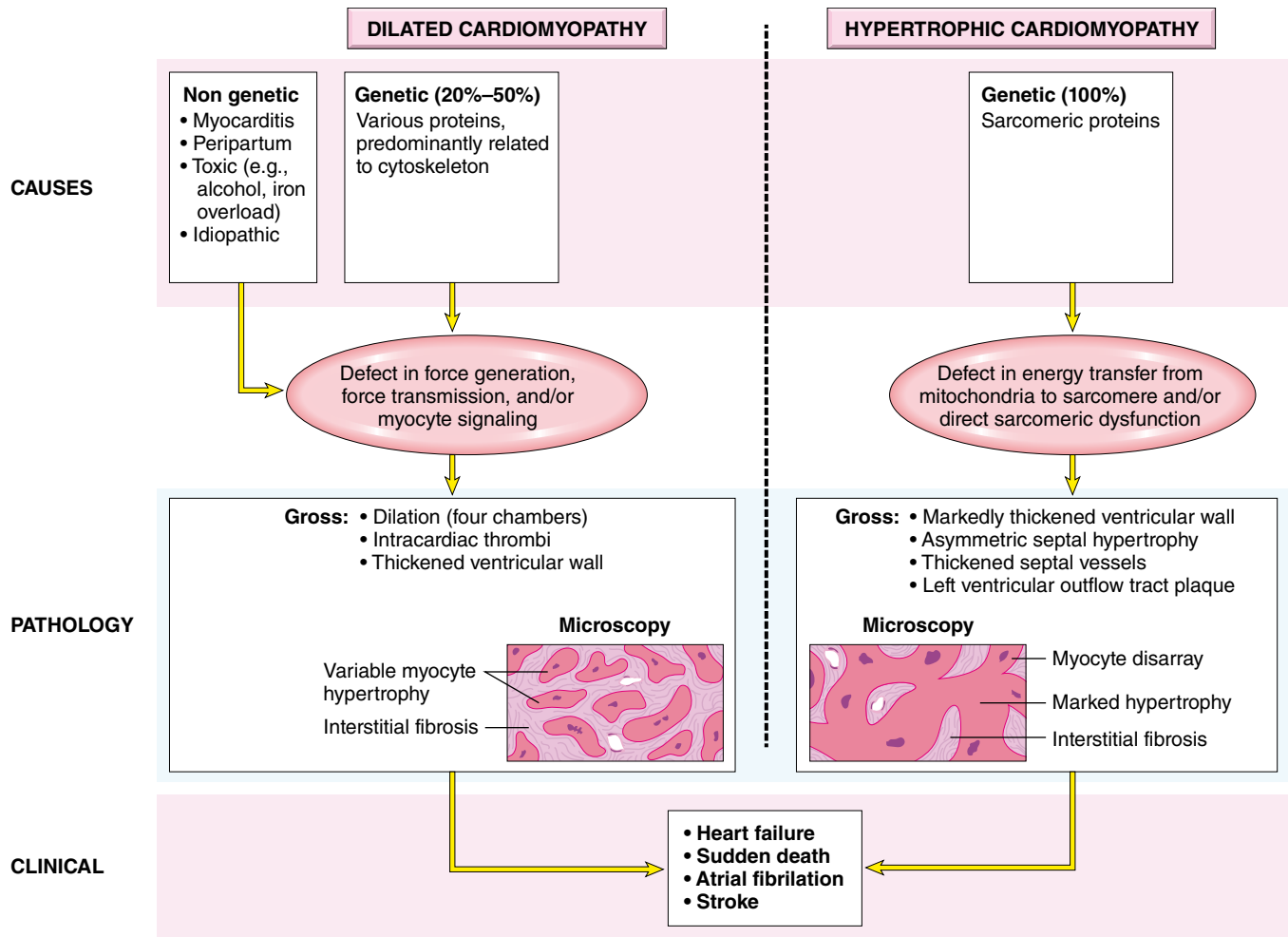


Fig. 12.32 Causes and consequences of dilated and hypertrophic cardiomyopathy. LV, Left ventricle.

Clinical Features

HCM is characterized by reduced stroke volume due to impaired diastolic filling—a consequence of a reduced chamber size and the reduced compliance of the massively hypertrophied left ventricle. In addition, approximately 70% of patients with HCM eventually develop dynamic obstruction to the left ventricular outflow as the anterior mitral leaflet moves toward the ventricular septum during systole. The compromised cardiac output in conjunction with a secondary increase in pulmonary venous pressure explains the exertional dyspnea seen in these patients. Auscultation discloses a harsh systolic ejection murmur caused by the ventricular outflow obstruction. Because of the massive hypertrophy, high left ventricular chamber pressure, and frequently thick-walled intramural arteries, focal myocardial ischemia commonly results, even in the absence of concomitant epicardial CAD. Major clinical problems in HCM are atrial fibrillation, mural thrombus formation leading to embolization and possible stroke, cardiac failure, ventricular arrhythmias, and, not infrequently, sudden death, especially with certain mutations. Indeed, HCM is one of the most common causes of sudden death in young athletes.

The natural history of HCM is highly variable. Most patients benefit from pharmacologic intervention (e.g., β -adrenergic blockade) to decrease heart rate and contractility. Implantable cardioverter

defibrillators are warranted for HCM patients at risk for ventricular arrhythmias. Outflow tract obstruction due to the septal muscle mass can be mitigated by surgical excision or by carefully controlled septal infarction through a catheter-based infusion of alcohol.

Restrictive Cardiomyopathy

Restrictive cardiomyopathy is characterized by a primary decrease in ventricular compliance, resulting in impaired ventricular diastolic filling. Because the contractile (systolic) function of the left ventricle is usually unaffected, the functional abnormality can be confused with that of constrictive pericarditis or HCM. Restrictive cardiomyopathy can be idiopathic or associated with distinct disorders that affect the myocardium, principally amyloidosis (described later), sarcoidosis, radiation-induced fibrosis, metastatic tumors, or the accumulation of metabolites from inborn errors of metabolism.

The gross morphologic features are nonspecific; although biatrial dilation is commonly observed due to restricted ventricular filling and pressure overloads, the ventricles are of approximately normal size (or slightly enlarged), the cavities are not dilated, and the myocardium is largely unremarkable. Microscopically, there can be patchy or diffuse interstitial fibrosis, varying from minimal to extensive.

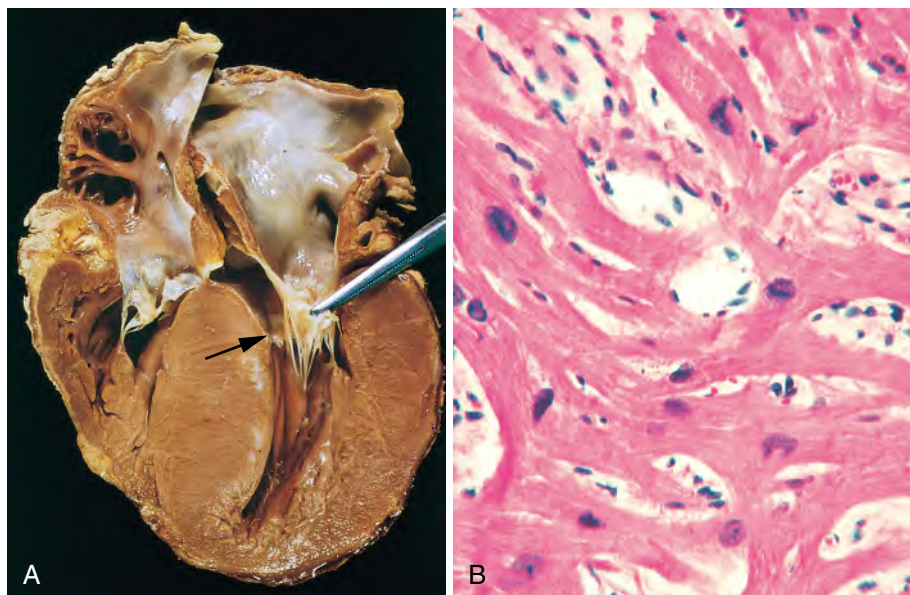


Fig. 12.33 Hypertrophic cardiomyopathy with asymmetric septal hypertrophy. (A) The septal muscle bulges into the left ventricular outflow tract, and the left atrium is enlarged. The anterior mitral leaflet has been retracted away from the septum to reveal a fibrous endocardial plaque (arrow) (see text). (B) Histologic appearance demonstrating myocyte disarray, extreme hypertrophy, and exaggerated myocyte branching, as well as the characteristic interstitial fibrosis.

Cardiac imaging is increasingly used to characterize myocardial “infiltrates,” and endomyocardial biopsy can occasionally suggest a specific etiology.

Three other restrictive conditions merit brief mention.

- *Endomyocardial fibrosis* is principally a disease of children and young adults in Africa and other tropical areas, characterized by fibrosis of the ventricular endocardium and subendocardium that extends from the apex upward, often eventually involving the tricuspid and mitral valves. The fibrous tissue markedly diminishes the volume and compliance of affected chambers and so causes a restrictive functional defect. Ventricular mural thrombi sometimes develop, and indeed much of the endocardial fibrosis may result from thrombus organization. Endomyocardial fibrosis is linked to nutritional deficiencies and/or inflammation related to parasitic infections (e.g., hypereosinophilia); worldwide, it is the most common form of restrictive cardiomyopathy.
- *Loeffler endomyocarditis* also results in endomyocardial fibrosis, typically with large mural thrombi, and an overall morphology similar to the tropical disease. However, in addition to the cardiac changes, there is often peripheral eosinophilia and eosinophilic infiltrates in multiple organs, including the heart. The release of toxic products of eosinophils, especially major basic protein, probably causes endomyocardial necrosis, followed by scarring of the necrotic area, layering of the endocardium by thrombus, and finally organization of the thrombus. Some patients with Loeffler endomyocarditis have a myeloid neoplasm associated with eosinophilia caused by acquired chromosomal rearrangements involving several different tyrosine kinase genes (Chapter 13). Treatment of such patients with tyrosine kinase inhibitors has resulted in hematologic remissions and resolution of the endomyocardial lesions.
- *Endocardial fibroelastosis* is an uncommon heart disease characterized by fibroelastic thickening that typically involves the left ventricular endocardium. It is most common in the first 2 years of

life; in one-third of cases, it is accompanied by aortic valve obstruction or other congenital cardiac anomalies. Endocardial fibroelastosis may represent a common morphologic endpoint of several different insults including viral infections (e.g., intrauterine exposure to mumps) or mutations in the gene for *tafazzin*, a protein that is required for mitochondrial inner membrane integrity. Whatever the cause, diffuse involvement may be responsible for rapid and progressive cardiac decompensation and death.

Amyloidosis

Amyloidosis is an important form of restrictive cardiomyopathy resulting from the extracellular accumulation of protein fibrils that form insoluble β -pleated sheets (Chapter 6). *Cardiac amyloidosis* can appear as a consequence of systemic amyloidosis (e.g., due to plasma cell neoplasms or inflammation-associated amyloid) or can be largely restricted to the heart, particularly in the aged (*senile cardiac amyloidosis*). Cardiac amyloidosis characteristically occurs in individuals 70 years of age and older, and it has a far better prognosis than systemic amyloidosis; the amyloid deposits are largely composed of *wild type transthyretin* (ATTRwt), a normal serum protein synthesized in the liver that transports thyroxine and retinol-binding protein. Mutant forms of hereditary transthyretin (hTTR) can accelerate the cardiac amyloid deposition; by an ancient founder effect, 3% to 4% of coastal West Africans have a transthyretin mutation substituting isoleucine for valine at position 122 that produces a particularly amyloidogenic protein that is responsible for an autosomal dominant familial form of transthyretin amyloidosis.

Cardiac amyloidosis most frequently produces a restrictive cardiomyopathy, but it can also be asymptomatic, manifest as dilation or arrhythmias, or mimic ischemic or valvular disease. The varied presentations depend on the predominant location of the deposits, for example, interstitium, conduction system, coronary vasculature, or valves.

MORPHOLOGY

In cardiac amyloidosis, the heart varies in consistency from normal to firm and rubbery. The chambers are usually of normal size but can be dilated and have thickened walls. Small, semitranslucent nodules resembling drips of wax may be seen on the atrial endocardial surface, particularly on the left. Histologically, hyaline eosinophilic deposits of amyloid may be found in the interstitium, conduction tissue, valves, endocardium, pericardium, and small intramural coronary arteries (Fig. 12.34). Amyloid accumulations can be distinguished from other deposits by special stains such as Congo red or sulfated Alcian blue; the former produces classic **apple-green birefringence** when viewed under polarized light (see Fig. 12.34B). Intramural arteries and arterioles may have sufficient amyloid in their walls to compress and occlude their lumens, inducing myocardial ischemia (small-vessel disease).

Myocarditis

Myocarditis is a diverse group of pathologic entities in which infectious microorganisms and/or a primary inflammatory process cause myocardial injury. Myocarditis should be distinguished from conditions such as IHD, in which myocardial inflammation is secondary to other causes.

Pathogenesis

In the United States, viral infections are the most common cause of myocarditis. Coxsackie viruses A and B and other enteroviruses probably account for most of the cases. Other less common etiologic agents include cytomegalovirus, HIV, and influenza (Table 12.12). In some (but not all) cases, the offending agent can be implicated by serologic studies or by identifying viral nucleic acid sequences in myocardial biopsies. Depending on the pathogen and the host, viruses can potentially cause myocardial injury either as a direct cytopathic effect, or by eliciting a destructive immune response. Inflammatory cytokines produced in response to microbes or myocardial injury can also cause myocardial dysfunction that is out of proportion to the degree of actual myocyte damage. In some cases of active SARS-CoV-2 infection, the heart is dysfunctional due to myocarditis, but this is rare, and more commonly the clinically important COVID-19 heart pathology shows focal myocardial necrosis, often due to microthrombi.

TABLE 12.12 Major Causes of Myocarditis

Infections
• Viruses (e.g., Coxsackievirus, echovirus, influenza, HIV, cytomegalovirus) • Chlamydiae (e.g., <i>Chlamydothila psittaci</i>) • Rickettsiae (e.g., <i>Rickettsia typhi</i>, typhus fever) • Bacteria (e.g., <i>Corynebacterium diphtheriae</i>, <i>Neisseria meningococcus</i>, <i>Borrelia</i> [Lyme disease]) • Fungi (e.g., <i>Candida</i>) • Protozoa (e.g., <i>Trypanosoma cruzi</i> [Chagas disease], toxoplasmosis) • Helminths (e.g., trichinosis)
Immune-Mediated Reactions
• Postviral • Poststreptococcal (rheumatic fever) • Systemic lupus erythematosus • Drug hypersensitivity (e.g., methyldopa, sulfonamides) • Transplant rejection • Immune checkpoint inhibitor therapies
Unknown
• Sarcoidosis • Giant cell myocarditis

HIV, Human immunodeficiency virus.

Nonviral agents are also important causes of infectious myocarditis, particularly the protozoan *Trypanosoma cruzi*, the agent of Chagas disease. Chagas disease is endemic in some regions of South America, and myocardial involvement is seen in most infected individuals. About 10% of patients die during an acute attack; others develop chronic immune-mediated myocarditis that may progress to cardiac insufficiency in 10 to 20 years. Trichinosis (*Trichinella spiralis*) is the most common helminthic disease associated with myocarditis. Parasitic diseases, including toxoplasmosis, and bacterial infections such as Lyme disease and diphtheria, can also cause myocarditis. In the case of diphtheritic myocarditis, the myocardial injury is a consequence of diphtheria toxin release by *Corynebacterium diphtheriae* (Chapter 8). Myocarditis occurs in approximately 5% of patients with Lyme disease, a systemic illness caused by the bacterial spirochete *Borrelia burgdorferi* (Chapter 8); it manifests primarily as a self-limited conduction system

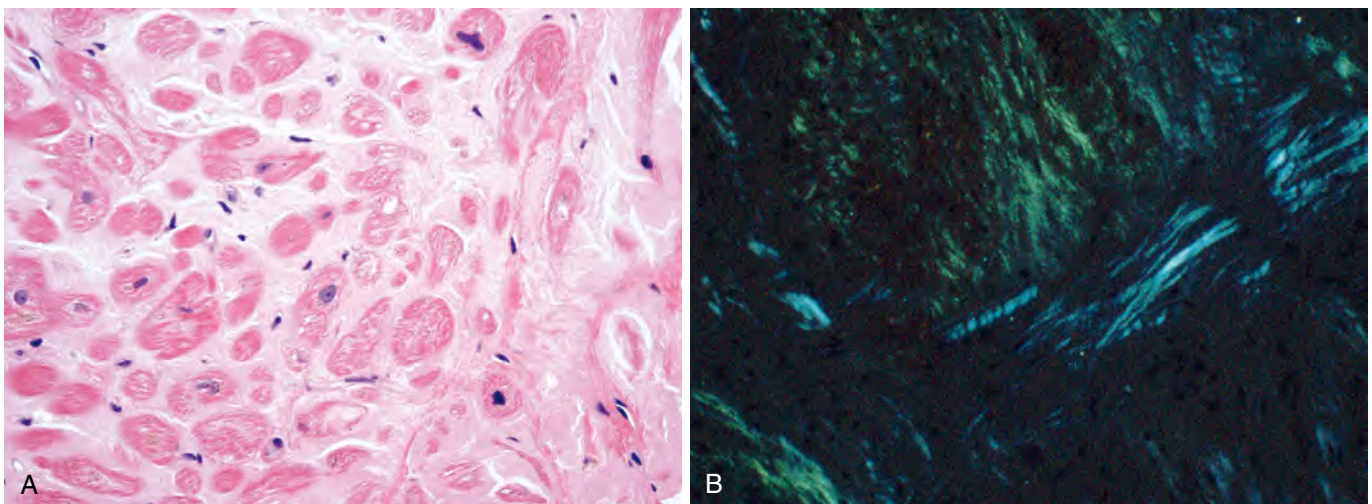


Fig. 12.34 Cardiac amyloidosis. (A) Hematoxylin and eosin stain, showing amyloid appearing as amorphous pink material around myocytes. (B) Congo red stain viewed under polarized light, in which amyloid shows characteristic apple-green birefringence (compared with collagen, pale blue).

disorder that may require a temporary pacemaker. AIDS-associated myocarditis may be attributable to direct effects of HIV or to an opportunistic pathogen but in some instances is present without a clear etiologic agent.

There are also noninfectious causes of myocarditis. Broadly speaking they are either immunologically mediated (hypersensitivity myocarditis) or idiopathic conditions with distinctive morphology (giant cell myocarditis) suspected to be of immunologic origin (see Table 12.12). More recently, immune checkpoint inhibitors administered for treatment of cancer have occasionally led to often fatal lymphocytic myocarditis. Finally, there is much debate about myocarditis as a consequence of mRNA COVID-19 vaccination. It should be noted that biopsy-proven cases are exceptionally rare, and the incidence of such a vaccine-associated myocarditis may be indistinguishable from baseline population rates.

MORPHOLOGY

Grossly, the heart in myocarditis may appear normal or dilated; some hypertrophy may be present depending on disease duration. In advanced stages, the ventricular myocardium is flabby and often mottled by either pale foci or minute hemorrhagic lesions. Mural thrombi may be present.

Active myocarditis is characterized by an interstitial inflammatory infiltrate associated with degenerating or apoptotic myocytes (Fig. 12.35); larger zones of myocyte drop-out or necrosis with interstitial edema may reflect vascular injury and secondary ischemic damage. A diffuse, mononuclear, predominantly lymphocytic infiltrate is most common (Fig. 12.35A). Endomyocardial biopsies can be diagnostic but are often spuriously negative because inflammation of the myocardium is patchy. If the patient survives the acute phase of myocarditis, the inflammatory lesions either resolve, leaving no residual changes, or heal by progressive fibrosis.

Hypersensitivity myocarditis is characterized by perivascular infiltrates composed of lymphocytes, macrophages, and a high proportion of eosinophils (Fig. 12.35B). A morphologically distinctive form of myocarditis, called **giant cell myocarditis**, is characterized by a widespread inflammatory infiltrate containing multinucleate giant cells (fused macrophages) interspersed with lymphocytes, eosinophils, plasma cells, and macrophages. Focal to frequently extensive myocyte damage is present (Fig. 12.35C). This variant likely represents the fulminant end of the myocarditis spectrum and carries a poor prognosis.

The myocarditis of **Chagas disease** is distinctive by virtue of the parasitization of scattered myofibers by trypanosomes accompanied by a mixed inflammatory infiltrate of neutrophils, lymphocytes, macrophages, and occasional eosinophils (Fig. 12.35D).

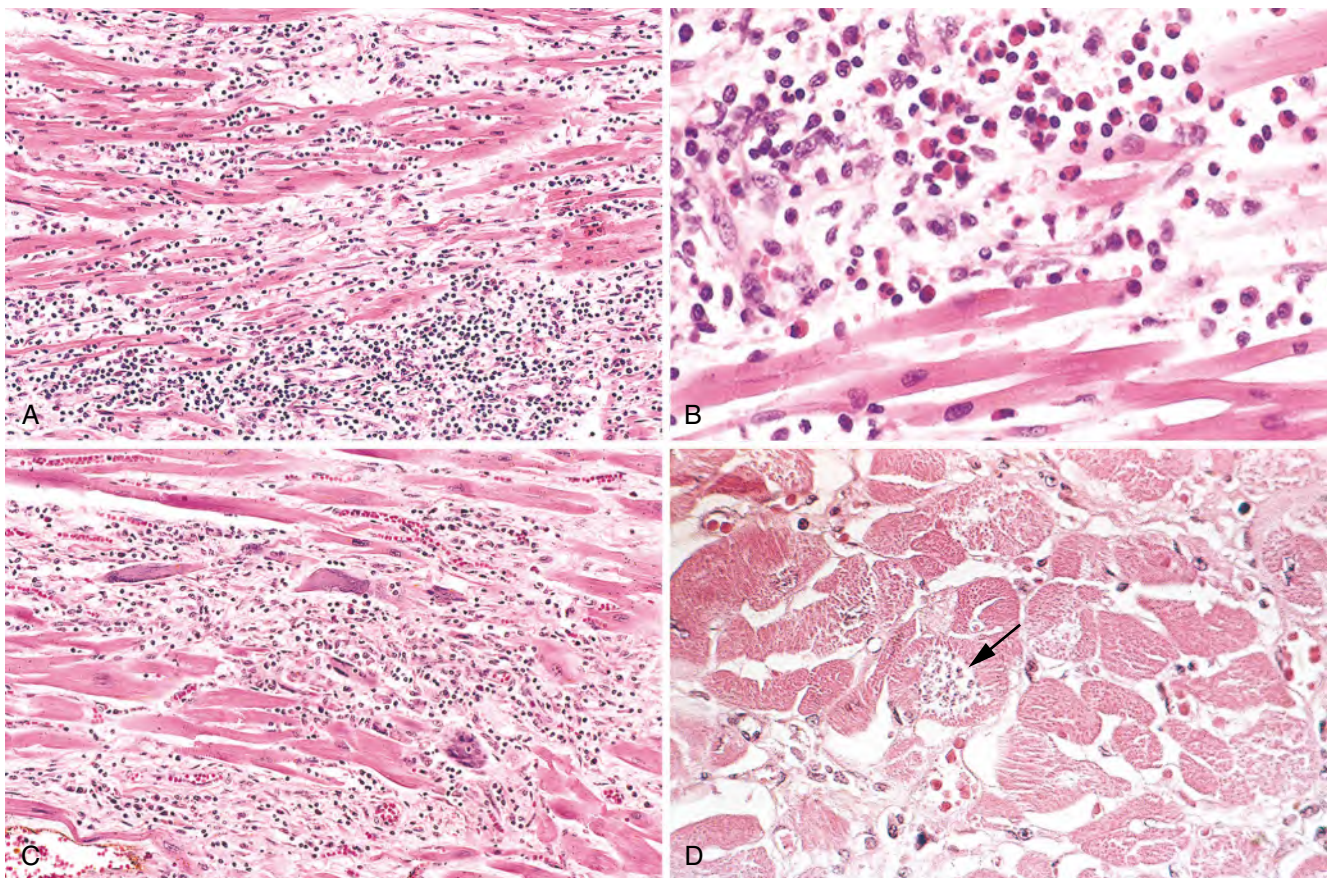


Fig. 12.35 Myocarditis. (A) Lymphocytic myocarditis, associated with myocyte injury. (B) Hypersensitivity myocarditis, characterized by interstitial inflammatory infiltrate composed largely of eosinophils and mononuclear inflammatory cells, predominantly localized to perivascular and expanded interstitial spaces. (C) Giant cell myocarditis, showing extensive loss of muscle associated with an inflammatory infiltrate containing lymphocytes, macrophages, and multinucleated giant cells (fused macrophages). (D) Myocarditis of Chagas disease. A myofiber distended with trypanosomes (arrow) is present along with individual myofiber necrosis and modest inflammation.

Clinical Features

The clinical spectrum of myocarditis is broad. At one end, the disease may be asymptomatic. Even with symptomatic disease—typically marked by fatigue, dyspnea, palpitations, precordial discomfort, and fever—most patients make a complete recovery without sequelae. At the other extreme is the precipitous onset of heart failure or arrhythmias, mimicking those of acute MI, occasionally followed by sudden death. As noted previously, DCM may develop as a late complication.

Other Causes of Myocardial Disease

Cardiotoxic Drugs

Cardiac complications of cytotoxic cancer therapies are an important clinical problem. Cardiotoxicity may be associated with many different conventional chemotherapeutic agents, as well as tyrosine kinase inhibitors. The anthracyclines *doxorubicin* and *daunorubicin* are the chemotherapeutic agents most often associated with toxic myocardial injury; they cause DCM and heart failure by direct damage to cardiomyocytes. Anthracycline toxicity is dose dependent, with the cardiotoxicity risk increasing when cumulative lifetime doses exceed 250 mg/m².

Many other therapeutic agents, including lithium, phenothiazines, and chloroquine, can idiosyncratically induce myocardial injury and sometimes sudden death. Common findings in affected myocardium include myofiber swelling, cytoplasmic vacuolization, and fatty change. Discontinuing the offending agent often leads to prompt resolution, without apparent sequelae. Occasionally, however, more extensive damage produces myocyte death that can evolve to a DCM.

Radiation

Radiation therapy to the thorax can cause damage throughout the heart. Manifestations include accelerated atherosclerosis, valvular stenosis, pericardial fibrosis, and/or a restrictive cardiomyopathy due to interstitial fibrosis.

KEY CONCEPTS

Cardiomyopathy

- Cardiomyopathies are intrinsic cardiac muscle diseases that may be genetic or due to other well-defined causes.
- There are three general pathophysiologic categories of cardiomyopathy: dilated (90%), hypertrophic, and restrictive (least common).
- Dilated cardiomyopathy (DCM) results in systolic (contractile) dysfunction. Causes include myocarditis, toxic exposures (e.g., alcohol), and pregnancy. In up to 50% of cases, autosomal dominant loss of function mutations affecting genes encoding cytoskeleton, sarcolemma, or nuclear envelope are causal, with titin-truncation mutations representing up to 20% of cases of DCM.
- Hypertrophic cardiomyopathy (HCM) results in diastolic (relaxation) dysfunction. Virtually all cases are due to autosomal dominant gain of function mutations in the proteins comprising the contractile apparatus.
- Restrictive cardiomyopathy results in a stiff, noncompliant myocardium and can be due to deposition (e.g., amyloid), increased interstitial fibrosis (e.g., due to radiation), or endomyocardial scarring.
- Myocarditis is myocardial damage caused by inflammatory infiltrates secondary to infections or immune reactions. Viral infections are the most common causes in the United States. Clinically, myocarditis can be asymptomatic, give rise to acute heart failure, or evolve into DCM.

PERICARDIAL DISEASE

The most important pericardial disorders involve fluid accumulation, inflammation, fibrous constriction, or some combination of these

processes, often in association with other cardiac pathology or a systemic disease.

Pericardial Effusion and Hemopericardium

Normally, the pericardial sac contains less than 50 mL of thin, clear, straw-colored fluid. Under various circumstances, the parietal pericardium may be distended by serous fluid (pericardial effusion), blood (hemopericardium), or pus (purulent pericarditis). With long-standing cardiac enlargement or with slowly accumulating fluid, the pericardium has time to remodel to accommodate the larger volume. This permits a slowly accumulating pericardial effusion to become quite substantial without restricting cardiac function. Thus, with chronic effusions of less than 500 mL in volume, the only clinical significance is a characteristic globular enlargement of the heart shadow on chest radiographs. In contrast, rapidly developing fluid collections of as little as 200 to 300 mL (e.g., due to hemopericardium caused by a ruptured MI or aortic dissection) can severely compress the thin-walled atria and venae cavae, or even the ventricles; cardiac filling is thereby compromised, producing fatal cardiac tamponade.

Pericarditis

Pericardial inflammation can occur secondary to a variety of cardiac, thoracic, or systemic disorders, metastases from remote neoplasms, or cardiac surgical procedures. The major causes of pericarditis are listed in Table 12.13. Most evoke acute pericarditis, but a few, such as tuberculosis and fungi, produce only chronic reactions.

Acute Pericarditis

Serous pericarditis is characteristically produced by noninfectious inflammatory diseases, including rheumatic fever, SLE, and scleroderma, as well as tumors and uremia. An infection in the tissues contiguous to the pericardium—for example, a bacterial pleuritis—may incite sufficient irritation of the parietal pericardial serosa to cause a sterile serous effusion that can progress to serofibrinous pericarditis and ultimately to a suppurative reaction. In some cases, a well-defined viral infection elsewhere—upper respiratory tract, lung, parotid gland—antedates the pericarditis and serves as the primary focus of infection. Infrequently, usually in young adults, viral pericarditis occurs as an apparent primary infection that may be accompanied by

TABLE 12.13 Causes of Pericarditis

Infectious Agents

- Viruses
- Pyogenic bacteria
- Tuberculosis
- Fungi
- Other parasites

Presumably Immunologically Mediated

- Rheumatic fever
- Systemic lupus erythematosus
- Scleroderma
- Postcardiotomy
- Postmyocardial infarction (Dressler) syndrome
- Drug hypersensitivity reaction

Miscellaneous

- Myocardial infarction
- Uremia
- Cardiac surgery
- Neoplasia
- Trauma
- Radiation

myocarditis (myopericarditis). Tumors can cause a serous pericarditis by lymphatic invasion or direct contiguous extension into the pericardium. Histologically, serous pericarditis elicits a mild inflammatory infiltrate in the epipericardial fat consisting predominantly of lymphocytes; neoplastic cells may also be seen in effusions associated with metastatic tumors. Organization into fibrous adhesions rarely occurs.

Fibrinous and serofibrinous pericarditis are the most frequent types of pericarditis; these are composed of serous fluid variably admixed with a fibrinous exudate. Common causes include acute MI (see Fig. 12.17D), postinfarction Dressler syndrome (an autoimmune response appearing weeks after an MI), uremia, rheumatic fever, SLE, and trauma. A fibrinous reaction also follows routine cardiac surgery. Radiation used to treat breast, lung, or mediastinal neoplasms can cause pericarditis, pericardial effusion, and chronic pericardial disorders.

Symptoms of fibrinous pericarditis characteristically include pain (sharp, pleuritic, and position dependent) and fever; congestive failure may also be present. A loud pericardial friction rub may be the most striking clinical finding but is usually absent if sufficient fluid accumulates to separate the two layers of the pericardium.

Purulent or suppurative pericarditis reflects an active infection caused by microbial invasion of the pericardial space; this can occur through:

- *Direct extension* from neighboring infections, such as an empyema of the pleural cavity, lobar pneumonia, mediastinal infections, or extension of a ring abscess through the myocardium or aortic root
- *Seeding from the blood*
- *Lymphatic extension*
- *Direct introduction* during cardiectomy

The exudate ranges from a thin cloudy fluid to thick pus up to 500 mL in volume; tuberculous pericarditis can exhibit foci of caseation. The serosal surfaces are reddened, granular, and coated with the exudate (Fig. 12.36). Microscopically there is an acute inflammatory reaction,

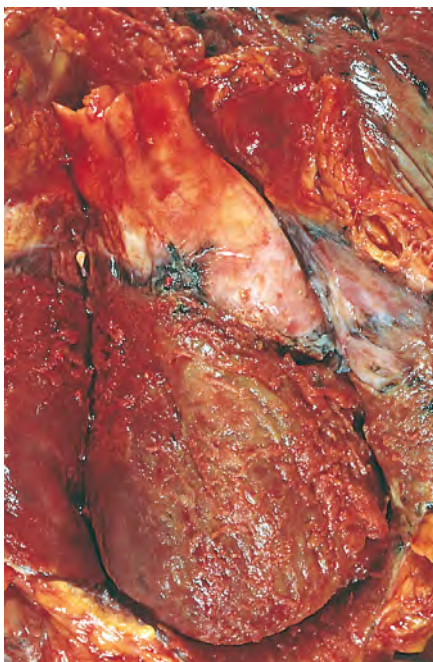


Fig. 12.36 Acute suppurative pericarditis arising from direct extension of an adjacent pneumonia. Extensive purulent exudate is evident.

which sometimes extends into surrounding structures (*mediastinopericarditis*). Complete resolution is infrequent, and organization by scarring is the usual outcome. The intense inflammatory response and the subsequent scarring frequently produce constrictive pericarditis (see later). Clinical findings in the active phase resemble those seen in fibrinous pericarditis, but infection leads to more marked systemic symptoms, including spiking fevers and rigors.

Hemorrhagic pericarditis has an exudate composed of blood mixed with a fibrinous or suppurative effusion. It is most commonly caused by the spread of a malignant neoplasm to the pericardial space; cytologic examination of fluid removed through a pericardial tap often reveals neoplastic cells. Hemorrhagic pericarditis can also be found in bacterial infections, in persons with an underlying bleeding diathesis, and in tuberculosis. Hemorrhagic pericarditis often follows cardiac surgery and is occasionally responsible for significant blood loss or even tamponade, requiring reoperation. Hemorrhagic pericarditis due spread of a malignancy generally has a poor prognosis, but exceptions exist in the case of highly treatable cancers, such as non-Hodgkin lymphoma.

Chronic or Healed Pericarditis

Organization of pericardial inflammation may produce unobtrusive plaque-like fibrous thickenings of the serosal membranes (“*soldiers’ plaque*”) or thin, delicate adhesions that rarely cause impairment of cardiac function. In other cases, fibrosis in the form of mesh-like stringy adhesions can largely obliterate the virtual space in the pericardial sac (adhesive pericarditis); in most instances, this has no effect on cardiac function. Two forms are worthy of some discussion.

- *Adhesive mediastinopericarditis* may follow infectious pericarditis, previous cardiac surgery, or mediastinal irradiation. The pericardial sac is obliterated, and adherence of the external aspect of the parietal layer to surrounding structures strains cardiac function. With each systolic contraction, the heart pulls not only against the parietal pericardium but also against the attached surrounding structures. Systolic retraction of the rib cage and diaphragm and pulsus paradoxus may be observed. The increased workload occasionally causes significant cardiac hypertrophy and dilation.
- In *constrictive pericarditis*, the heart is encased in a dense, fibrous, or fibrocalcific scar that limits diastolic expansion and cardiac output, features that mimic a restrictive cardiomyopathy. A prior history of pericarditis may or may not be present. The fibrous scar can be up to 1 cm in thickness, obliterating the pericardial space and sometimes calcifying; in extreme cases, it can resemble a plaster mold (*concretio cordis*). Because of the dense enclosing scar, cardiac hypertrophy and dilation cannot occur. Cardiac output may be reduced at rest, but more importantly the heart has little if any capacity to increase its output in response to increased systemic demands. Signs of constrictive pericarditis include distant or muffled heart sounds, elevated jugular venous pressure, and peripheral edema. Treatment consists of surgical resection of the shell of constricting fibrous tissue (pericardiectomy).

TUMORS OF THE HEART

Primary Cardiac Tumors

Primary cardiac tumors are uncommon; moreover, most are (fortunately) benign. The five most common tumors have no malignant potential and account for almost 90% of all primary heart tumors. In descending order of frequency (combined pediatric and adult populations), these are myxomas, fibromas, lipomas, papillary fibroelastomas, and rhabdomyomas.

Primary malignant tumors of the heart are exceedingly rare, and usually prove to be angiosarcoma or undifferentiated sarcomas with

MDM2 oncogene amplification. Sarcomas of the heart are not clinically or morphologically distinctive from sarcomas arising in other locations.

Fibromas and lipomas resemble their counterparts elsewhere; only the myxomas, papillary fibroelastomas, and rhabdomyomas merit further mention here.

Myxomas are the most common primary tumor of the adult heart. These are benign neoplasms arising from primitive multipotent mesenchymal cells. Although sporadic myxomas do not show consistent genetic alterations, two syndromes associated with myxomas are recognized. *McCune-Albright syndrome* is caused by gain of function mutations in the gene *GNAS1*, which encodes the $G\alpha$ subunit of a G protein, leading to enhance G protein signaling. *Carney complex* is caused by loss of function mutations in *PRKARIA*, which encodes a regulatory subunit of a cyclic AMP-dependent protein kinase, leading to increase cyclic AMP-dependent signaling. About 90% of myxomas arise in the atria, with a left-to-right ratio of approximately 4:1.

MORPHOLOGY

The tumors are usually single but can rarely be multiple. The region of the fossa ovalis in the atrial septum is the favored site of origin. Myxomas range from small (<1 cm) to large (≥ 10 cm) and can be sessile or pedunculated lesions (Fig. 12.37A). They vary from globular hard masses mottled with hemorrhage to soft, translucent, papillary, or villous lesions having a gelatinous appearance. The pedunculated form is often sufficiently mobile to move during systole into the atrioventricular valve opening, causing intermittent obstruction that may be position dependent.

Histologically, myxomas are composed of stellate or globular *myxoma cells* embedded within an abundant acid mucopolysaccharide ground substance (Fig. 12.37B). Peculiar vessel-like or gland-like structures are characteristic. Hemorrhage and mononuclear inflammation are usually present.

The major clinical manifestations of myxoma are due to valvular “ball-valve” obstruction, embolization, or a syndrome of constitutional symptoms, such as fever and malaise. Sometimes fragmentation and systemic embolization calls attention to these lesions. Constitutional symptoms are probably due to the elaboration by some myxomas of the cytokine interleukin-6, a mediator of the acute-phase response. Echocardiography provides the opportunity to identify these masses noninvasively. Surgical

removal is usually curative; rarely, presumably because of incomplete excision, the neoplasm recurs months to years later.

Papillary fibroelastomas are curious, sea-anemone-shaped endocardial tumors; they are usually incidental but can embolize and thereby become clinically important. Clonal cytogenetic abnormalities have been reported, suggesting that fibroelastomas are benign neoplasms. They resemble the much smaller, usually trivial, Lambl excrescences that project from sites of valve apposition.

MORPHOLOGY

Papillary fibroelastomas are usually (>80%) located on valves, particularly the ventricular surfaces of semilunar valves and the atrial surfaces of atrioventricular valves. Each lesion, typically 0.5 to 2 cm in diameter, consists of a distinctive cluster of hair-like projections up to 1 cm in length. Histologically, the projections are covered by a surface endothelium surrounding a core of myxoid connective tissue with abundant mucopolysaccharide matrix and elastic fibers.

Rhabdomyomas are the most frequent primary tumor of the pediatric heart; they are commonly discovered in the first years of life during an evaluation for valve dysfunction or flow obstruction. Approximately one-half of cardiac rhabdomyomas are due to sporadic mutations; the other 50% of cases are associated with tuberous sclerosis (Chapter 28), which is caused by mutations in the tuberous sclerosis complex *TSC1* or *TSC2* tumor suppressor genes. The *TSC1* and *TSC2* proteins (hamartin and tuberlin, respectively) function in a complex that inhibits the activity of the mammalian target of rapamycin (mTOR), a kinase that stimulates cell growth and regulates cell size. *TSC1* or *TSC2* expression is often absent in tuberous sclerosis-associated rhabdomyomas, providing a mechanism for myocyte overgrowth. Because rhabdomyomas often regress spontaneously, they are considered by some to be hamartomas rather than true neoplasms.

MORPHOLOGY

Rhabdomyomas are gray-white myocardial masses that can be small or up to several centimeters in diameter. They are usually multiple and involve the ventricles preferentially, protruding into the lumen. Microscopically, they are composed of bizarre, markedly enlarged myocytes with large collections of glycogen. Routine histologic processing removes the glycogen and artifactually reduces the abundant cytoplasm to thin strands that stretch from the nucleus to the surface membrane, an appearance referred to as “**spider**” cells.

Metastatic Neoplasms

Metastatic tumors to the heart occur in roughly 5% of patients dying of cancer; the most frequent are lung and breast carcinomas, melanomas, leukemias, and lymphomas. Metastases can reach the heart and pericardium by retrograde lymphatic extension from the mediastinum (mostly carcinomas), by hematogenous seeding (many tumors), by direct contiguous extension (of intrathoracic tumors), or by direct venous extension (tumors of the kidney or liver). Clinical symptoms are most often associated with pericardial spread, which can cause symptomatic pericardial effusions or a mass-effect that is sufficient to restrict cardiac filling. Myocardial metastases are usually clinically silent or have non-specific features, such as a generalized defect in ventricular contractility or compliance; occasionally they may cause arrhythmias. Bronchogenic carcinoma or malignant lymphoma can infiltrate the mediastinum extensively, causing encasement, compression, or invasion of the superior

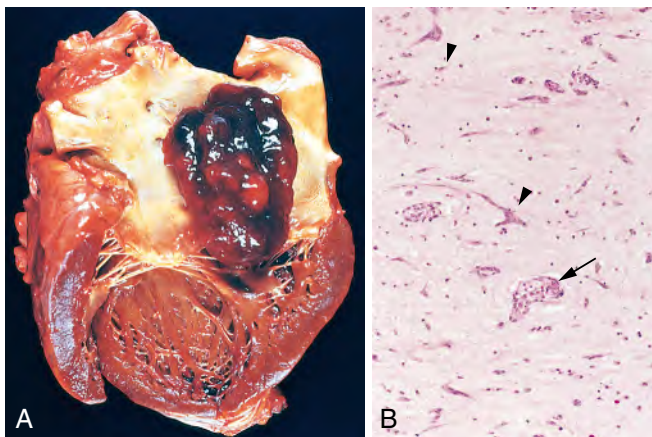


Fig. 12.37 Atrial myxoma. (A) A large sessile lesion arises from the region of the fossa ovalis and extends into the mitral valve orifice. (B) Abundant amorphous extracellular matrix contains scattered multinucleate myxoma cells (arrowheads) in various groupings, including abnormal vessel-like formations (arrow).

vena cava with resultant obstruction of blood flow from the head and upper extremities (superior vena cava syndrome). Renal cell carcinoma often invades the renal vein and can grow as a continuous column of tumor up the inferior vena cava lumen and into the right atrium, blocking venous return to the heart.

CARDIAC TRANSPLANTATION

Cardiac transplantation is performed for severe, intractable heart failure of diverse causes—most commonly DCM and IHD; over 5000 surgeries per year are performed worldwide. Three factors are primarily responsible for the dramatic improvement in allograft outcomes since the first transplant in 1967: (1) more effective immunosuppressive therapy (including the use of calcineurin inhibitors, mTOR inhibitors, glucocorticoids, and other agents), (2) careful selection of candidates, and (3) timely histopathologic diagnosis of acute allograft rejection by endomyocardial biopsy.

Of the major complications, allograft rejection is the primary problem requiring surveillance; routine endomyocardial biopsy is the only

reliable means of diagnosing acute cardiac rejection before substantial myocardial damage has occurred and at a stage that is reversible in most instances.

Classic cellular rejection is characterized by interstitial lymphocytic inflammation with associated myocyte damage; the histology resembles myocarditis (Fig. 12.38A). There may also be interstitial edema due to vascular injury, and local cytokine elaboration can affect myocardial contractility without necessarily eliciting myocyte damage.

Increasingly, antibody-mediated rejection is also recognized as a mechanism of injury; donor-specific antibodies directed against major histocompatibility complex proteins lead to complement activation and the recruitment of Fc-receptor-bearing cells. These donor-specific antibodies cause allograft compromise by inducing endothelial cell injury leading to microvascular damage with thrombosis. Antibody-mediated rejection shows mild perivascular edema and scattered adherent intravascular inflammatory cells; it is often confirmed by immunohistochemical staining for complement component C4d, a long-lived C4 fragment that is released during complement cascade activation (see Fig. 12.38B–C).

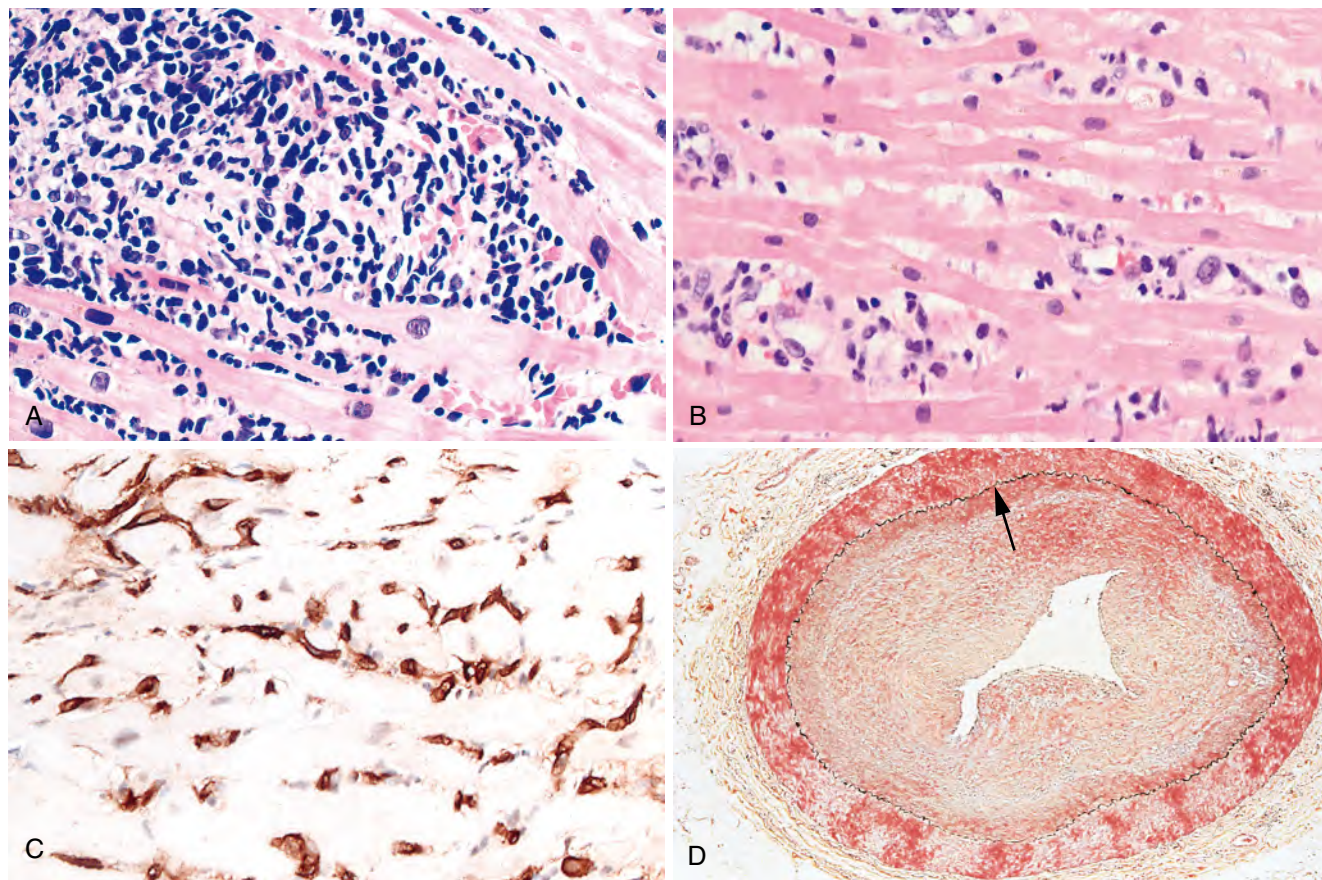


Fig. 12.38 Complications of heart transplantation. (A) Cellular allograft rejection, typified by lymphocytic infiltrate and cardiac myocyte damage. B and C, Antibody-mediated rejection. (B) Hematoxylin and eosin–stained section shows plump, activated endothelium with a sparse interstitial infiltrate; there is also perivascular edema. (C) Immunohistochemistry for the complement component C4d shows strong staining of capillaries (brown), reflecting complement activation and deposition. (D) Allograft vasculopathy, with marked diffuse concentric intimal thickening producing critical stenosis. The internal elastic lamina (arrow) and media are intact (Movat pentachrome stain, elastin black). (C, Courtesy Dr. Robert Padera, Brigham and Women's Hospital, Boston. D, From Salomon RN, Hughes CC, Schoen FJ, et al. Human coronary transplantation-associated arteriosclerosis. Evidence for chronic immune reaction to activated graft endothelial cells. *Am J Pathol.* 1991;138(4):791-798.)

Mild rejection may resolve spontaneously; prompt recognition of more severe episodes allows successful treatment by augmenting baseline levels of immunosuppression. Occasionally, aggressive anti-T cell or anti-B cell immunotherapy, with or without plasmapheresis, may be necessary.

Allograft vasculopathy is the single most important long-term limitation for cardiac transplantation. It is a late, progressive, diffusely stenosing intimal proliferation in allograft blood vessels, predominantly in the coronary arteries (see Fig. 12.38D), leading to ischemic injury. Within 5 years of transplantation, 50% of patients develop significant allograft vasculopathy, and virtually all patients have lesions within 10 years. The pathogenesis of allograft vasculopathy involves immunologic responses that induce local production of growth factors, which promote intimal smooth muscle cell recruitment and proliferation with ECM synthesis. Allograft vasculopathy is a particularly vexing problem because it can lead to silent MI (transplant patients have denervated hearts and do not typically experience angina), progressive CHF, or SCD.

Other postoperative complications include infections and malignancies, particularly Epstein-Barr virus-associated B-cell lymphomas that arise in the setting of chronic T-cell immunosuppression. Despite these problems, the overall outlook is reasonably good; the 1-year survival is 90%, and 5-year survival is greater than 70%.

CARDIAC DEVICES

Some of the most dramatic recent advances in cardiovascular medicine involve the use of mechanical devices. Thus, stents are now routinely deployed to maintain vascular patency (especially in the coronary arteries; Chapter 11), and endovascular conduit repairs are common approaches for treating abdominal aortic aneurysms. Implantable pacemakers and cardioverter defibrillators are indispensable for patients with cardiac rhythm disorders, and transcatheter aortic valve replacement (discussed earlier) has broadened the population that can benefit from valve replacement. Although impressive, these various devices are not without (somewhat predictable) risk. Mechanical failure can occur with any device, including strut fracture in stents, or lead fracture in pacing devices. Anything inserted into the cardiovascular system also has the potential to develop a thrombus that may become occlusive or embolize, and a foreign body is always a potential nidus for persistent (and difficult-to-treat) infection. Fortunately, the manifest benefits significantly outweigh the potential risks.

Ventricular Assist Devices

Perhaps the most impressive engineering advances have occurred in the development of ventricular assist devices (VADs). These allow the support of patients with cardiac failure caused by conditions ranging from transiently stunned myocardium after surgery, to MI, to progressive cardiomyopathy. Given the limited availability of hearts suitable for transplantation, VADs can provide important life-sustaining support for end-stage failure patients, allowing them to survive (and indeed to improve their overall conditioning status) until a suitable donor can be identified. Increasingly, VADs are also being implanted in patients who are not otherwise candidates for transplantation (e.g., a patient with a concurrent history of malignancy).

SELECTED READINGS

- Braunwald E. Cardiomyopathies: an overview. *Circ Res.* 2017;121:711-854. [The introductory article in an excellent series of nine articles covering many aspects of cardiomyopathy and myocarditis.]
- Buja L, Ottaviani G, Mitchell RN. Pathobiology of cardiovascular diseases: an update. *Cardiovasc Pathol.* 2019;42:44-53. [A review of cardiovascular pathobiology along with many useful references.]
- Cannata A, McDonagh TA. Heart failure with preserved ejection fraction. *New Engl J Med.* 2025;392:173-184. [A clinically oriented review of a frequently noted form of cardiac failure.]
- Crea F, Libby P. Acute coronary syndromes: the way forward from mechanisms to precision treatment. *Circulation.* 2017;136:1155-1166. [An excellent, well-written review on plaque erosion and rupture in acute coronary syndromes.]
- Garfinkel AC, Seidman JG, Seidman CE. Genetic pathogenesis of hypertrophic and dilated cardiomyopathy. *Heart Fail Clin.* 2018;14(2):139-146. [An excellent review of the genetic defects underlying many of the cardiomyopathies.]
- Kumar A, Avishay DM, Jones CR, et al. Sudden cardiac death: epidemiology, pathogenesis, and management. *Rev Cardiovasc Med.* 2021;22:147-158. [Good authoritative overview.]
- Libby P. *Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine.* 12th ed. Philadelphia: Elsevier; 2021. [An outstanding and authoritative text, with excellent sections on heart failure and atherosclerotic cardiovascular disease.]
- Liu A, Diller GP, Moons P, et al. Changing epidemiology of congenital heart disease: effect on outcomes and quality of care in adults. *Nat Rev Cardiol.* 2023;20(2):126-137. [An overview of trends in adult congenital heart disease.]
- Maleszewski JJ, Basso C, Bois MC. The 2021 WHO Classification of Tumors of the Heart. *J Thorac Oncol.* 2022;17:510-518. [The most recent consensus classification scheme for tumors of the heart and pericardium.]
- Moncla LM, Briand M, Bossé Y, et al. Calcific aortic valve disease: mechanisms, prevention and treatment. *Nat Rev Cardiol.* 2023;20:546-559. [Up-to-date review on the pathobiology of calcific degenerative valvular disease.]
- Morton SU, Quiat D, Seidman JG, Seidman CE. Genomic frontiers in congenital heart disease. *Nat Rev Cardiol.* 2022;19:26-42. [A nicely written discussion about the newest genetic insights regarding congenital heart disorders.]
- Nakamura M, Sadoshima J. Mechanisms of physiological and pathological cardiac hypertrophy. *Nat Rev Cardiol.* 2018;15:387-407. [Good introduction to cardiac hypertrophy and then detailed review of possible molecular mechanisms involved.]
- Olschewski A, Berghausen EM, Eichstaedt CA, et al. Pathobiology, pathology and genetics of pulmonary hypertension. *Int J Cardiol.* 2018;272S:4-10. [An excellent review of right-sided hypertension-associated pathology and pathobiology.]
- Sanchorawala V. Systemic light chain amyloidosis. *New Engl J Med.* 2024;390:2295-2307. [A good general review of amyloidosis.]
- Schoen FJ, Gotlib AI. Heart valve health, disease, replacement, and repair: a 25-year cardiovascular pathology perspective. *Cardiovasc Pathol.* 2016;25:341-352. [An excellent review on mechanisms of valve disease and therapeutic approaches.]
- Thygesen K, Alpert JS, Jaffe AS, et al. Fourth universal definition of myocardial infarction. *Circulation.* 2018;138:e618-e651. [An important consensus paper about definitions of myocardial infarction, with some discussion of pathobiology.]
- Tschöpe C, Ammirati E, Bozkurt, et al. Myocarditis and inflammatory cardiomyopathy: current evidence and future directions. *Nat Rev Cardiol.* 2021;18:169-193. [A very good perspective on the various forms of myocarditis and their underlying pathogenesis.]